

Where the Shortcut Sits

Location-dependent failure of region-of-interest masking as a shortcut defence

Full report: the six stages of the work

Abstract

Region-of-interest (ROI) masking — segmenting the lesion and discarding the background — is the most common defence against acquisition artifacts that act as shortcuts in medical image classifiers. Its rationale assumes that the artifact lies outside the ROI. This report asks what happens when it does not. It collects the six stages of the work in one document. Stage 1 re-implements the pilot study from its written specification, fixes the protocol and corrects the interval estimator. Stage 2 varies only the artifact’s overlap with the ROI and establishes the *location law*: masking helps when the artifact lies outside the ROI and loses its benefit — often turning harmful — once it overlaps it, because the retained artifact becomes what the model relies on. Stage 3 shows the same crossover with real artifacts in dermoscopy, thyroid and ovarian ultrasound and capsule endoscopy, and marks chest radiography as a boundary. Stage 4 shows that the crossover is not explained by which images carry the artifact. Stage 5 shows that it matters on unaltered data and at clinical operating points, replicates across encoders, sizes, fine-tuning, untrained models and new patients, and is explained by a two-parameter theory. Stage 6 develops and tests what to do instead of masking and concludes that the right remedy depends on what carries the shortcut. Every number is generated from the result files of the repository and traced by an automatic audit.

Contents

Overview	5
1 Stage 1 — The problem, the pilot and the protocol	7
1.1 Purpose of this stage	7
1.2 The clinical problem	7
1.2.1 Shortcut artifacts	7
1.2.2 The standard defence and its hidden assumption	7
1.2.3 Research question	7
1.3 Why the study is designed this way	8
1.4 Data	8
1.5 The protocol	9
1.5.1 Leakage: grouped versus random splits	10
1.5.2 Uncertainty: the corrected (crossed) bootstrap	11
1.5.3 Pre-registration	12
1.6 The pilot and its re-implementation	13
1.6.1 What the pilot tested	13
1.6.2 How it was re-implemented	13
1.7 Results	14

1.7.1	Replication overview	14
1.7.2	The controlled sweep (E1–E3, E8–E9)	16
1.7.3	Real hair (E10, E12–E15)	17
1.7.4	E12: the overlap-contrast design	18
1.7.5	Audit of the pilot’s written claims	18
1.7.6	The mismatches	18
1.8	What failed or changed in this stage	19
1.9	Anticipated questions	19
1.10	Conclusion	20
1.11	Questions for the supervisor	20
2	Stage 2 — The location law and its mechanism	21
2.1	Where this stage starts	21
2.2	Questions	21
2.2.1	Why these experiments, and what was rejected	21
2.3	Methods	22
2.4	Experiments	23
2.5	Results	23
2.5.1	The dermoscopy sweep: every arm, every overlap	23
2.5.2	Mechanism	24
2.5.3	Other modalities	28
2.5.4	Sensitivity to the interval estimator	31
2.6	What failed or changed	31
2.7	Anticipated questions	32
2.8	Conclusion	33
2.9	Questions for the supervisor	33
3	Stage 3 — Real artifacts across modalities	34
3.1	Where this stage starts	34
3.2	Questions	34
3.3	Methods	34
3.3.1	Cohorts	34
3.4	Experiments	36
3.5	Results	36
3.5.1	Cohorts and artifact masks	36
3.5.2	The crossover with real artifacts	38
3.5.3	The four location tests	39
3.5.4	Every arm, and the environments	40
3.5.5	What carries the hair shortcut?	41
3.5.6	The exception among encoders: DermLIP	42
3.5.7	Chest radiography: the boundary	42
3.5.8	Datasets considered and not used	43
3.5.9	Sensitivity to the interval estimator	44
3.6	What failed or changed	44

3.7	Anticipated questions	45
3.8	Conclusion	45
3.9	Questions for the supervisor	46
4	Stage 4 — Is it location, or which images carry the artifact?	47
4.1	Where this stage starts	47
4.2	Questions	47
4.3	Methods	47
4.4	Experiments	48
4.5	Results	49
4.5.1	The traps contain different images	49
4.5.2	Matching	49
4.5.3	Regression adjustment on the matched sample (A2)	50
4.5.4	The same artifact inside and outside the ROI	52
4.5.5	Robustness of the crossover	53
4.5.6	Sensitivity to the interval estimator	54
4.6	What failed or changed	55
4.7	Anticipated questions	56
4.8	Conclusion	56
4.9	Questions for the supervisor	57
5	Stage 5 — Practice, breadth and theory	58
5.1	Where this stage starts	58
5.2	Questions	58
5.3	Methods	58
5.4	Experiments	60
5.5	Results	60
5.5.1	Natural test sets	60
5.5.2	Clinical consequences: thyroid operating points	61
5.5.3	Replication breadth	62
5.5.4	The DermLIP exception	65
5.5.5	The theory, judged by held-out error	65
5.5.6	Sensitivity to the interval estimator	66
5.6	What failed or changed	67
5.7	Anticipated questions	68
5.8	Conclusion	68
5.9	Questions for the supervisor	68
6	Stage 6 — What to do instead of masking, and the state of the thesis	70
6.1	Where this stage starts	70
6.2	Questions	70
6.3	Method: from insert-then-erase to U-MtE	70
6.4	Experiments	72
6.5	Results	72

6.5.1	Remedies in the real traps	72
6.5.2	Which model is most robust?	74
6.5.3	The 16-test primary family	75
6.5.4	Remedies in the controlled sweeps	75
6.5.5	Erasure versus augmentation, protection, and other contrasts	76
6.5.6	How many directions must be erased?	76
6.5.7	Encoder size	77
6.5.8	Published baselines and alternatives	77
6.5.9	Robustness of the remedy tests to leakage groups	79
6.5.10	Fully fine-tuned networks	80
6.5.11	Unaltered data	80
6.5.12	The decision guide	81
6.6	Validating the artifact labels: the clinician audit	81
6.7	The thesis manuscript	82
6.7.1	Draft submission statements (for the author to confirm)	83
6.8	Limitations of the work as a whole	83
6.9	What failed or changed	83
6.10	Anticipated questions	84
6.11	Conclusion	85
6.12	Questions for the supervisor	85
6.13	Remaining work before submission	85
7	Conclusion	86
7.1	What the work shows	86
7.2	Contributions	87
7.3	Claims the work does not make	87
7.4	Limitations and open work	87
7.5	Take-home message	87

Overview

The question Does the benefit of ROI masking depend on where the artifact sits relative to the ROI — and if so, through which mechanism, is it causal, does it matter clinically, and what should replace masking when it fails?

How the report is organised Each of Sections 1–6 is one stage of the work, in the order in which it was carried out. Each stage opens with a one-sentence summary, states its questions and the reasons for its design, describes its methods and registrations, reports its results with figures and tables, lists what failed or changed, answers the questions a reader is likely to ask, draws its conclusion and lists the decisions it raises. Section 7 draws the conclusions of the work as a whole.

Table 1: The six stages at a glance.

Stage	Topic	Question	Answer in one line
1	Problem, pilot, protocol	Does the pilot reproduce, and is its protocol sound?	Yes for the central sweep result (76 of 87 numbers match); the interval estimator was too narrow and was replaced.
2	Location law and mechanism	Does masking’s benefit depend on overlap, and why?	Yes in 10 of 11 controlled sweeps; the retained artifact gains reliance ($\approx 2.1\times$ for the ruler).
3	Real artifacts	Does it hold for artifacts clinicians produce?	Crossover positive in 9 of 10 cohort–encoder pairs; hair is correlate-carried; chest radiography is a boundary.
4	Causal checks	Is it location, or which images carry the artifact?	Location: survives matching and adjustment (3/3) and the same-image transplant.
5	Practice, breadth, theory	Does it matter clinically, how general is it, can it be predicted?	Lower thyroid sensitivity at every operating point; replicates widely incl. new patients; theory error 0.052 vs 0.129.
6	Remedies and thesis	What to do instead of masking?	It depends on what carries the shortcut; no remedy transfers uniformly to unaltered data.

Notation

$Y \in \{0, 1\}$	diagnosis: 1 = melanoma (dermoscopy), malignant nodule (thyroid), tumour with a solid component (ovary), erosion (capsule), pneumothorax or finding (chest radiograph).
$A \in \{0, 1\}$	artifact present: ruler, hair, sonographer caliper, capsule debris, chest tube, device or drain — synthetic (pasted at a controlled position) or real.
ROI	region of interest: lesion, nodule, tumour, union of expert lesion boxes, lungs; drawn in green ; artifacts in red .
r	overlap $ \text{artifact} \cap \text{ROI} / \text{artifact} $. Trap A: $r \geq 0.5$ (artifact in the ROI); Trap B: $r < 0.1$ (outside).
Environments	training and correlated test $P(A=1 Y=1) = 0.9$, $P(A=1 Y=0) = 0.1$; reversed test 0.1/0.9; clean test: artifact independent of Y (real artifacts: half of each class) or absent (synthetic artifacts).
ERM, mask	linear head on the frozen features of the original image / of the ROI-masked image.
Crossover	$[\text{mask} - \text{ERM}]_{\text{outside}} - [\text{mask} - \text{ERM}]_{\text{inside}}$ of reversed-test AUROC: in the controlled sweeps and the transplant the <i>location interaction</i> on the same images; in the real traps Trap B – Trap A.
CI	95% crossed seed×image bootstrap interval (Section 1); analyses whose per-image predictions were not saved are shown as point estimates and marked “not re-estimated”.
$ \Delta p $	same-head counterfactual sensitivity: change of the predicted probability when the artifact is inserted into the same image.
SMD	standardised mean difference of a covariate between two groups ($ \text{SMD} > 0.1$: imbalanced).
Hard pairs	positive–negative pairs whose artifact status conflicts with the test set’s own artifact–label association.
OP1–OP5	thresholds fixed on validation data: maximal balanced accuracy; specificity ≥ 0.80 , ≥ 0.90 ; sensitivity ≥ 0.80 , ≥ 0.90 .
U-MtE	mask-then-erase: ROI masking followed by erasure of the feature subspace along which generic synthetic overlays move masked images.
$\min(\text{rev}, \text{corr})$	robustness summary: the lower of reversed and correlated AUROC († : the arm flips the shortcut).

1 Stage 1 — The problem, the pilot and the protocol

In one sentence. Region-of-interest (ROI) masking is the standard defence against acquisition artifacts that act as shortcuts, but its rationale assumes that the artifact lies outside the ROI; the pilot study suggested that masking turns from helpful to harmful when the artifact overlaps the lesion. This stage re-implements the pilot from its written specification, compares every reported number (76 of 87 match, 5 do not), fixes the experimental protocol, and replaces the pilot’s interval estimator, which was too optimistic, by a crossed seed×image bootstrap.

1.1 Purpose of this stage

This stage covers the first block of work: understanding the clinical problem, re-deriving the pilot study (experiments E1–E15) from scratch, and fixing a protocol that is safe against the two most common ways a shortcut study fools itself — data leakage between training and test images, and too-narrow uncertainty intervals. The pilot’s claims are treated as hypotheses: each reported number was re-computed by new code written from the pilot’s specification, and the two sets of numbers were compared one by one. Nothing in this stage depends on results that were not re-derived.

1.2 The clinical problem

1.2.1 Shortcut artifacts

Deep classifiers exploit any feature that predicts the label in the training data, including features that are not evidence of disease (Geirhos et al., 2020; DeGrave et al., 2021). In dermoscopy, surgical skin markings (Winkler et al., 2019) and ruler scale bars (Winkler et al., 2021) changed the melanoma predictions of a market-approved system; hair and dark corners are learned as well (Wang et al., 2024; Pewton et al., 2024). In chest radiography, pneumothorax classifiers detect the chest drain that treats the condition (Oakden-Rayner et al., 2020; Jiménez-Sánchez et al., 2023), and a pneumonia model learned the hospital that took the image (Zech et al., 2018). Such associations are produced by the clinical workflow itself: suspicious lesions are measured and marked, so the measuring device appears more often on malignant cases. They are therefore present in every retrospective dataset, and a model that relies on them fails exactly on the patients whose workflow differs — for example a melanoma that was photographed without a ruler (Brown et al., 2023).

1.2.2 The standard defence and its hidden assumption

The most common defence is to restrict the classifier to the ROI: segment the lesion and discard the background (“ROI masking”). It is attractive because it needs only a lesion mask, which many pipelines compute anyway, and no knowledge of the artifact. Its rationale is that the shortcut lives in the background. Rulers, hair and ink markings do not respect that assumption: they are placed on, across or next to the lesion. When an artifact overlaps the ROI, masking cannot remove it — and it removes everything *else* that competes with it.

1.2.3 Research question

Does the benefit of ROI masking depend on where the artifact sits relative to the ROI — and if so, by how much, through which mechanism, and does it matter for real artifacts?

For this stage the question is narrower: **does the pilot’s evidence for a location-dependent effect of masking survive an independent re-implementation, and is the protocol that produced it sound?**

1.3 Why the study is designed this way

- **Control the overlap first.** The pilot pasted a synthetic ruler at a controlled overlap with the lesion (0–100%, Figure 1). This isolates location from everything else: the same images, the same ruler, only its position changes. Starting with real artifacts only was rejected, because the images that carry an artifact on the lesion differ from those that carry it beside the lesion, so a location effect could not be separated from a population difference.
- **Frozen foundation-model features and linear heads.** This is how foundation models are typically used on small medical datasets; it makes every arm comparable (same features, only the head or the input changes); and it is cheap enough to repeat over seeds, overlaps and arms. End-to-end fine-tuning multiplies the cost by about two orders of magnitude and was not part of the pilot.
- **A trap with a reversed test.** Training with a strong, known artifact–label association (0.9/0.1) and testing on the reversed association (0.1/0.9) measures reliance on the artifact directly: a model that uses it must fall on the reversed test (Figure 3). The correlated and clean tests measure what a mitigation costs. A single natural test set was rejected as the main design because its shortcut is too weak to measure a location effect precisely.
- **AUROC as the primary metric.** It does not depend on a decision threshold, so a mitigation cannot look good merely by moving the threshold. Thresholds are still fixed (on clean validation data) for secondary metrics.
- **An artifact-free comparison group for real artifacts (E13).** The pilot’s first real-hair design (E12) compared in-lesion hair directly with out-of-lesion hair; E13 instead builds two traps that share one hair-free group. Section 1.7.4 shows why the difference matters.

1.4 Data

Controlled cohort (ISIC 2018) ISIC 2018 Task 1/2 images with manual lesion masks (Codella et al., 2019); melanoma labels from the union of the ISIC 2016 and 2017 ground truth (2017 takes precedence; three images with conflicting labels dropped). A 65×19 px ruler (at 518 px) must be placeable at 0, 25, 50, 75 and 100% overlap with the lesion within 10 percentage points; 2,437 images qualify. The excluded images have very small lesions (a ruler cannot lie fully inside them) and more melanomas; their placements are frozen in `frozen/placements_518.65x19.json` and reused by every run.

Real-hair cohort (ISIC 2019) 25,331 dermoscopic images from three hospitals (HAM10000 (Tschandl et al., 2018), BCN20000 (Combalia et al., 2019), MSK), melanoma versus all other diagnoses. Hair masks: the published semi-automatic hair/ruler masks of Wegley et al. (2026). Lesion masks: HAM10000 manual masks where available, otherwise a U-Net trained only on ISIC 2018 Task 1 (Table 7). Leakage groups: lesion identifier $\cup$ perceptual hash.

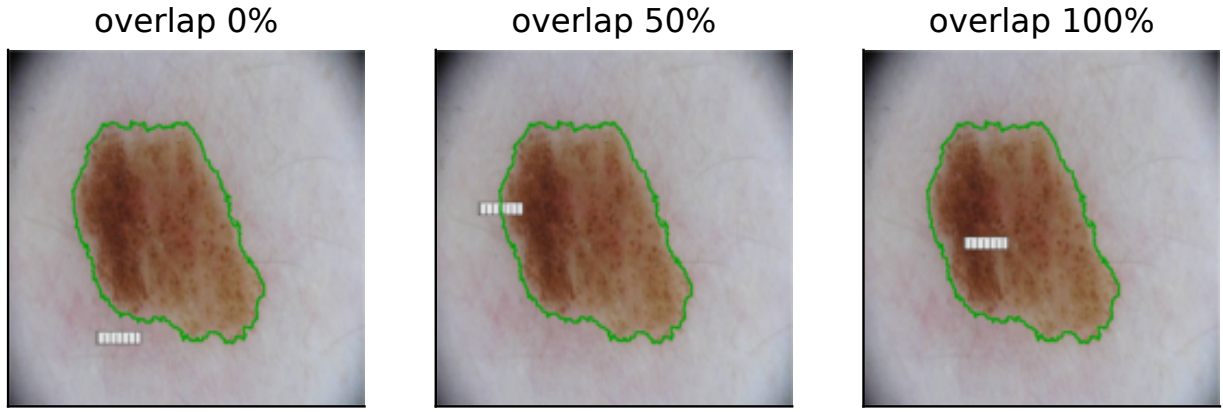


Figure 1: The controlled sweep: the same ISIC 2018 image with the synthetic ruler at 0, 50 and 100% overlap with the lesion (**green**: manual lesion mask). ISIC images CC BY-NC 4.0.



Figure 2: The real hair traps (ISIC 2019): hair on the lesion (Trap A) and beside it (Trap B); the green line is the lesion outline. The hair masks are not drawn because the published masks carry no licence that allows redistribution. ISIC images CC BY-NC 4.0.

1.5 The protocol

Table 2 lists every component with the reason it was chosen; Figure 3 shows the four test environments.

Table 2: The protocol fixed in this stage.

Component	Choice	Why
Environments	training and correlated test $P(A=1 Y=1) = 0.9$, $P(A=1 Y=0) = 0.1$; reversed test 0.1/0.9; clean test: no artifact (synthetic) or artifact independent of the label (real)	a strong, known shortcut whose use is exposed when the association is reversed
Primary metric	reversed-test AUROC; clean and correlated AUROC reported beside it	threshold-free; a model that uses the artifact must fall on the reversed test
Encoders	frozen DINOv2 ViT-B/14 at 518 px (CLS token), DINOv2 at 224 px, DermLIP (dermatology CLIP model) at 224 px	how foundation models are used on small medical data; every arm sees the same features
Heads	logistic regression (liblinear, class-balanced); $C \in \{0.01, 0.1, 1, 10\}$ and the decision threshold (maximal balanced accuracy) chosen on clean validation data only	no test or artifact information enters model selection
Arms	ERM; ROI masking; oracle inpainting of the artifact; group-balanced training; DFR; paired and unpaired LEACE; GroupDRO; prevalence calibration	masking is compared with methods that target the artifact itself
Splits	grouped by lesion identifier $\cup$ perceptual hash (Hamming ≤ 8); no group crosses a split	near-duplicates across a split inflate shortcut measurements (Sec. 1.5.1)
Seeds	3 seeds (42, 123, 456) for the synthetic sweeps; 5 seeds $\times$ 5 grouped folds for the real hair traps	training and split variation are both part of the uncertainty
Uncertainty	crossed seed $\times$ image bootstrap, 10,000 replicates, percentile 95% intervals	seeds share test images (Sec. 1.5.2)
Registration	design, claims and support criteria committed to the repository before each verdict-producing run	separates confirmation from exploration

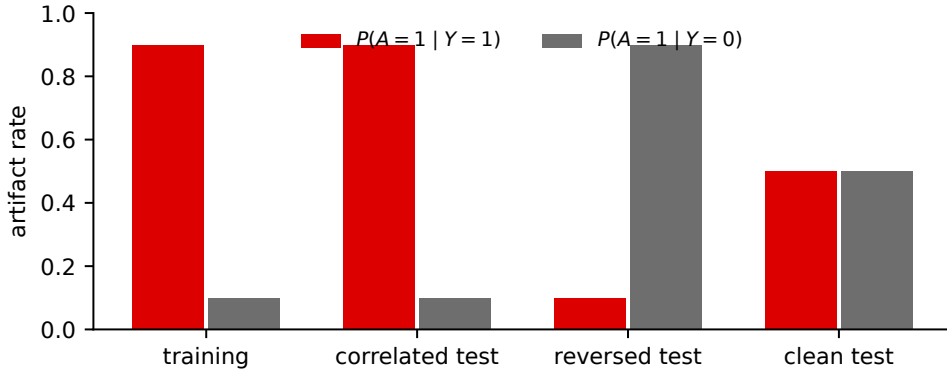


Figure 3: The trap environments. The training set and the correlated test share a strong artifact-label association; the reversed test inverts it; the clean test removes it. A model that uses the artifact gains on the correlated test and collapses on the reversed one.

Arms *ERM*: logistic head on the unmasked image. *ROI masking*: pixels outside the lesion mask are set to the image mean before feature extraction, at training and at test time. *Oracle inpainting*: the known artifact pixels are inpainted (Telea, 2004) — an upper bound that needs the artifact mask. *Group-balanced*: the four artifact $\times$ label cells are equally weighted. *DFR*: the last layer is re-fitted on a group-balanced subset of the validation fold (Kirichenko et al., 2023). *LEACE*: closed-form linear erasure of the artifact concept (Belrose et al., 2023), either *paired* (original versus artifact-removed versions of the same images) or *unpaired* (image-level artifact labels). *GroupDRO* (Sagawa et al., 2020). *Prevalence calibration* (Kina and Petersen, 2026). Balanced training, DFR, GroupDRO and unpaired LEACE need image-level artifact labels; inpainting and paired LEACE need artifact masks; masking needs only the lesion mask.

1.5.1 Leakage: grouped versus random splits

Near-duplicate photographs of one lesion are common in ISIC. If one copy is in training and another in test, the test measures memorisation. Every split therefore groups images by lesion identifier and perceptual hash (64-bit pHash, Hamming distance ≤ 8) and checks that no group crosses a split. To

measure what this protects against, the controlled sweep was repeated with five random image-level splits, everything else fixed.

Result Random splits leave near-duplicate images on both sides of the split and inflate the measured ERM shortcut gap by 18% (no overlap) and 10% (full overlap), and shift both masking effects (Table 3); every analysis therefore uses grouped splits.

Table 3: Leakage: grouped versus random image-level splits (controlled sweep, DINOv2@518).

Overlap	Split	Near-duplicate pairs across split	ERM gap (corr – rev)	mask – ERM (rev)
0%	grouped	0	0.241	+0.184
	random (mean of 5)	93	0.284	+0.223
100%	grouped	0	0.421	–0.129
	random (mean of 5)	93	0.465	–0.088

1.5.2 Uncertainty: the corrected (crossed) bootstrap

Every comparison is a paired difference in AUROC between two arms on the same test images, averaged over training seeds (or cross-validation folds). The pilot’s estimator resampled the seeds and then, *independently within each sampled seed*, the test images. In these designs, however, every seed scores the *same* test images: the controlled sweep uses one fixed test split, and in the hair traps the pooled cross-validation folds of every seed cover the same image pool. Resampling the images independently per seed then treats one test set as several and divides the test-sampling variance by the number of seeds, so the intervals are too narrow.

Why it matters With three seeds and one test set, the per-seed estimator behaves as if the test set were three times larger than it is. The problem is not specific to this project: any study that averages several training seeds on one test set and resamples the images per seed has it.

The crossed estimator Each replicate resamples the seeds with replacement and draws one Poisson(1) weight per test *image*, shared by every seed (and both arms) in which that image appears; the per-seed statistic is recomputed with those weights and averaged (`wtss.stats_crossed`; 10,000 replicates; percentile intervals). The image weights make the test-set variance count once, as it should; the seed resampling keeps the training variance. The estimator was registered before any interval was recomputed (85da735, 2026-09-27 23:43; applied everywhere under c45d482, 2026-09-28 00:35).

Consequences for the pilot On the 178 pilot intervals that both estimators can compute, the crossed interval is wider by a median factor of 1.45 (interquartile range 1.32–1.61; Figure 4, Table 4). 7 intervals that excluded zero with the pilot’s estimator no longer do, and 1 now excludes zero (Table 5). Point estimates are identical: only the uncertainty changed.

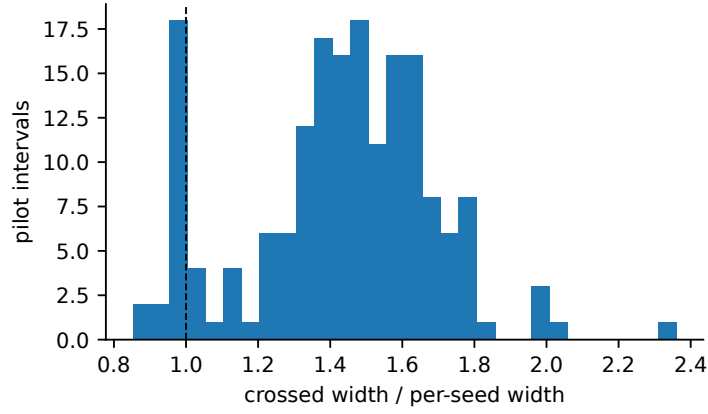


Figure 4: Width of the crossed interval divided by the width of the pilot's per-seed interval, for every pilot contrast (same predictions).

Table 4: The pilot's analyses under both interval estimators (same predictions, same point estimates; 10,000 replicates each). Width ratio > 1: the crossed interval is wider.

Result group	Intervals	Excluded with the estimator	0 only per-seed	Excluded with the estimator	0 only crossed	Median width ratio (crossed / per-seed)
analysis/E14_area_ratio	4	0		0		1.63
spec_e13/dermlip224_spec	29	1		0		1.45
spec_e13/dino518_e12_contrast	3	0		0		1.44
spec_e13/dino518_spec	28	0		1		1.54
synthetic/isc2018	114	6		0		1.44
all	178	7		1		1.45

Table 5: The pilot contrasts whose verdict (interval excludes 0 or not) depends on the estimator. Every other contrast keeps its verdict.

Result	Contrast	Crossed estimate [95% CI]	Verdict vs per-seed estimator
synthetic/isc2018/ dino518_ruler_fixed_corr_main	0.25 / mask / erm / test_rev	-0.059 [-0.127, +0.008]	no longer excludes 0
synthetic/isc2018/ dino518_ruler_fixed_occlusion_occlusion	0.0 / mask / erm / test_uncorr	+0.036 [-0.016, +0.086]	no longer excludes 0
synthetic/isc2018/ dino518_ruler_fixed_occlusion_occlusion	0.5 / mask / erm / test_uncorr	+0.041 [-0.010, +0.090]	no longer excludes 0
synthetic/isc2018/ dino518_ruler_fixed_occlusion_occlusion	0.75 / mask / erm / test_uncorr	+0.037 [-0.013, +0.086]	no longer excludes 0
synthetic/isc2018/ dino518_ruler_fixed_occlusion_occlusion	1.0 / mask / erm / test_uncorr	+0.039 [-0.010, +0.091]	no longer excludes 0
synthetic/isc2018/ dino224_ruler_fixed_corr_main	0.25 / mask / erm / test_rev	-0.049 [-0.117, +0.018]	no longer excludes 0
spec_e13/dino518_spec	trapA / mask / clean / all / mask / erm	-0.023 [-0.045, -0.001]	now excludes 0
spec_e13/dermlip224_spec	trapB / leace_unpaired / clean / all / leace_unpaired / erm	-0.019 [-0.046, +0.007]	no longer excludes 0

1.5.3 Pre-registration

Before every verdict-producing experiment, a document stating the design, the claims and what counts as support is committed to the repository; results are committed afterwards, and a claim is reported whichever way it goes. The pilot itself preceded this rule; its written specification (`docs/REPLICATION_SPEC.md`, first commit 2699d51, 2026-09-26 15:47) serves as its registration for the replication. **Caveat:** commit times are set by the machine that makes the commit and are

not independent proof of order; registrations that are also pushed to GitHub before running carry GitHub’s time stamp as an external reference.

1.6 The pilot and its re-implementation

1.6.1 What the pilot tested

post hoc (the pilot preceded pre-registration; its specification was written down afterwards and re-implemented). Table 6 lists the fifteen experiments. The pilot made five claims: (i) masking removes an artifact that lies outside the lesion and makes the model worse than doing nothing when it lies inside; (ii) the harm is governed by how much of the artifact survives the mask (retention), not by occlusion or lost context; (iii) methods that need only an image-level artifact label (balanced training, DFR) fix both cases; (iv) linear erasure is insufficient for real artifacts under the sound (paired) protocol; (v) all of this holds for real hair with pre-registered criteria on two encoders.

Table 6: The pilot experiments (specification: `docs/REPLICATION_SPEC.md`).

Exp.	What it tests
E1	controlled ruler sweep, DINOv2@518: mask – ERM at 0–100% overlap; location interaction
E2	the same at 224px (DINOv2) and with the dermatology model DermLIP
E3	appearance randomisation (colour, opacity, ticks, scale of the ruler)
E4	occlusion control: ruler present but uncorrelated with the label (50/50)
E5	mask dilation: restoring ruler pixels around the lesion (retention)
E6	same-head counterfactual sensitivity $ \Delta p $ to inserting the ruler
E7	lesion-size strata (artifact-to-lesion area ratio)
E8	oracle inpainting and a prediction-consistency repair
E9	bridge: group-balanced training, DFR, GroupDRO, paired LEACE at 100% overlap
E10	hair decodability from the embedding; paired LEACE on real hair
E11	atlas of real artifacts (hair, ink, vignetting) in ISIC 2019 / HAM10000; vignetting trap
E12	overlap-contrast trap: in-lesion versus out-of-lesion hair, no artifact-free group
E13	real hair traps (Trap A in-lesion, Trap B out-of-lesion, shared hair-free group), ISIC 2019
E14	area-ratio strata of the real hair trap
E15	E13 with DermLIP

1.6.2 How it was re-implemented

registered before results (specification `docs/REPLICATION_SPEC.md`, 2699d51, 2026-09-26 15:47). The pilot’s code was not available for most experiments (E9–E15), so every experiment was re-implemented from the written specification in a new code base (`src/wtss`). A number *matches* (MATCH) if it has the same sign, the same verdict on whether its interval excludes zero, and differs by at most 0.02; SIGN+CI if sign and verdict agree but the difference is larger; MISMATCH otherwise. Where the specification was silent or impossible to follow, the choice made is recorded with its reason (Table 7). Two choices matter most: the hair-free group was defined as at most 30 hair pixels in the native mask — the only threshold that reproduces the pilot’s outcome-free cell counts — and, on the full ISIC 2019 cohort, pHash distance ≤ 8 chains unrelated lesions into one giant component, so the real-hair traps apply it within each trap pool, as the pilot’s counts imply.

Table 7: Where the re-implementation deviates from the pilot’s written specification, and why.

#	Specification	Re-implementation	Reason and effect
1	GPU V100 32 GB	RTX 3090 24 GB	Frozen features; per-seed AUROCs of the rerun match the archive within 0.0005 (E1, E8, E9).
2	Feature extraction per (seed, env)	per *view* (image $\times$ method $\times$ overlap), environments assembled by indexing	Mathematically identical (presence is a function of image/label/seed/env only); $\sim 10\times$ fewer backbone passes.
3	pHash ≤ 8 on the full ISIC 2019 cohort (repo-default analysis)	spec E13 run: pHash ≤ 8 within each matched trap pool (as the spec’s counts imply); exploratory runs: lesion_id $\cup$ pHash ≤ 2 on all 25,331	On all 25,331 images pHash ≤ 8 chains unrelated lesions into one component of 10,974 images (same-lesion precision at distance 8: 2%).
4	Hair-free A = 0 group (threshold not stated in spec)	≤ 30 hair/ruler pixels in the native-resolution archive mask	Calibrated on outcome-free counts only: reproduces A0.Y0 709 (spec 708), A0.Y1 157 (157) and all six source proportions to 3 decimals.
5	Archive mask file names	matched on zero-padded numeric id	Archive names are irregular (<code>_downsampled_mvig</code> , <code>_downsam_inkmark</code> , typos, truncated ids). 24 truncated ink ids flag their 10 candidate images as ink-uncertain (exploratory runs only).
6	Images without HAM/BCN lesion_id	source = MSK	Spec lists three sources; ISIC 2019’s non-HAM/BCN images are MSK (incl. <code>_downsampled</code>).
7	U-Net (Ronneberger 64–1024, Task 1 only)	same architecture (+BatchNorm), 256 px, 40 epochs, Adam 1e-4, BCE+Dice, 20% held-out pHash groups	Training hyper-parameters are not in the spec.
8	Derm1M from archive	fresh clone of the official repo	Archived copy lacks <code>bpe_simple_vocab_16e6.txt.gz</code> (.gz files were not archived); DermLIP cannot load without it.
9	§3.1 “leakage inflated the gap by 27%”	not reproduced from the archive	Pre-correction pilot used the identical split; matched-resolution gap ratio 0.96–0.97. A controlled leakage experiment is run instead (<code>scripts/run_leakage_experiment.py</code>).

1.7 Results

1.7.1 Replication overview

Of the 87 numbers the pilot reported, 76 MATCH, 6 agree in sign and significance but differ by more than 0.02, and 5 MISMATCH (Table 8). Before the interval estimator was corrected the ledger read 73 / 12 / 3 of 88; the difference is entirely due to the wider intervals (Section 1.7.6).

Table 8: Replication ledger: every number the pilot reported, its re-derived value with the crossed 95% interval, and the verdict. The pilot’s own intervals came from the per-seed estimator (Sec. 1.5.2) and are not reproduced; only its point values are shown.

Exp.	Quantity	Pilot (point)	Replication (crossed CI)	Verdict
E1	mask – ERM rev @0%	+0.184	+0.184 [+0.117, +0.250]	MATCH
E1	mask – ERM rev @25%	-0.058	-0.059 [-0.127, +0.008]	MISMATCH
E1	mask – ERM rev @50%	-0.097	-0.097 [-0.164, -0.030]	MATCH
E1	mask – ERM rev @75%	-0.111	-0.111 [-0.181, -0.043]	MATCH
E1	mask – ERM rev @100%	-0.128	-0.129 [-0.195, -0.064]	MATCH
E1	interaction 0% – 100% (mask)	+0.312	+0.313 [+0.258, +0.370]	MATCH
E2	dino224 mask – ERM @0%	+0.364	+0.361 [+0.275, +0.444]	MATCH
E2	dino224 mask – ERM @25%	-0.040	-0.049 [-0.117, +0.018]	MATCH
E2	dino224 mask – ERM @50%	-0.076	-0.081 [-0.143, -0.019]	MATCH
E2	dino224 mask – ERM @75%	-0.102	-0.106 [-0.166, -0.044]	MATCH
E2	dino224 mask – ERM @100%	-0.092	-0.095 [-0.156, -0.035]	MATCH
E2	dermlip224 mask – ERM @0%	+0.160	+0.161 [+0.102, +0.222]	MATCH
E2	dermlip224 mask – ERM @25%	-0.291	-0.289 [-0.366, -0.215]	MATCH
E2	dermlip224 mask – ERM @50%	-0.300	-0.298 [-0.371, -0.225]	MATCH
E2	dermlip224 mask – ERM @75%	-0.313	-0.309 [-0.378, -0.239]	MATCH
E2	dermlip224 mask – ERM @100%	-0.268	-0.263 [-0.348, -0.180]	MATCH
E3	variable ruler mask – ERM @0%	+0.147	+0.146 [+0.082, +0.211]	MATCH
E3	variable ruler mask – ERM @50%	-0.094	-0.094 [-0.168, -0.021]	MATCH
E3	variable ruler mask – ERM @100%	-0.155	-0.155 [-0.228, -0.084]	MATCH
E3	variable ruler balanced – ERM @0%	+0.103	+0.102 [+0.068, +0.140]	MATCH
E3	variable ruler balanced – ERM @50%	+0.171	+0.171 [+0.132, +0.213]	MATCH
E3	variable ruler balanced – ERM @100%	+0.222	+0.222 [+0.181, +0.265]	MATCH

Exp.	Quantity	Pilot (point)	Replication (crossed CI)	Verdict
E4	occlusion (50/50) mask – ERM @0%	+0.035	+0.036 [-0.016, +0.086]	MISMATCH
E4	occlusion (50/50) mask – ERM @50%	+0.033	+0.041 [-0.010, +0.090]	MATCH
E4	occlusion (50/50) mask – ERM @100%	+0.039	+0.039 [-0.010, +0.091]	MISMATCH
E5	dilate0 – ERM @50%	-0.097	-0.097 [-0.164, -0.031]	MATCH
E5	dilate10 – ERM @50%	-0.138	-0.138 [-0.203, -0.074]	MATCH
E5	dilate25 – ERM @50%	-0.153	-0.153 [-0.221, -0.089]	MATCH
E5	dilate50 – ERM @50%	-0.142	-0.142 [-0.206, -0.083]	MATCH
E5	dilate0 – ERM @100%	-0.128	-0.129 [-0.195, -0.066]	MATCH
E5	dilate10 – ERM @100%	-0.143	-0.144 [-0.206, -0.082]	MATCH
E5	dilate25 – ERM @100%	-0.144	-0.144 [-0.209, -0.082]	MATCH
E5	dilate50 – ERM @100%	-0.138	-0.138 [-0.200, -0.082]	MATCH
E6	$ \Delta p $ erm @100%	+0.235	+0.235	MATCH
E6	$ \Delta p $ mask @100%	+0.490	+0.491	MATCH
E6	$ \Delta p $ mask @0%	+0.000	+0.000	MATCH
E6	$ \Delta p $ erm @0%	+0.128	+0.128	MATCH
E6	$ \Delta p $ erm @50%	+0.195	+0.195	MATCH
E7	mask – ERM, large lesions @0%	+0.249	+0.249	MATCH
E7	mask – ERM, medium lesions @0%	+0.214	+0.215	MATCH
E7	mask – ERM, small lesions @0%	+0.150	+0.150	MATCH
E7	mask – ERM, large lesions @50%	+0.087	+0.087	MATCH
E7	mask – ERM, medium lesions @50%	-0.129	-0.129	MATCH
E7	mask – ERM, small lesions @50%	-0.325	-0.326	MATCH
E7	mask – ERM, large lesions @100%	+0.064	+0.063	MATCH
E7	mask – ERM, medium lesions @100%	-0.158	-0.158	MATCH
E7	mask – ERM, small lesions @100%	-0.364	-0.364	MATCH
E14	in-lesion hair median area ratio	+0.033	+0.033	MATCH
E14	Trap A mask – ERM, area-ratio tertile T1	-0.064	-0.057 [-0.095, -0.022]	MATCH
E14	Trap A mask – ERM, area-ratio tertile T2	-0.057	-0.046 [-0.082, -0.013]	MATCH
E14	Trap A mask – ERM, area-ratio tertile T3	-0.059	-0.037 [-0.071, -0.005]	SIGN+CI
E10	hair decodability test AUROC	+0.943	+0.943 [+0.919, +0.964]	MATCH
E10	detector mean IoU	+0.220	+0.222	MATCH
E10	detector median IoU	+0.180	+0.184	MATCH
E10	LEACE A-decoder after erasure	+0.500	+0.500	MATCH
E10	clean melanoma AUROC after erasure	+0.814	+0.815	MATCH
E10	melanoma-head $ \Delta p $ after erasure	+0.058	+0.057	MATCH
E10	LEACE projection rank	+1.000	+1.000	MATCH
E12	overlap-contrast trap mask – ERM (spec: null)	+0.047	-0.103 [-0.121, -0.085]	MISMATCH
E8/E9	inpaint – ERM @50%	+0.211	+0.211 [+0.175, +0.249]	MATCH
E8/E9	inpaint – ERM @100%	+0.244	+0.243 [+0.205, +0.284]	MATCH
E8/E9	inpaint_consistency – ERM @50%	+0.046	+0.044 [-0.007, +0.092]	MATCH
E8/E9	inpaint_consistency_lam0 – ERM @50%	+0.036	+0.035 [-0.015, +0.086]	MATCH
E8/E9	leace – ERM @100%	+0.294	+0.294 [+0.249, +0.339]	MATCH
E8/E9	balanced – ERM @100%	+0.281	+0.281 [+0.236, +0.329]	MATCH
E8/E9	dfr – ERM @100%	+0.279	+0.279 [+0.220, +0.339]	MATCH
E8/E9	groupdro – ERM @100%	+0.197	+0.197 [+0.155, +0.240]	MATCH
E13	trapA mask – ERM (rev)	-0.064	-0.049 [-0.083, -0.018]	MATCH
E13	trapB mask – ERM (rev)	+0.124	+0.098 [+0.060, +0.136]	SIGN+CI
E13	trapA inpaint – ERM (rev)	+0.101	+0.098 [+0.084, +0.112]	MATCH
E13	trapB inpaint – ERM (rev)	+0.108	+0.100 [+0.083, +0.117]	MATCH
E13	trapA balanced – ERM (rev)	+0.213	+0.213 [+0.183, +0.243]	MATCH
E13	trapB balanced – ERM (rev)	+0.234	+0.204 [+0.166, +0.245]	SIGN+CI
E13	trapA dfr – ERM (rev)	+0.202	+0.196 [+0.157, +0.235]	MATCH
E13	trapB dfr – ERM (rev)	+0.192	+0.196 [+0.147, +0.251]	MATCH
E13	trapA leace-paired – ERM (rev)	+0.058	+0.074 [+0.062, +0.086]	MATCH
E13	trapB leace-paired – ERM (rev)	+0.078	+0.078 [+0.059, +0.097]	MATCH
E13	trapA leace_unpaired – ERM (rev)	+0.316	+0.320 [+0.280, +0.361]	MATCH
E13	trapA mask – ERM, HAM only	-0.030	-0.011 [-0.064, +0.043]	MISMATCH
E13	trapA mask – ERM, BCN only	-0.078	-0.067 [-0.110, -0.025]	MATCH
E15	trapA mask – ERM (rev)	-0.069	-0.104 [-0.143, -0.067]	SIGN+CI
E15	trapA balanced – ERM (rev)	+0.126	+0.146 [+0.122, +0.170]	MATCH
E15	trapA dfr – ERM (rev)	+0.114	+0.116 [+0.090, +0.145]	MATCH
E15	trapA leace-paired – ERM (rev)	+0.031	+0.034 [+0.026, +0.042]	MATCH
E15	trapA leace_unpaired – ERM (rev)	+0.192	+0.203 [+0.165, +0.243]	MATCH

1.7.2 The controlled sweep (E1–E3, E8–E9)

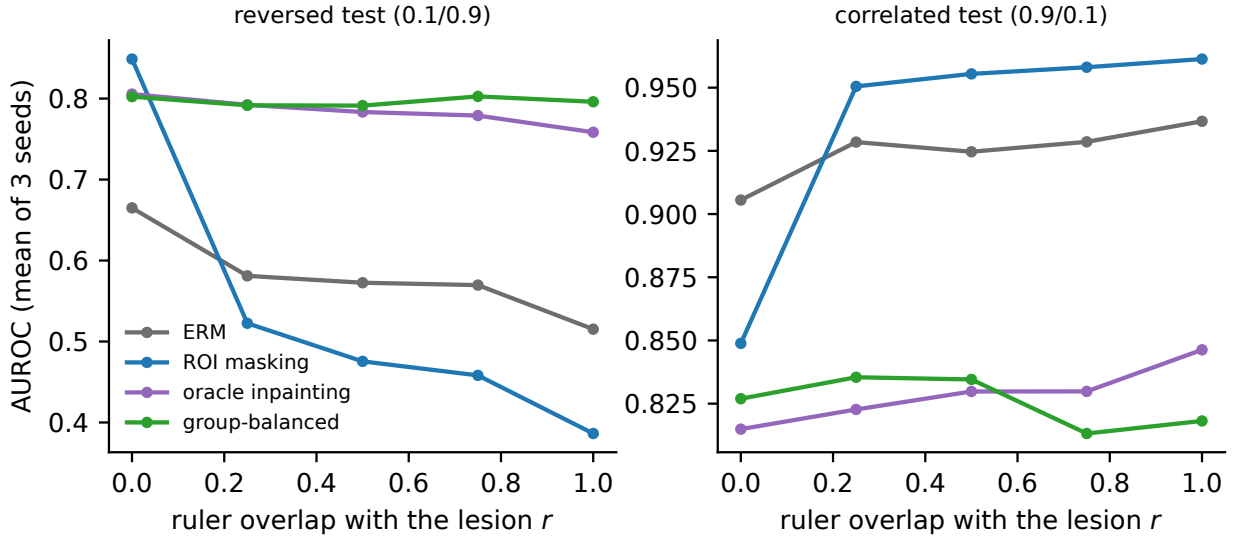


Figure 5: The pilot’s central result, re-derived: reversed-test (left) and correlated-test (right) AUROC of four arms as the ruler moves into the lesion (ISIC 2018, DINOv2 @518, mean of 3 seeds). Masking helps most when the ruler lies outside the lesion and falls below ERM once it overlaps it; its correlated-test AUROC rises with overlap — the masked model relies on the ruler it can still see.

Masking changes the reversed-test AUROC by $+0.184$ [$+0.117, +0.250$] when the ruler lies outside the lesion and by -0.129 [$-0.195, -0.064$] when it lies fully inside; the location interaction is $+0.313$ [$+0.258, +0.370$] (Figure 5). Two numbers show what happens. At 0% overlap the masked model’s correlated and reversed AUROC coincide (0.849 and 0.849): the ruler has been removed and nothing of the shortcut remains. At 100% overlap the masked model’s correlated AUROC (0.961) is *higher* than ERM’s (0.937) while its reversed AUROC collapses to 0.386 (ERM 0.515): after masking, the model uses the ruler *more*.

The reversal is not a property of one encoder or one rendering. At 224px DINOv2 gives $+0.361$ [$+0.275, +0.444$] and -0.095 [$-0.156, -0.035$] (interaction $+0.456$ [$+0.390, +0.525$]); the dermatology model DermLIP gives $+0.161$ [$+0.102, +0.222$] and -0.263 [$-0.348, -0.180$] ($+0.424$ [$+0.335, +0.517$]); a ruler whose colour, opacity, tick pattern and scale are randomised per image gives an interaction of $+0.302$ [$+0.248, +0.357$].

Methods that target the artifact rather than the background all help at full overlap: oracle inpainting $+0.243$ [$+0.205, +0.284$], group balancing $+0.281$ [$+0.236, +0.329$], DFR $+0.279$ [$+0.220, +0.339$], LEACE $+0.294$ [$+0.249, +0.339$] and GroupDRO $+0.197$ [$+0.155, +0.240$]. A training-time repair that asks the model to give the same prediction with and without the ruler (“consistency”, E8) gives $+0.044$ [$-0.007, +0.092$] at 50% overlap, and $+0.035$ [$-0.015, +0.086$] without its penalty ($\lambda = 0$): neither is significant, and the penalty adds nothing to the augmentation.

The pilot’s mechanism experiments (E4–E7) replicate numerically (Table 8): with the ruler present but uninformative, masking does not harm (E4); dilating the mask so that more ruler pixels survive increases the harm at 50% overlap (E5); the masked model’s prediction reacts about twice as strongly to inserting the ruler as ERM’s at full overlap (E6); and the harm is largest for small lesions (E7). What these observations imply about the mechanism is not interpreted in this stage; only their reproducibility is.

1.7.3 Real hair (E10, E12–E15)

Cohort Under the specification’s rules the matched cells are: hair-free group 706 benign / 157 melanoma (pilot 708 / 157); Trap A hair-bearing 2,655 / 942 (pilot 2,903 / 1,144); Trap B 4,412 / 969 (pilot 4,093 / 809); 172 reversed-test melanomas per trap and seed, as in the pilot. The hair-free cells reproduce the pilot’s counts; the hair-bearing cells differ by 5–15% because the lesion masks of non-HAM images come from a re-trained U-Net, which moves some images across the overlap thresholds (a swap of mask source changes the overlap stratum of 6.9% of hair-bearing images and moves none from Trap A to Trap B).

E13 (DINOv2@518) Table 9. Masking *harms* when the hair lies on the lesion (-0.049 [$-0.083, -0.018$]) and helps when it lies beside it ($+0.098$ [$+0.060, +0.136$]); the crossover is $+0.147$ [$+0.111, +0.185$]. The masked Trap-A model still separates the correlated test (0.917) far better than the reversed one (0.460): it uses the hair it can still see. Every artifact-targeting arm helps in both traps; group balancing and DFR, which need only image-level hair labels, help most (C4). Paired LEACE recovers about a third of the balanced gain (C5). Unpaired LEACE “helps” by *flipping* the shortcut — its correlated AUROC (0.624) falls far below its reversed one (0.830) — which is why the pilot rejected it as unsound. The harm is concentrated in BCN images (-0.067 [$-0.110, -0.025$]; HAM -0.011 [$-0.064, +0.043$]). Masking also costs clean-test AUROC in Trap A (-0.023 [$-0.045, -0.001$]).

Table 9: Real hair traps (DINOv2@518; E13): AUROC in the three test environments (mean over 5 seeds) and change in reversed-test AUROC versus ERM (crossed 95% CI). Trap A: hair on the lesion; Trap B: hair beside it; both share the hair-free group.

Arm	Trap A: clean / corr / rev	Trap A: Δ rev vs ERM	Trap B: clean / corr / rev	Trap B: Δ rev vs ERM
ERM	0.734 / 0.920 / 0.510	—	0.716 / 0.894 / 0.510	—
ROI masking	0.712 / 0.917 / 0.460	-0.049 [$-0.083, -0.018$]	0.701 / 0.795 / 0.608	$+0.098$ [$+0.060, +0.136$]
oracle inpainting of hair	0.766 / 0.906 / 0.607	$+0.098$ [$+0.084, +0.112$]	0.738 / 0.865 / 0.610	$+0.100$ [$+0.083, +0.117$]
group-balanced	0.785 / 0.850 / 0.722	$+0.213$ [$+0.183, +0.243$]	0.752 / 0.800 / 0.714	$+0.204$ [$+0.166, +0.245$]
DFR	0.715 / 0.740 / 0.705	$+0.196$ [$+0.157, +0.235$]	0.683 / 0.669 / 0.706	$+0.196$ [$+0.147, +0.251$]
paired LEACE	0.762 / 0.913 / 0.584	$+0.074$ [$+0.062, +0.086$]	0.737 / 0.876 / 0.588	$+0.078$ [$+0.059, +0.097$]
unpaired LEACE (image-level A)	0.726 / 0.624 / 0.830	$+0.320$ [$+0.280, +0.361$]	<i>n/a</i>	<i>n/a</i>

E15 (DermLIP@224) Table 10. Masking harms in Trap A (-0.104 [$-0.143, -0.067$]) as the pilot reported, but — a finding the pilot did not test — it harms in Trap B as well (-0.090 [$-0.135, -0.046$]), so DermLIP shows no crossover ($+0.014$ [$-0.037, +0.063$]). With DermLIP, removing the background costs disease signal wherever the hair lies (clean AUROC -0.066 [$-0.089, -0.043$] in Trap A).

Table 10: Real hair traps (DermLIP@224; E13/E15): AUROC in the three test environments (mean over 5 seeds) and change in reversed-test AUROC versus ERM (crossed 95% CI). Trap A: hair on the lesion; Trap B: hair beside it; both share the hair-free group.

Arm	Trap A: clean / corr / rev	Trap A: Δ rev vs ERM	Trap B: clean / corr / rev	Trap B: Δ rev vs ERM
ERM	0.801 / 0.917 / 0.665	—	<i>n/a</i>	<i>n/a</i>
ROI masking	0.735 / 0.884 / 0.561	-0.104 [$-0.143, -0.067$]	<i>n/a</i>	<i>n/a</i>
oracle inpainting of hair	<i>n/a</i>	<i>n/a</i>	<i>n/a</i>	<i>n/a</i>
group-balanced	0.836 / 0.875 / 0.811	$+0.146$ [$+0.122, +0.170$]	<i>n/a</i>	<i>n/a</i>
DFR	0.802 / 0.835 / 0.782	$+0.116$ [$+0.090, +0.145$]	<i>n/a</i>	<i>n/a</i>
paired LEACE	0.814 / 0.917 / 0.699	$+0.034$ [$+0.026, +0.042$]	<i>n/a</i>	<i>n/a</i>
unpaired LEACE (image-level A)	0.756 / 0.648 / 0.868	$+0.203$ [$+0.165, +0.243$]	<i>n/a</i>	<i>n/a</i>

E10: is hair visible to the encoder? A linear probe decodes hair presence from the DINOv2 embedding with AUROC 0.943 [0.919, 0.964] ($n = 356$ test images); in that set hair is *less* common

on melanomas (0.411) than on benign lesions (0.608). Paired LEACE on 478 images with manual hair masks (354 training pairs) needs only one direction: the hair decoder falls from 0.963 to chance (0.500), clean melanoma AUROC is unchanged (0.812 to 0.815), yet the melanoma head’s reaction to hair only falls from 0.071 to 0.057. A classical morphological hair detector (DullRazor-style) is poor: mean IoU 0.222 (median 0.184; precision 0.316, recall 0.472) — which is why the real traps use the published masks rather than a detector.

E14: how large is real hair relative to the lesion? In Trap A the in-lesion hair covers a median 0.033 of the lesion area, similar to the synthetic ruler. Masking harms in every tertile of this ratio (-0.057 [$-0.095, -0.022$], -0.046 [$-0.082, -0.013$], -0.037 [$-0.071, -0.005$]); the pilot’s “same harm at every ratio” pattern replicates, with the smallest harm in the tertile with the most hair.

1.7.4 E12: the overlap-contrast design

E12 labelled hair on the lesion as $A = 1$ and hair beside it as $A = 0$, with no hair-free group, and the pilot reported a null result. The re-implementation finds masking clearly harmful (-0.103 [$-0.121, -0.085$]), while balancing ($+0.160$ [$+0.144, +0.177$]) and DFR ($+0.179$ [$+0.148, +0.211$]) help. This is what the design predicts whatever the truth, because in E12 location *is* the label proxy: masking removes exactly the out-of-lesion hair that defines one side of the contrast and keeps the in-lesion hair that defines the other. The design cannot measure a location effect; E13’s two traps with a shared hair-free group can. The pilot’s null E12 result is not supported, and the design is not used further.

1.7.5 Audit of the pilot’s written claims

Besides the numbers, the pilot’s report made statements that were checked against the archived material (docs/THESIS_CLAIMS_AUDIT.md):

- **“Leakage inflated the shortcut gap by 27%”** — **not supported** as stated. The pre-correction pilot used the identical split; the 27% compared runs at different resolutions. The controlled experiment of Section 1.5.1 supports the direction with a magnitude of 10–18%.
- **Archived numbers** — verified: all 21 bridge intervals of the controlled sweep were recomputed from the archived predictions to 10^{-16} , and a re-run from raw images reproduced 252 per-seed AUROCs within 0.0005 (a GPU precision effect).
- **Lesion U-Net** — the pilot’s U-Net code was missing and was re-implemented. The specification’s architecture, used for the hair traps, reaches a held-out Dice of 0.89 (pilot 0.88); a ResNet-34 U-Net used for exploratory checks reaches 0.94.
- **Mask-source sensitivity** — the pilot reported that about 11% of images change overlap stratum when the mask source is swapped; the re-implementation finds 6.9%, and no image moves between Trap A and Trap B.
- **E11 (artifact atlas and vignetting trap)** was not re-implemented: ink appears on the lesion in too few images (the traps exclude ink), and vignetting is a frame artifact that lies outside the lesion by construction, so it cannot inform the in-ROI question.

1.7.6 The mismatches

1. **E1, masking at 25% overlap.** The pilot’s interval excluded zero; the crossed interval includes it. The harm is significant from 50% overlap onward.

2. **E4, occlusion control at 0% and 100% overlap.** The pilot reported small positive effects of masking with the ruler present but uninformative; with the crossed estimator they are not significant. The conclusion — occlusion by the mask does not harm — is unchanged.
3. **E12.** Not null (above); a structural problem of the design.
4. **E13, HAM-only stratum.** The pilot reported harm in HAM images; the replication does not ($-0.011 [-0.064, +0.043]$). The harm is carried by BCN images.

None of these affects the pilot’s central claim — that masking reverses from helpful to harmful as the ruler moves into the lesion — which replicates with every encoder and ruler rendering.

1.8 What failed or changed in this stage

- The pilot’s interval estimator was too narrow whenever seeds share test images; it was replaced, and three pilot significance claims (E1 at 25%, E4 at 0% and 100%) no longer hold.
- E12’s null result and the “27%” leakage figure are not supported; the HAM-only harm of E13 does not replicate.
- DermLIP shows no hair crossover in the replication: it is harmed by masking at both hair locations.
- The hair-bearing cells of E13 differ by 5–15% from the pilot’s because the lesion U-Net had to be re-trained.

1.9 Anticipated questions

Why re-implement instead of re-running the pilot’s code? Most of the pilot’s code for E9–E15 was not in the archive, and re-running code verifies only that it runs. Re-implementing from a written specification tests whether the *description* is sufficient to reproduce the result, which is what a reader of a paper can do.

Why is the reversed test the primary metric, not the clean test? The clean test measures disease recognition without the shortcut, so it cannot show whether a model *uses* the artifact. The reversed test penalises reliance on the artifact directly. Both are always reported, together with the correlated test, so that a method that simply inverts the shortcut (as unpaired LEACE does) is visible.

Why 0.9/0.1? It is strong enough that every encoder learns the shortcut, which is the regime where mitigation matters, and weak enough that both classes contain artifact-free and artifact-bearing images in training. The same values are used everywhere, so effects are comparable.

Why frozen encoders and linear heads? Every arm then uses identical features; differences between arms are caused by the input (masked or not) or the head, not by optimisation noise. It is also the common way foundation models are deployed on small medical datasets.

Why choose C and the threshold on clean validation data? Choosing them on data with the shortcut would reward the shortcut; choosing them on reversed data would leak the test condition into training.

Why group by perceptual hash and not by patient? ISIC 2018 does not publish patient identifiers; lesion identifiers and near-duplicate detection are the strongest grouping available. The leakage experiment shows that this grouping matters (10–18% gap inflation otherwise).

Why a crossed bootstrap and not simply more seeds? More seeds reduce training variance but do not create new test images. The quantity of interest is a difference on a finite test set; its uncertainty must include the sampling of that set once, which is what the shared image weights do.

Why do the E13 cell counts differ from the pilot’s? The hair-free group reproduces exactly. The hair-bearing cells depend on the lesion masks of non-HAM images, which come from a re-trained U-Net; images near the 0.1 and 0.5 overlap thresholds move between strata. No image moves between Trap A and Trap B.

Why does unpaired LEACE look best, and why is it rejected? It removes the direction that separates images with and without hair *in the training data*, where hair is strongly associated with the label; that direction contains much of the diagnosis, and the model ends up predicting the opposite of the shortcut (correlated AUROC far below reversed). Paired LEACE compares the same image with and without hair and removes only the hair.

Is 2,437 images enough? For the controlled sweep, yes: the location interaction is large relative to its interval (+0.313 [+0.258, +0.370]). Individual overlap cells are less precise, which is why the 25% cell lost significance.

1.10 Conclusion

Supported. The pilot’s central result replicates from its written specification: masking helps when the ruler lies outside the lesion and harms when it overlaps it (interaction +0.313 [+0.258, +0.370]), with two encoders, two resolutions and a randomised ruler; with real hair on DINOv2 masking harms in-lesion and helps out-of-lesion (crossover +0.147 [+0.111, +0.185]); methods that need only an image-level artifact label fix both traps; paired linear erasure is insufficient. **Not supported.** The pilot’s null E12 result; the “27%” leakage figure; significance at 25% overlap and in the occlusion control; harm in HAM images; a hair crossover with DermLIP. **Fixed for the rest of the work.** Grouped splits, clean-validation model selection, pre-registration, and the crossed bootstrap.

1.11 Questions for the supervisor

1. Is the reversed-environment AUROC acceptable as the primary metric for the thesis, or should a clinical metric (sensitivity at a fixed specificity) be primary from the start?
2. How should the thesis present the estimator correction: as a methods detail, or as a contribution in its own right (it affects any study that averages several training seeds on one test set)?
3. Should the E12 design and its failure be reported in the thesis (as a warning about contrast designs) or only in the appendix?
4. Is it acceptable that the pilot’s registration is its written specification, committed after the pilot ran?

2 Stage 2 — The location law and its mechanism

In one sentence. When only the artifact’s overlap with the region of interest (ROI) is varied, ROI masking helps while the artifact lies outside the ROI and loses its benefit — in 7 of 11 sweeps it hurts — once the artifact overlaps it; the reversal comes from the artifact *surviving* the mask and becoming what the model relies on, not from occlusion; and it recurs in thyroid and ovarian ultrasound, capsule endoscopy and chest radiography, with one principled exception (an encoder that ignores an artifact outside the lungs).

2.1 Where this stage starts

Stage 1 re-derived the pilot and fixed the protocol: trap environments, frozen foundation-model features, linear heads chosen on clean validation data, grouped splits, pre-registration and the crossed bootstrap. It showed that the pilot’s central observation replicates: in the controlled dermoscopy sweep, masking changes reversed-test AUROC from +0.184 [+0.117, +0.250] at 0% overlap to −0.129 [−0.195, −0.064] at 100% overlap. Stage 1 deliberately did not interpret the pilot’s mechanism experiments. This stage asks the two questions that follow: *why* does masking reverse, and *how general* is the reversal — does it depend on dermoscopy, on the ruler, on DINOv2?

2.2 Questions

- Q2a** Is the benefit of masking a function of the artifact’s overlap r with the ROI, falling from $r = 0$ to $r = 1$? We call a positive location interaction $[\text{mask} - \text{ERM}]_{r=0} - [\text{mask} - \text{ERM}]_{r=1}$ on the reversed test the *location law*.
- Q2b** Which mechanism produces the fall? Three candidates were stated in advance: *occlusion* (masking removes useful context whatever the artifact does), *lost disease signal* (the background carries diagnostic information), and *retention with amplified reliance* (the part of the artifact inside the ROI survives masking, its competitors are removed, and the head relies on it more).
- Q2c** Does the law hold beyond one encoder, one artifact rendering and one modality — and where does it fail?

2.2.1 Why these experiments, and what was rejected

- *An occlusion control instead of an argument from intuition.* If masking harmed because it throws away context, it would harm even when the artifact carries no label information (cf. Fong and Vedaldi, 2019). The control keeps the ruler at the same positions but makes it independent of the label.
- *Dilation as a dose of retention.* Growing the mask by a margin keeps more of the ruler visible without changing the lesion. If retention drives the harm, harm should grow with retention when retention is incomplete (50% overlap) and stay flat when the ruler is already fully retained (100% overlap) — even though dilation then *lowers* the ruler’s share of the visible pixels.
- *A same-head counterfactual instead of saliency maps.* Saliency maps are qualitative and unstable for frozen encoders. Instead, the ruler is inserted into and removed from the *same* test image and the change in each trained head’s predicted probability, $|\Delta p|$, is read: a direct, per-image measure of reliance on the artifact.
- *Lesion size as a natural dose.* The same ruler covers a larger share of a small lesion. If what matters is how dominant the retained artifact is among the remaining evidence, harm should be largest for small lesions. No threshold ratio was pre-specified, because the pilot showed none.
- *Other modalities with their own artifact.* A caliper drawn like a sonographer’s, a debris blob drawn like capsule contamination and a tube drawn like a drain, each pasted at controlled overlap on

real images of that modality with that modality’s own ROI masks. This tests whether the law is a property of dermoscopy or of masking.

- *Rejected:* measuring reliance through head weights (they are not comparable between masked and unmasked feature spaces), and a single “average” overlap (it would hide the sign change).

2.3 Methods

Controlled dermoscopy sweep As in Stage 1: ISIC 2018 pilot cohort, a 65×19 px ruler at a frozen position per image and overlap $r \in \{0, 0.25, 0.5, 0.75, 1\}$; DINOv2 ViT-B/14 at 518 px (Oquab et al., 2024) (CLS features), logistic heads, seeds 42, 123, 456; primary outcome reversed-test AUROC.

Arms ERM; ROI masking; oracle inpainting of the ruler; group-balanced reweighting; DFR (Kirichenko et al., 2023); paired LEACE (Belrose et al., 2023); GroupDRO (Sagawa et al., 2020). All arms share features and seeds; only the input or the head changes.

Occlusion control The ruler is pasted at the same positions in 50% of images independently of the label; AUROC is measured on a 50/50 test.

Mask dilation The lesion mask is dilated by 0, 10, 25 or 50 px before masking. *Retention* is the share of ruler pixels that remain visible; *ruler share* is the ruler’s share of all visible pixels.

Same-head counterfactual For each artifact-bearing image of the reversed test and each trained head, the image is scored with and without the pasted artifact (the masked head sees the masked version of both); $|\Delta p|$ is the absolute change in predicted probability. The mask – ERM difference of mean $|\Delta p|$ is bootstrapped with seeds and images crossed.

Lesion-size strata Test images are split into tertiles of lesion area; the median ruler-to-lesion area ratio is reported per tertile.

Replications DINOv2 at 224 px, DermLIP at 224 px (Yan et al., 2025), and a ruler whose colour, tick pattern and size are randomised per image (“stress”).

Other modalities Each sweep uses only images of that cohort in which the real artifact is absent, so that the pasted artifact is the only one:

- **Thyroid ultrasound** (TN3K images with TNCD malignancy labels and manual nodule masks (Gong et al., 2021); 1,663 caliper-free images): a synthetic sonographer caliper (dotted line with “+” ends) in the same 65×19 box as the ruler; DINOv2 and MedSigLIP-448. Split: official TN3K test images as test; the rest 80/20 by leakage group.
- **Ovarian ultrasound** (MMOTU OTU_2d (Zhao et al., 2022); benign cystic lesions versus tumours with a solid component; 343 caliper-free images, 64 in test): the same caliper; DINOv2. Small, so wide intervals were expected.
- **Capsule endoscopy** (SEE-AI (Yokote et al., 2023); erosion versus polyp-like lesions; ROI = the union of the expert lesion boxes; 796 low-contamination frames, split 60/20/20 by frame-block group): a textured yellow-green debris blob with bubbles, 48×48 px; DINOv2 and MedSigLIP.
- **Chest radiography** (NIH ChestX-ray14 pneumothorax (Wang et al., 2017); lung masks from a PSPNet (Cohen et al., 2022); patient-grouped split): a synthetic tube; RAD-DINO @518 (Pérez-García et al., 2025) (a chest-radiograph foundation model) and DINOv2.

Figure 7 shows three of these sweeps on real images.

Statistics Every interval is a 95% percentile interval from 10,000 crossed bootstrap replicates. “Significant” means the 95% interval excludes zero; it is not a multiplicity-adjusted test.

2.4 Experiments

Table 11 lists each experiment of this stage with its registration and support criterion. The dermoscopy analyses (E1–E7) were specified in the pilot specification (2699d51, 2026-09-26 15:47); the other-modality sweeps were registered with their cohorts before any model was fitted. The DINOv2 chest-tube sweep was added after the RAD-DINO result and is labelled post hoc.

Table 11: Experiments of this stage: registration status, the registration document (docs/PREREGISTRATION_<name>.md or the replication specification), the commit that first contains it, and what was stated in advance to count as support. Commit times are set by the committing machine and are not independent proof of order.

Experiment	Status	Registration	First commit (local time)	Support criterion
Controlled dermoscopy sweep (E1)	supported registered	REPLICATION_SPEC	2699d51, 15:47	2026-09-26 interaction CI > 0; mask-ERM CI > 0 at 0% and < 0 at 100%
Replications: DINOv2@224, DermLIP, randomised ruler	supported registered	REPLICATION_SPEC	2699d51, 15:47	2026-09-26 same signs and verdicts as E1
Occlusion control (E4)	supported registered	REPLICATION_SPEC	2699d51, 15:47	2026-09-26 no harm on the 50/50 test at any overlap
Mask dilation (E5)	supported registered	REPLICATION_SPEC	2699d51, 15:47	2026-09-26 harm grows with retention at 50%; flat at 100%
Same-head counterfactual $ \Delta p $ (E6, T3)	supported registered	REVIEW2 (T3)	29cd0da, 18:56	2026-09-27 mask > ERM at in-ROI overlap, CI > 0
Lesion-size tertiles (E7)	partly supported pilot analysis, replicated	REPLICATION_SPEC	2699d51, 15:47	2026-09-26 harm largest in small lesions
Thyroid caliper sweep	supported registered	THYROID_TRAPS	ea57661, 22:46	2026-09-26 interaction CI > 0
Ovarian caliper sweep	supported registered	OVARY_TRAPS	2517600, 07:29	2026-09-27 interaction CI > 0
Capsule debris sweep	supported registered	CAPSULE_TRAPS	fcbf8d9, 00:40	2026-09-27 interaction CI > 0
Chest tube sweep (RAD-DINO)	supported registered	CXR_DEVICE_TRAPS	4d1a924, 18:07	2026-09-26 interaction CI > 0
Chest tube sweep (DINOv2)	not supported post hoc	—	—	descriptive only
Crossed-bootstrap intervals for every sweep (A1)	supported registered	FINAL (A1)	c45d482, 00:35	2026-09-28 a claim is stated only if its crossed CI excludes 0

2.5 Results

2.5.1 The dermoscopy sweep: every arm, every overlap

Masking changes the reversed-test AUROC by +0.184 [+0.117, +0.250] when the ruler lies outside the lesion, by −0.097 [−0.164, −0.030] at 50% overlap and by −0.129 [−0.195, −0.064] at full overlap (Table 12); the location interaction is +0.313 [+0.258, +0.370]. Masking therefore lowers it significantly at 50, 75 and 100% overlap. Every other arm helps at every overlap, and their benefit *grows* with overlap — the opposite of masking: balancing’s location interaction is −0.143 [−0.171, −0.119], inpainting’s −0.103 [−0.127, −0.080]. They target the artifact, not the background, and the ERM shortcut they remove is larger when the ruler sits on the lesion. On the correlated test the masked model at full overlap scores 0.961 against 0.937 for ERM: the masked model exploits the ruler more.

Table 12: Controlled dermoscopy sweep (ISIC 2018, DINOv2 ViT-B/14 @518, three seeds): change in reversed-test AUROC versus ERM, crossed CIs.

	0% overlap	50%	100%
ERM (reversed AUROC)	0.665	0.573	0.515
ROI masking – ERM	+0.184 [+0.117, +0.250]	−0.097 [−0.164, −0.030]	−0.129 [−0.195, −0.064]
oracle inpainting – ERM	+0.141 [+0.109, +0.175]	+0.211 [+0.175, +0.249]	+0.243 [+0.205, +0.284]
group-balanced – ERM	+0.137 [+0.099, +0.179]	+0.219 [+0.177, +0.265]	+0.281 [+0.236, +0.329]
DFR – ERM	+0.124 [+0.075, +0.179]	+0.221 [+0.166, +0.281]	+0.279 [+0.220, +0.339]
paired LEACE – ERM	+0.167 [+0.126, +0.211]	+0.237 [+0.197, +0.280]	+0.294 [+0.249, +0.339]
GroupDRO – ERM	+0.080 [+0.044, +0.118]	+0.152 [+0.112, +0.193]	+0.197 [+0.155, +0.240]

Replications The sweep reverses with DINOv2 at 224 px (interaction +0.456 [+0.390, +0.525]), with DermLIP (+0.424 [+0.335, +0.517]) and with a randomised ruler (+0.302 [+0.248, +0.357]) (Table 13).

Table 13: Replications of the controlled sweep (reversed-test AUROC, crossed CIs).

Backbone / artifact	mask – ERM at 0%	at 100%	Location interaction
DINOv2 @224	+0.361 [+0.275, +0.444]	−0.095 [−0.156, −0.035]	+0.456 [+0.390, +0.525]
DermLIP @224	+0.161 [+0.102, +0.222]	−0.263 [−0.348, −0.180]	+0.424 [+0.335, +0.517]
DINOv2 @518, randomised ruler	+0.146 [+0.082, +0.211]	−0.155 [−0.228, −0.084]	+0.302 [+0.248, +0.357]

2.5.2 Mechanism

Occlusion does not explain the harm With the ruler present but uninformative, masking changes it significantly at none of the five overlaps (every CI includes zero) (Table 14); inpainting, which removes only the ruler, changes nothing either. Masking by itself does not cost the reversed-test AUROC it loses in the sweep.

Table 14: Occlusion control: ruler present but uncorrelated with the label (50/50); AUROC change on the 50/50 test.

Overlap	mask – ERM	inpaint – ERM
0%	+0.036 [−0.016, +0.086]	−0.001 [−0.005, +0.003]
25%	+0.022 [−0.030, +0.073]	−0.010 [−0.034, +0.002]
50%	+0.041 [−0.010, +0.090]	−0.002 [−0.007, +0.003]
75%	+0.037 [−0.013, +0.086]	−0.003 [−0.008, +0.001]
100%	+0.039 [−0.010, +0.091]	−0.004 [−0.010, +0.002]

Harm follows the visibility of the artifact At 50% overlap, dilating the mask from 0 to 25 px raises retention from 0.500 to 0.946 and moves the masking effect from −0.097 [−0.164, −0.031] to −0.153 [−0.221, −0.089]. At 100% overlap the ruler is already retained (0.999) and the effect stays flat (−0.129 [−0.195, −0.066] without and −0.138 [−0.200, −0.082] with a 50 px margin), although dilation *lowers* the ruler’s share of the visible pixels threefold (0.055 to 0.018). Harm therefore tracks whether the ruler is visible, not how much of the image it occupies once visible (Table 15). This also rules out “lost context” as the sole cause: dilation *restores* context, yet harm grows at 50% overlap. The intervals of neighbouring margins overlap; the pattern, not each step, is the finding.

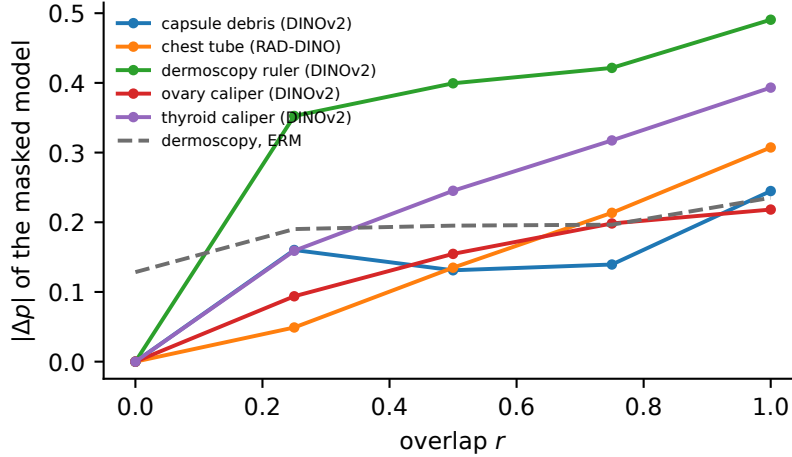
Table 15: Mask dilation: restoring context around the lesion restores ruler pixels (retention) and changes harm.

Overlap	Margin (px)	Ruler retained	Ruler share of visible pixels	mask – ERM (rev)
50%	0	0.500	0.028	−0.097 [−0.164, −0.031]
	10	0.725	0.029	−0.138 [−0.203, −0.074]
	25	0.946	0.026	−0.153 [−0.221, −0.089]
	50	1.000	0.018	−0.142 [−0.206, −0.083]
75%	0	0.750	0.042	−0.111 [−0.181, −0.044]
	10	0.940	0.037	−0.152 [−0.217, −0.091]
	25	1.000	0.028	−0.160 [−0.226, −0.096]
	50	1.000	0.018	−0.140 [−0.201, −0.079]
100%	0	0.999	0.055	−0.129 [−0.195, −0.066]
	10	1.000	0.040	−0.144 [−0.206, −0.082]
	25	1.000	0.028	−0.144 [−0.209, −0.082]
	50	1.000	0.018	−0.138 [−0.200, −0.082]

The masked model relies more on the retained artifact On the same images, inserting the ruler at full overlap changes the masked head’s prediction by 0.491 on average and the ERM head’s by 0.235 — about 2.1 times as much; the paired difference is +0.255 [+0.221, +0.287] (Table 16, Figure 6). At 0% overlap the masked head cannot see the ruler (0.000) whereas ERM still reacts to it (0.128). This is the mechanism in one measurement: masking removes the competing evidence, the retained artifact becomes the most informative remaining feature, and the head leans on it. Across the 5 sweeps with counterfactual scores the masked head is significantly more sensitive at full overlap in 3: ovarian calipers (0.218 versus 0.032; +0.188 [+0.133, +0.250]) and capsule debris (0.245 versus 0.057; +0.186 [+0.130, +0.260]) besides the ruler. For thyroid calipers the two heads are equally sensitive (+0.025 [−0.016, +0.065]) — ERM already relies heavily on a caliper inside the nodule — and with RAD-DINO the masked head is *less* sensitive to an in-lung tube (−0.127 [−0.136, −0.118]).

Table 16: Same-head counterfactual sensitivity to inserting the artifact (reversed test, artifact-bearing images), crossed CIs of the paired difference.

Sweep	Overlap	$ \Delta p $ ERM	$ \Delta p $ mask	mask – ERM
dermoscopy ruler (DINOv2)	0%	0.128	0.000	−0.128 [−0.136, −0.121]
	25%	0.190	0.353	+0.162 [+0.136, +0.188]
	50%	0.195	0.399	+0.204 [+0.177, +0.230]
	75%	0.197	0.422	+0.225 [+0.195, +0.255]
	100%	0.235	0.491	+0.255 [+0.221, +0.287]
thyroid caliper (DINOv2)	0%	0.285	0.000	−0.285 [−0.317, −0.255]
	25%	0.353	0.159	−0.195 [−0.271, −0.131]
	50%	0.351	0.245	−0.106 [−0.176, −0.046]
	75%	0.359	0.317	−0.041 [−0.078, −0.005]
	100%	0.369	0.393	+0.025 [−0.016, +0.065]
ovary caliper (DINOv2)	0%	0.016	0.000	−0.016 [−0.020, −0.013]
	25%	0.027	0.094	+0.067 [+0.040, +0.098]
	50%	0.027	0.155	+0.129 [+0.090, +0.174]
	75%	0.031	0.198	+0.168 [+0.125, +0.217]
	100%	0.032	0.218	+0.188 [+0.133, +0.250]
capsule debris (DINOv2)	0%	0.033	0.000	−0.033 [−0.041, −0.025]
	25%	0.079	0.160	+0.081 [+0.001, +0.149]
	50%	0.053	0.131	+0.077 [+0.031, +0.140]
	75%	0.058	0.139	+0.081 [+0.058, +0.109]
	100%	0.057	0.245	+0.186 [+0.130, +0.260]
chest tube (RAD-DINO)	0%	0.005	0.000	−0.005 [−0.006, −0.005]
	25%	0.095	0.049	−0.046 [−0.051, −0.041]
	50%	0.260	0.135	−0.126 [−0.134, −0.117]
	75%	0.355	0.214	−0.141 [−0.151, −0.130]
	100%	0.434	0.307	−0.127 [−0.136, −0.118]

Figure 6: Mean same-head sensitivity $|\Delta p|$ to inserting the artifact: masked model per sweep (solid) and ERM for the dermoscopy sweep (dashed), by overlap.

Lesion size Table 17. The ruler covers a median 0.012 of a large lesion’s area, 0.031 of a medium and 0.106 of a small one. The harm at full overlap is significant for small (−0.364 [−0.509, −0.217]) and medium lesions (−0.158 [−0.295, −0.027]) and not for large ones (+0.063 [−0.015, +0.144], which does not differ significantly from zero); at 50% overlap it is significant only for small lesions (−0.326 [−0.471, −0.176]; medium −0.129 [−0.283, +0.027]). What holds is an ordering — harm is

largest where the artifact is large relative to the lesion — with wide intervals, because each tertile has few melanomas. No fixed ratio threshold is claimed.

Table 17: Lesion-size strata of the controlled sweep (reversed-test AUROC, crossed CIs).

Lesion tertile	Ruler/lesion area (median)	Melanomas in test	mask – ERM at 0%	50%	100%
large	0.012	36	+0.249 [+0.155, +0.346]	+0.087 [+0.006, +0.172]	+0.063 [−0.015, +0.144]
medium	0.031	21	+0.215 [+0.033, +0.391]	−0.129 [−0.283, +0.027]	−0.158 [−0.295, −0.027]
small	0.106	16	+0.150 [+0.036, +0.273]	−0.326 [−0.471, −0.176]	−0.364 [−0.509, −0.217]

Synthesis Occlusion alone does not harm; restoring context does not remove the harm; harm follows whether the artifact is visible after masking, and the masked model’s reliance on a visible artifact is larger than ERM’s. The mechanism is *retention with amplified reliance*: masking deletes the out-of-ROI evidence — including other cues that competed with the artifact — and leaves the in-ROI artifact as a larger share of what predicts the label in training, so the head weights it more heavily.

2.5.3 Other modalities

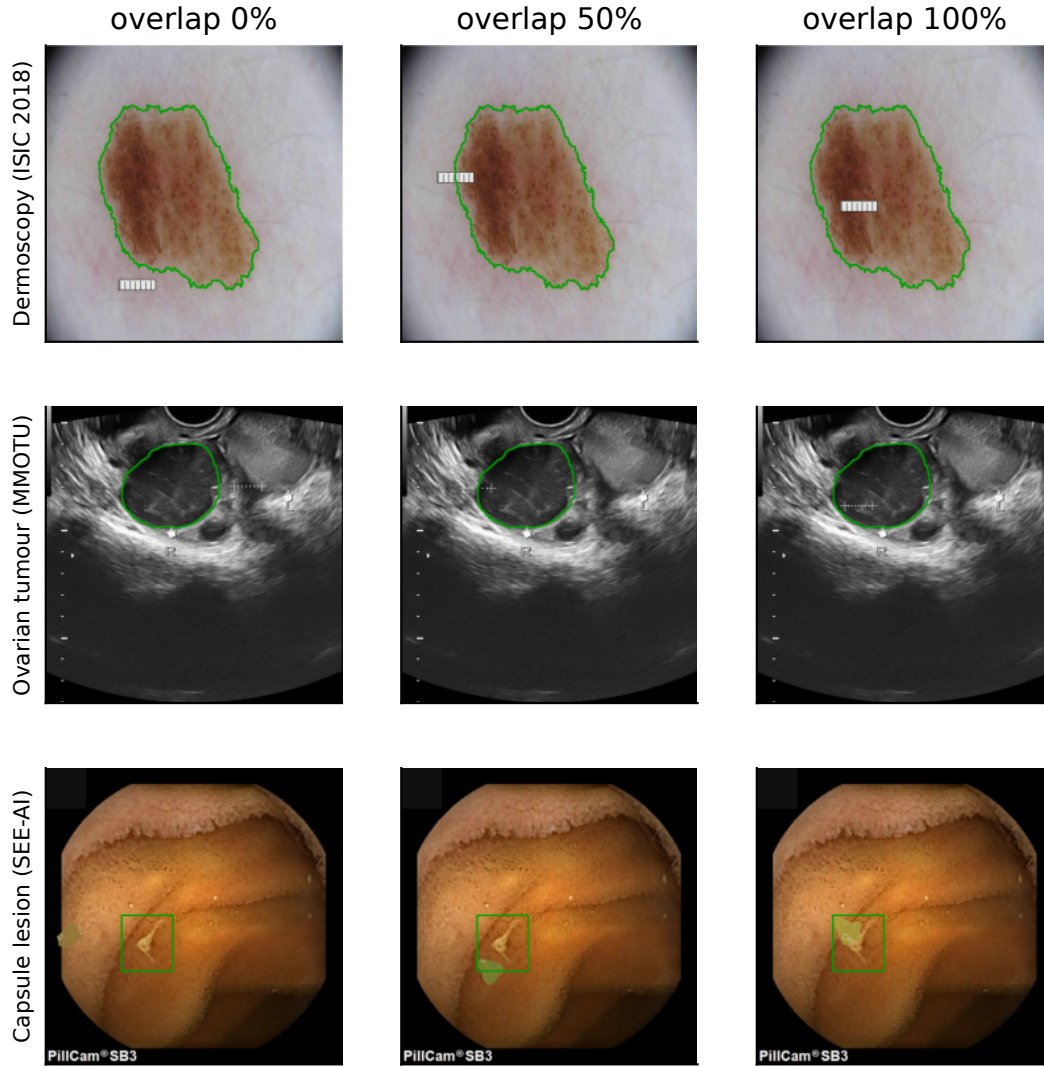


Figure 7: Real images with the synthetic artifact at 0, 50 and 100% overlap (green: ROI; the capsule ROI is the annotated bounding box). Top: ISIC 2018 (CC BY-NC 4.0); middle: MMOTU ovarian ultrasound (CC BY 4.0); bottom: SEE-AI capsule endoscopy (CC BY 4.0). Thyroid (TN3K) and NIH images are not shown because their redistribution terms are unclear.

Table 18 and Figure 8 repeat the sweep in four further modalities; Table 19 gives every baseline arm. Across all 11 controlled sweeps (four dermoscopy, seven others) the location interaction is positive with an interval above zero in 10 and negative in 1. Whether the falling benefit turns into harm at full overlap depends on the setting: masking harms significantly at full overlap in 7 sweeps, still helps in 2 and makes no significant difference in 2.

Table 18: Controlled sweeps in the other modalities (synthetic caliper, debris, tube pasted at controlled overlap; reversed-test AUROC, crossed CIs).

Sweep	mask – ERM at 0%	at 100%	Location interaction
Thyroid caliper, DINOv2	+0.402 [+0.315, +0.489]	−0.030 [−0.105, +0.042]	+0.433 [+0.362, +0.507]
Thyroid caliper, MedSigLIP	+0.514 [+0.430, +0.600]	+0.044 [−0.011, +0.104]	+0.470 [+0.388, +0.556]
Ovarian caliper, DINOv2	+0.041 [−0.096, +0.175]	−0.197 [−0.357, −0.035]	+0.238 [+0.144, +0.340]
Capsule debris, DINOv2	+0.116 [+0.052, +0.191]	−0.112 [−0.209, −0.019]	+0.228 [+0.144, +0.324]
Capsule debris, MedSigLIP	+0.374 [+0.248, +0.509]	+0.216 [+0.118, +0.317]	+0.158 [+0.012, +0.314]
Chest tube, RAD-DINO	−0.002 [−0.010, +0.005]	+0.111 [+0.095, +0.127]	−0.113 [−0.129, −0.097]
Chest tube, DINOv2	+0.254 [+0.230, +0.278]	−0.083 [−0.112, −0.057]	+0.337 [+0.312, +0.364]

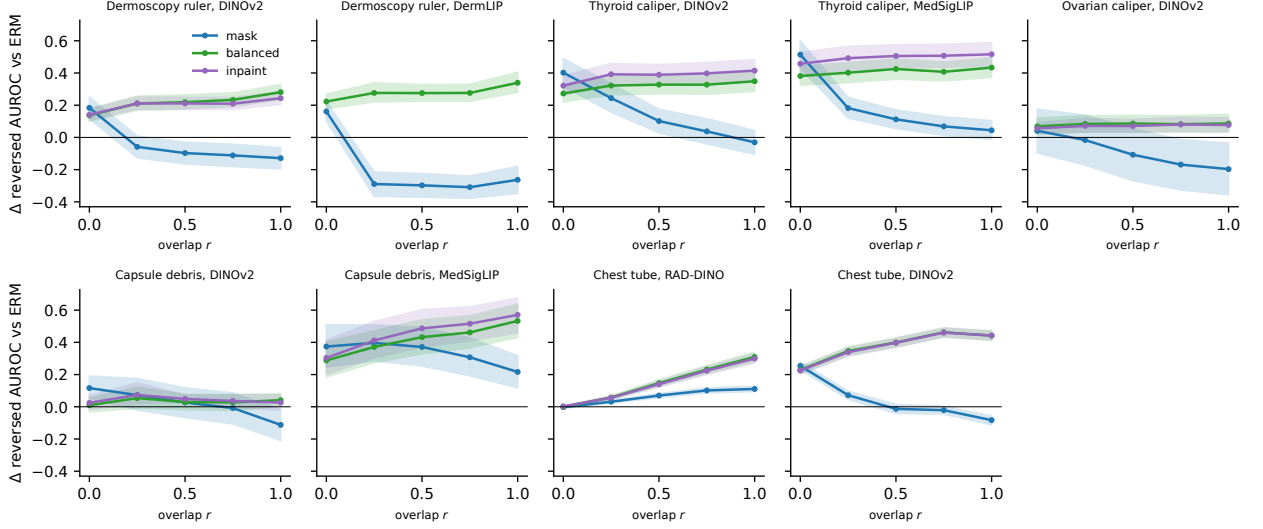


Figure 8: Change in reversed-test AUROC versus ERM as a function of overlap, for masking (blue), group balancing (green) and oracle inpainting (purple), in every controlled sweep (shaded: crossed 95% CI). Masking falls with overlap in every panel except the RAD-DINO chest tube.

- **Thyroid calipers.** Masking helps strongly with the caliper outside the nodule (DINOv2 +0.402 [+0.315, +0.489]; MedSigLIP +0.514 [+0.430, +0.600]) and the benefit vanishes at full overlap (−0.030 [−0.105, +0.042]; +0.044 [−0.011, +0.104]): the law holds as a loss of benefit, not as harm. Nodules are large relative to the caliper, and the ultrasound background carries little disease information, so masking costs little besides leaving the caliper.
- **Ovarian calipers.** Masking harms at full overlap (−0.197 [−0.357, −0.035]) and its benefit outside the tumour is not significant (+0.041 [−0.096, +0.175]): MMOTU tumours fill much of the frame, so little background is removed, and the test set is small (64 images).
- **Capsule debris.** With DINOv2 the sweep reverses (+0.116 [+0.052, +0.191] to −0.112 [−0.209, −0.019]). With MedSigLIP masking still helps at full overlap (+0.216 [+0.118, +0.317]) but less than outside (+0.374 [+0.248, +0.509]; interaction +0.158 [+0.012, +0.314]): the box ROI contains much normal mucosa, so masking removes little of the lesion’s context.
- **Chest tube — the boundary.** With RAD-DINO the ERM model ignores a tube outside the lungs (reversed 0.913 versus clean 0.915 at no overlap), so masking has nothing to remove at $r = 0$; at full overlap ERM collapses (0.614) and the masked model less so (0.725). The interaction is negative. With DINOv2, which does use the extra-pulmonary tube (reversed 0.629 versus clean 0.859), the law holds (+0.337 [+0.312, +0.364]). The law’s premise is that the encoder uses the out-of-ROI artifact; when it does not, masking cannot help outside and the law does not apply.
- **The artifact-targeting arms** (inpainting, balancing, DFR, paired LEACE) help at both overlaps in every modality except where the intervals are wide (ovary, DINOv2 capsule; Table 19).

None of them shows the positive location interaction of masking; most show a small negative one, because ERM relies more on an artifact that sits on the lesion and there is more for them to remove.

Table 19: All baseline arms in the controlled sweeps of the other modalities: ERM’s clean and reversed AUROC and each arm’s change in reversed-test AUROC versus ERM at no and at full overlap (crossed CIs). The last column is the arm’s own location interaction ($[\cdot]_{r=0} - [\cdot]_{r=1}$); an arm that does not depend on where the artifact sits has an interaction near zero.

Sweep	Arm	$r = 0$	$r = 1$	Location interaction
Thyroid caliper, DI-NOv2	ERM (AUROC: clean; reversed)	0.666; 0.417	0.643; 0.325	—
	masking — ERM	+0.402 [+0.315, +0.489]	−0.030 [−0.105, +0.042]	+0.433 [+0.362, +0.507]
	oracle inpainting — ERM	+0.321 [+0.261, +0.383]	+0.415 [+0.348, +0.484]	−0.093 [−0.124, −0.065]
	group-balanced — ERM	+0.273 [+0.220, +0.328]	+0.349 [+0.286, +0.415]	−0.076 [−0.106, −0.048]
	DFR — ERM	+0.299 [+0.228, +0.374]	+0.384 [+0.298, +0.472]	−0.085 [−0.123, −0.050]
	paired LEACE — ERM	+0.324 [+0.264, +0.386]	+0.420 [+0.353, +0.488]	−0.096 [−0.129, −0.066]
Thyroid caliper, MedSigLIP	ERM (AUROC: clean; reversed)	0.742; 0.320	0.723; 0.260	—
	masking — ERM	+0.514 [+0.430, +0.600]	+0.044 [−0.011, +0.104]	+0.470 [+0.388, +0.556]
	oracle inpainting — ERM	+0.457 [+0.388, +0.527]	+0.516 [+0.442, +0.589]	−0.059 [−0.089, −0.031]
	group-balanced — ERM	+0.381 [+0.324, +0.442]	+0.433 [+0.373, +0.494]	−0.052 [−0.084, −0.020]
	DFR — ERM	+0.437 [+0.370, +0.505]	+0.484 [+0.409, +0.561]	−0.047 [−0.086, −0.008]
	paired LEACE — ERM	+0.460 [+0.389, +0.531]	+0.517 [+0.442, +0.587]	−0.057 [−0.089, −0.027]
Ovarian caliper, DI-NOv2	ERM (AUROC: clean; reversed)	0.668; 0.656	0.677; 0.635	—
	masking — ERM	+0.041 [−0.096, +0.175]	−0.197 [−0.357, −0.035]	+0.238 [+0.144, +0.340]
	oracle inpainting — ERM	+0.057 [+0.025, +0.092]	+0.077 [+0.034, +0.122]	−0.021 [−0.047, +0.004]
	group-balanced — ERM	+0.069 [+0.020, +0.121]	+0.087 [+0.034, +0.143]	−0.018 [−0.048, +0.009]
	DFR — ERM	+0.013 [−0.079, +0.102]	+0.047 [−0.049, +0.142]	−0.034 [−0.077, +0.005]
	paired LEACE — ERM	+0.056 [+0.023, +0.093]	+0.073 [+0.031, +0.119]	−0.017 [−0.044, +0.011]
Capsule debris, DI-NOv2	ERM (AUROC: clean; reversed)	0.853; 0.822	0.849; 0.797	—
	masking — ERM	+0.116 [+0.052, +0.191]	−0.112 [−0.209, −0.019]	+0.228 [+0.144, +0.324]
	oracle inpainting — ERM	+0.024 [−0.016, +0.069]	+0.026 [−0.020, +0.074]	−0.002 [−0.026, +0.024]
	group-balanced — ERM	+0.009 [−0.032, +0.053]	+0.041 [+0.004, +0.081]	−0.032 [−0.073, +0.001]
	DFR — ERM	+0.030 [−0.034, +0.090]	+0.049 [−0.016, +0.114]	−0.019 [−0.041, +0.001]
	paired LEACE — ERM	+0.035 [+0.014, +0.061]	+0.059 [+0.030, +0.094]	−0.023 [−0.043, −0.006]
Capsule debris, MedSigLIP	ERM (AUROC: clean; reversed)	0.902; 0.586	0.887; 0.302	—
	masking — ERM	+0.374 [+0.248, +0.509]	+0.216 [+0.118, +0.317]	+0.158 [+0.012, +0.314]
	oracle inpainting — ERM	+0.303 [+0.201, +0.414]	+0.570 [+0.459, +0.675]	−0.267 [−0.388, −0.145]
	group-balanced — ERM	+0.288 [+0.185, +0.402]	+0.533 [+0.430, +0.636]	−0.245 [−0.359, −0.130]
	DFR — ERM	+0.294 [+0.186, +0.416]	+0.575 [+0.453, +0.695]	−0.281 [−0.398, −0.160]
	paired LEACE — ERM	+0.319 [+0.212, +0.433]	+0.608 [+0.499, +0.711]	−0.289 [−0.405, −0.171]
Chest tube, RAD-DINO	ERM (AUROC: clean; reversed)	0.915; 0.913	0.907; 0.614	—
	masking — ERM	−0.002 [−0.010, +0.005]	+0.111 [+0.095, +0.127]	−0.113 [−0.129, −0.097]
	oracle inpainting — ERM	+0.003 [+0.002, +0.004]	+0.299 [+0.271, +0.329]	−0.297 [−0.326, −0.267]
	group-balanced — ERM	+0.000 [−0.004, +0.004]	+0.309 [+0.280, +0.339]	−0.309 [−0.338, −0.281]
	DFR — ERM	−0.014 [−0.022, −0.007]	+0.280 [+0.248, +0.313]	−0.294 [−0.328, −0.262]
	paired LEACE — ERM	+0.003 [+0.002, +0.004]	+0.287 [+0.259, +0.316]	−0.284 [−0.314, −0.255]
Chest tube, DI-NOv2	ERM (AUROC: clean; reversed)	0.859; 0.629	0.852; 0.410	—
	masking — ERM	+0.254 [+0.230, +0.278]	−0.083 [−0.112, −0.057]	+0.337 [+0.312, +0.364]
	oracle inpainting — ERM	+0.225 [+0.204, +0.247]	+0.442 [+0.413, +0.470]	−0.216 [−0.235, −0.198]
	group-balanced — ERM	+0.229 [+0.206, +0.251]	+0.442 [+0.414, +0.470]	−0.214 [−0.231, −0.196]
	DFR — ERM	+0.218 [+0.197, +0.240]	+0.433 [+0.406, +0.460]	−0.214 [−0.234, −0.196]
	paired LEACE — ERM	+0.233 [+0.212, +0.255]	+0.452 [+0.424, +0.481]	−0.219 [−0.238, −0.202]

2.5.4 Sensitivity to the interval estimator

As in Stage 1, both estimators were computed on the same predictions of the other-modality sweeps (Table 20): the crossed intervals are wider by a median factor of 1.45; 11 of 180 contrasts lose significance and none gains it (Table 21). Every statement in this stage uses the crossed intervals, so these changes are already reflected in the text: in particular, MedSigLIP’s small benefit at full thyroid overlap and the ovarian harm at 50% overlap are not claimed.

Table 20: The sweeps of the other modalities under the per-seed and the crossed estimator (same predictions; baseline arms only).

Result group	Intervals	Excluded with the estimator	0 only per-seed	Excluded with the crossed estimator	0 only	Median width ratio (crossed / per-seed)
synthetic/capsule	60	9		0		1.72
synthetic/nih_ptx	30	0		0		1.44
synthetic/ovary	30	1		0		1.59
synthetic/thyroid	60	1		0		1.37
all	180	11		0		1.45

Table 21: Contrasts of the other-modality sweeps whose verdict depends on the estimator.

Result	Contrast	Crossed estimate [95% CI]	Verdict vs per-seed estimator
synthetic/thyroid/ medsiglip448.caliper_corr_main	1.0 / mask / erm / test_rev	+0.044 [−0.011, +0.104]	no longer excludes 0
synthetic/capsule/ dino518_debris_corr_main	0.0 / inpaint / erm / test_rev	+0.024 [−0.016, +0.069]	no longer excludes 0
synthetic/capsule/ dino518_debris_corr_main	0.75 / inpaint / erm / test_rev	+0.036 [−0.005, +0.074]	no longer excludes 0
synthetic/capsule/ dino518_debris_corr_main	1.0 / inpaint / erm / test_rev	+0.026 [−0.020, +0.074]	no longer excludes 0
synthetic/capsule/ dino518_debris_corr_main	0.5 / balanced / erm / test_rev	+0.030 [−0.018, +0.076]	no longer excludes 0
synthetic/capsule/ dino518_debris_corr_main	0.75 / balanced / erm / test_rev	+0.028 [−0.023, +0.076]	no longer excludes 0
synthetic/capsule/ dino518_debris_corr_main	0.0 / dfr / erm / test_rev	+0.030 [−0.034, +0.090]	no longer excludes 0
synthetic/capsule/ dino518_debris_corr_main	0.5 / dfr / erm / test_rev	+0.048 [−0.014, +0.110]	no longer excludes 0
synthetic/capsule/ dino518_debris_corr_main	0.75 / dfr / erm / test_rev	+0.055 [−0.010, +0.122]	no longer excludes 0
synthetic/capsule/ dino518_debris_corr_main	1.0 / dfr / erm / test_rev	+0.049 [−0.016, +0.114]	no longer excludes 0
synthetic/ovary/ dino518_caliper_corr_main	0.5 / mask / erm / test_rev	−0.107 [−0.269, +0.054]	no longer excludes 0

2.6 What failed or changed

- **Harm at full overlap is not universal.** The robust statement is the *interaction* (10 of 11); harm at full overlap is significant in 7 of 11 sweeps.
- **Two lesion-size cells are not significant** (large lesions at full overlap, medium lesions at 50%): harm is ordered by lesion size, not monotone in every cell.
- **Chest radiography with RAD-DINO contradicts the law**, as a failure of its premise. It was registered as a confirmation and is reported as a boundary case.
- **The counterfactual mechanism holds in 3 of 5 sweeps:** not for thyroid calipers (equal sensitivity) and reversed for the chest tube with RAD-DINO.
- **The ovarian sweep has no significant benefit outside the ROI**, so its interaction rests on the harm side.

- **Reproducibility of the capsule split.** The capsule sweep’s split is produced by a seeded shuffle of the group list; rebuilding the cohort (a re-derived contamination mask moved a few frames across the 5% threshold) re-dealt the split, and only 32 of 71 artifact-bearing test frames of an earlier build recur. Point estimates of that sweep moved by up to about 0.2, while the location interaction kept its sign and significance. The split is now frozen with the cohort file.

2.7 Anticipated questions

Why not simply show saliency maps? They show where a model looks, not how much its decision depends on the artifact, and they are unstable for frozen encoders with linear heads. The same-head counterfactual answers the causal question directly on the same image.

If masking removes context, why is the occlusion control not harmed? Because in the occlusion control the ruler is uninformative: the head does not need it, so keeping it costs nothing, and the context lost by masking is small for this task. The harm in the sweep appears only when the retained ruler predicts the label in training.

Why does dilation make things worse at 50% overlap? Because it makes more of the ruler visible (retention $0.50 \rightarrow 0.95$) while adding only a thin rim of context. The shortcut becomes stronger; the head uses it more.

Why does the harm not grow at 100% overlap when the mask is dilated? The ruler is already fully visible; dilation only adds skin. If harm were about the ruler’s share of the image it would fall; it does not.

Why do the artifact-targeting arms help *more* when the artifact is inside? ERM relies more on a ruler on the lesion (its reversed AUROC is lower at $r = 1$), so removing that reliance gains more. Their gain does not depend on the ROI.

Why does thyroid show loss of benefit rather than harm? The ultrasound background contains little disease information, so masking loses almost nothing besides failing to remove the caliper; ERM is already caliper-driven, and masking leaves it so. Harm needs masking to remove disease evidence as well.

Why is the chest tube a boundary and not a failure? The law compares what masking removes outside and inside the ROI. RAD-DINO does not use the out-of-lung tube at all (reversed $\approx$ clean AUROC), so there is nothing to remove outside and the comparison has no content. DINOv2 uses the tube and shows the law.

Are 3 seeds enough? For the interactions, yes: the intervals include the variation of the fixed test set, and 10 of 11 exclude zero. Individual cells in small cohorts (ovary) are imprecise and are interpreted only through the interaction.

Could the synthetic artifacts be unrealistic? Yes, and that is their purpose: they isolate location. Whether the law holds for artifacts that clinicians actually produce is a separate question that synthetic sweeps cannot answer.

2.8 Conclusion

Supported. The location law in its interaction form (Q2a) holds in 10 of 11 controlled sweeps across five modalities and four encoders. The mechanism (Q2b) is retention with amplified reliance: occlusion alone does not harm, harm follows the visibility of the artifact under dilation, and the masked head’s same-image reliance on an in-ROI artifact is about 2.1 times that of ERM for the ruler (+0.255 [+0.221, +0.287]) and significantly higher in 3 of 5 sweeps. **Qualified.** Harm at full overlap is setting-dependent; the lesion-size gradient is an ordering. **Not supported.** The law for an encoder that ignores the out-of-ROI artifact (chest tube, RAD-DINO). The practical reading is: masking removes what lies outside the ROI and concentrates the model on what lies inside; whether that helps depends on where the shortcut sits.

2.9 Questions for the supervisor

1. Should the thesis state the law in its interaction form only (robust) and treat harm at full overlap as setting-dependent, or keep harm as the headline for dermoscopy, where it holds with every encoder?
2. The counterfactual mechanism holds in 3 of 5 sweeps. Is the thyroid result (the unmasked model is already as reliant on an in-nodule caliper as the masked one) best presented as a limitation or as a second route to the same outcome?
3. Is it acceptable to present chest radiography with RAD-DINO as a boundary of the law rather than as a failed confirmation, given that it was registered as a confirmation?

3 Stage 3 — Real artifacts across modalities

In one sentence. With real, unedited artifacts — hair in dermoscopy, calipers in thyroid and ovarian ultrasound, debris in capsule endoscopy — ROI masking helps far more when the artifact lies outside the ROI than inside it: the crossover is positive in 9 of 10 cohort–encoder pairs and in all four pre-specified location tests after Holm correction; the dermatology encoder DermLIP and real chest-radiograph devices are the exceptions, and the hair shortcut turns out to be carried largely by correlates of hair rather than by its pixels.

3.1 Where this stage starts

Stage 1 showed with real hair (E13) that masking harms when hair lies on the lesion and helps when it lies beside it. Stage 2 established the location law with synthetic artifacts whose position was controlled, in five modalities, and identified its mechanism (retention with amplified reliance). Synthetic artifacts answer *whether* location can matter; they cannot show that it matters for the artifacts clinicians actually produce, whose location is set by the workflow and read from masks. This stage builds real-artifact traps in three more cohorts, validates the automatic artifact masks that define “inside” and “outside”, replicates across encoder families, asks what actually carries the hair shortcut, and examines chest radiography, where the law might not hold.

3.2 Questions

- Q3a** With real artifacts and the same protocol, is ROI masking less useful — or harmful — when the artifact lies inside the ROI than when it lies outside it (a positive crossover)?
- Q3b** Does this hold across modalities, artifacts and encoder families?
- Q3c** Are the artifact masks that define inside and outside good enough to support the answer?
- Q3d** What carries a real shortcut: the artifact’s pixels, or things that go with the artifact?

Design: two traps sharing one artifact-free group A real artifact’s location is fixed by the image, so it cannot be moved. Each cohort is split into images whose artifact lies mostly inside the ROI ($r \geq 0.5$, Trap A) and images whose artifact lies outside it ($r < 0.1$, Trap B); images with $0.1 \leq r < 0.5$ belong to neither trap. Each trap is completed by the *same* artifact-free images and receives the environments of Stage 1. This is the E13 design of Stage 1, which replaced the pilot’s overlap-contrast design (E12) because in E12 location itself was the label proxy.

Why these cohorts Each has (i) an artifact that clinicians produce and that is associated with the diagnosis through the workflow (calipers are placed on nodules that are measured; debris is more frequent on some lesion types), (ii) public ROI masks or boxes, (iii) a way to locate the artifact without manual annotation, and (iv) enough images for an artifact-free group. Table 34 lists the cohorts that failed one of these conditions.

3.3 Methods

3.3.1 Cohorts

- **Dermoscopy** — ISIC 2019 (Combalia et al., 2019; Tschandl et al., 2018) as in Stage 1: published semi-automatic hair masks (Wegley et al., 2026); lesion masks from HAM10000 or the U-Net trained on ISIC 2018 Task 1 (Codella et al., 2019); leakage groups lesion identifier $\cup$ perceptual hash.
- **Thyroid ultrasound** — TN3K images (Gong et al., 2021) with TNCD benign/malignant labels (3,493 images, 1,210 malignant) and manual nodule masks. Calipers located by a rule-based

detector (dotted measurement lines = ≥ 7 evenly spaced bright dots; “+” calipers by template matching), tuned on 36 images and frozen before its final audit. $A = 0$: no detected caliper pixel; $A = 1$: ≥ 15 detected pixels. Groups: pHash ≤ 8 (TN3K publishes no patient identifiers).

- **Ovarian ultrasound** — MMOTU OTU_2d (Zhao et al., 2022) (CC BY 4.0): benign cystic lesions (chocolate cyst, serous cystadenoma, simple cyst) versus lesions with a solid component (teratoma, theca cell tumour, mucinous cystadenoma, high-grade serous carcinoma); normal ovaries excluded (1,202 images). The frozen thyroid detector, unchanged. Groups: pHash ≤ 8 .
- **Capsule endoscopy** — SEE-AI (Yokote et al., 2023) (CC BY 4.0), single-class frames: erosion versus polyp-like lesions; ROI = union of the expert lesion boxes. Contamination (debris, bubbles, bile) masks from the published expert masks where available (22 frames) and otherwise from a linear probe on DINOv2 patch tokens trained on 2,160 expert-masked frames. $A = 0$: contamination $< 3\%$ of the frame; $A = 1$: $\geq 10\%$. Groups: blocks of 100 consecutive frame numbers $\cup$ pHash ≤ 2 (no video identifiers are published).
- **Chest radiography** — NIH ChestX-ray14 (Wang et al., 2017); lungs from a PSPNet (Cohen et al., 2022). (i) *Device traps*: images linked to RANZCR-CLiP device annotations; Trap A = central venous catheter inside the lungs, Trap B = endotracheal tube outside them; four findings with ≥ 40 reversed-test positives; restricted to AP films (intubated patients are imaged supine, so view could not be matched). (ii) *Chest drains*: 29,687 pneumothorax-labelled images; drains from NEATX human labels and a drain detector (precision 0.95); drains lie inside the lungs, so this is an in-ROI trap only. Patient-grouped folds.

Protocol Frozen encoders: DINOv2 ViT-B/14 @518 for every cohort; MedSigLIP-448 (a medical vision–language model) for thyroid, ovary and capsule; ConvNeXt-Base @384 (a CNN) for thyroid and capsule; DermLIP for dermoscopy; RAD-DINO for chest radiographs. Logistic heads chosen on clean validation data; 5 seeds (42, 123, 456, 789, 2026) $\times$ 5 grouped folds, fold predictions pooled per seed. Within each label the artifact-bearing images are subsampled to the source mix of the artifact-free group (source matching; vacuous for single-source cohorts). Gates fixed before fitting: every matched cell ≥ 25 images and ≥ 40 reversed-test positives. Arms as in Stage 1.

Statistics Crossed seed $\times$ image bootstrap, 10,000 replicates; the crossover’s interval resamples the same images in both traps. The four location tests (P1, one per cohort, DINOv2) are Holm-corrected; everything else is reported with unadjusted 95% intervals.

3.4 Experiments

Table 22: Experiments of this stage with registration status, the registration document (docs/PREREGISTRATION_<name>.md), its first commit and the support criterion stated in advance. Commit times are set by the committing machine and are not independent proof of order.

Experiment	Status	Registration	First commit (local time)	Support criterion
Dermoscopy hair traps (E13, DINOv2)	supported registered	ISIC2019.TRAPS	76e25d4, 13:14	2026-09-26 crossover CI > 0
Dermoscopy hair, DermLIP	supported registered	ISIC2019.TRAPS	76e25d4, 13:14	2026-09-26 crossover CI > 0
Thyroid caliper traps	supported registered	THYROID_TRAPS	ea57661, 22:46	2026-09-26 crossover CI > 0
Ovarian caliper traps	supported registered	OVARY_TRAPS	2517600, 07:29	2026-09-27 crossover CI > 0
Capsule debris traps	supported registered	CAPSULE_TRAPS	fcfb8d9, 00:40	2026-09-27 crossover CI > 0
Chest-radiograph device traps	supported registered	CXR_DEVICE_TRAPS	4d1a924, 18:07	2026-09-26 crossover CI > 0
Location tests P1 (four cohorts, Holm)	partly supported each test registered; Holm grouping post hoc	STATISTICAL_PLAN	4e9d44c, 09:37	2026-09-27 Holm $p < 0.05$, positive estimate
Chest drains (NIH)	supported registered	CXR_DRAIN	3d0cffc, 03:29	2026-09-27 mask — ERM reported with CI, no direction
Crossed-bootstrap intervals (A1)	supported registered	FINAL (A1)	c45d482, 00:35	2026-09-28 a claim is stated only if its crossed CI excludes 0

3.5 Results

3.5.1 Cohorts and artifact masks

Table 23: Real-artifact cohorts and pool sizes (images, positive/negative diagnosis) before the environments are sampled. Trap A: artifact mostly inside the ROI ($r \geq 0.5$); Trap B: outside ($r < 0.1$); both share the artifact-free group.

Cohort	Task	Artifact source)	(mask ROI	Artifact-free pos/neg	Trap A pos/neg	Trap B pos/neg
ISIC 2019 dermoscopy (HAM10000, BCN20000, MSK)	melanoma vs other	hair masks)	(published lesion: HAM manual / U-Net	157/706	942/2655	969/4412
TN3K ultrasound, TNCD labels	malignant vs benign nodule	calipers (rule detector)	nodule (manual)	749/1052	292/858	99/227
MMOTU 2D ultrasound	solid-component vs cystic tumour	calipers (rule detector)	tumour (manual)	178/170	276/341	103/91
SEE-AI capsule endoscopy	erosion vs polyp-like lesion	debris (patch probe)	lesion box (expert)	343/164	412/404	1009/231

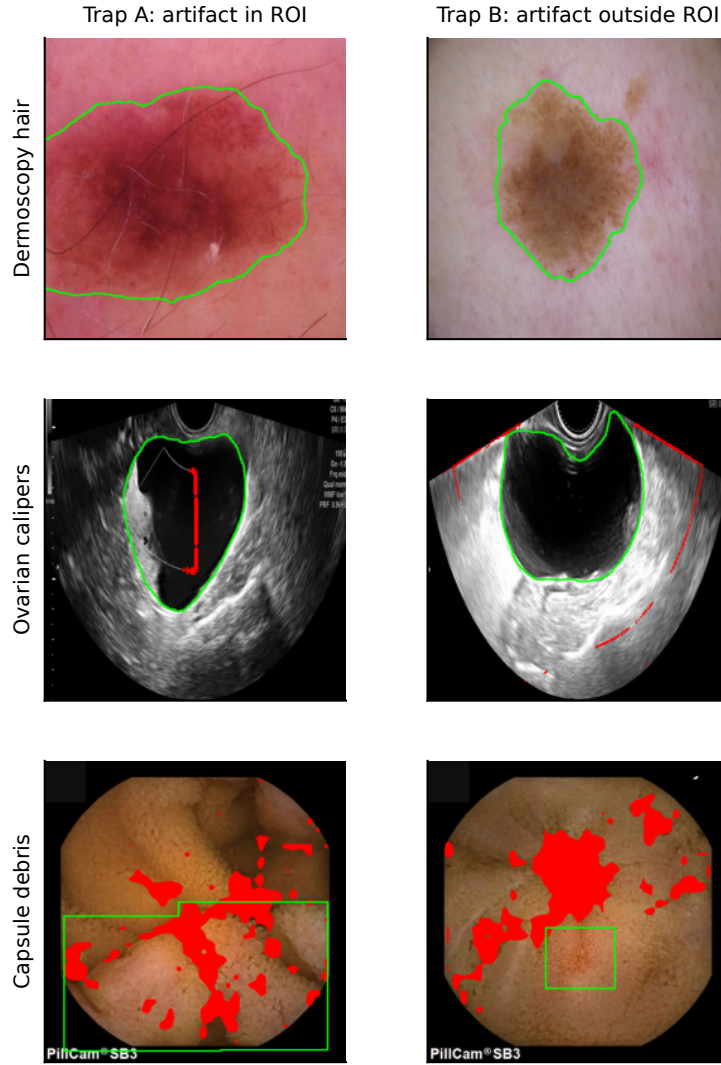


Figure 9: Real artifacts inside (Trap A) and outside (Trap B) the ROI. **Green**: ROI; **red**: automatic artifact mask. Dermoscopy is shown with the lesion outline only (the published hair masks carry no redistribution licence). Thyroid images are not shown (data licence not stated). Automatic overlap of the shown images: dermoscopy 0.534 and 0.000; ovary 1.000 and 0.000; capsule 0.740 and 0.005. ISIC CC BY-NC 4.0; MMOTU, SEE-AI CC BY 4.0.

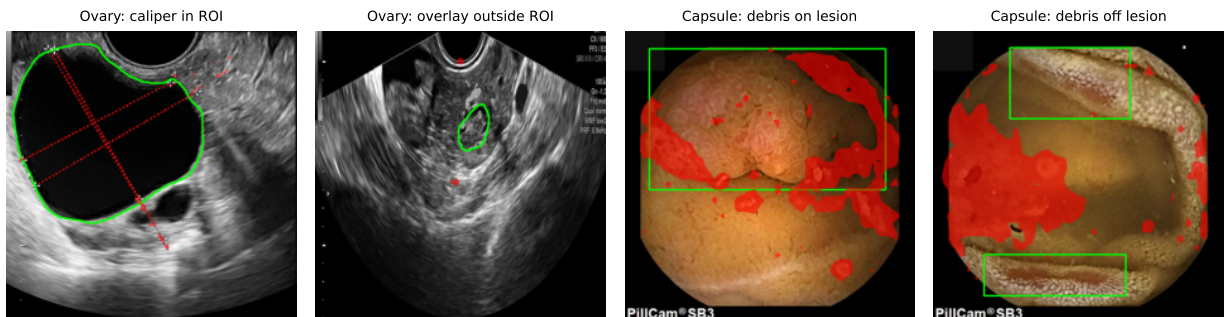


Figure 10: Further real examples with the automatic masks: ovarian calipers crossing the tumour and an overlay outside it; capsule debris on and off the lesion boxes (MMOTU, SEE-AI; CC BY 4.0).

Mask validity Table 24. Lesion masks are accurate (U-Net held-out Dice 0.886; against HAM manual masks 0.895). The capsule probe reaches pixel accuracy 0.922 and IoU 0.625 against held-out expert masks. The thyroid detector’s detections lie on a real caliper in 38 of 40 audited images, and 38–39 of 40 caliper-free images are clean. For the ovary, every detection is an on-screen overlay but only 31 of 40 are calipers (the others are dotted zoom arcs and focus markers), and the “marker-free” group is caliper-free but not overlay-free. The morphological hair detector of the pilot is poor (IoU 0.222), which is why the hair traps use the published masks.

Why label noise cannot create the result An image mislabelled as artifact-bearing (or artifact-free) weakens the constructed association between A and Y equally in both traps; it biases every effect toward zero. To create a crossover, errors would have to depend on location *and* on the label, which none of the detectors uses.

Table 24: Validity of the automatic masks that define artifact presence and location. Visual audits: docs/ARTIFACT_MASK_AUDIT.md and docs/THYROID_MARKER_AUDIT.md (visual audits during the analysis, not by a clinician).

Mask	Source	Check	Result
ISIC 2019 lesion masks (BCN, MSK)	U-Net trained on ISIC 2018 Task 1	held-out Dice (mean) / Dice vs HAM manual masks	0.886 / 0.895
Hair (DullRazor-type detector, E10)	morphological	IoU / precision / recall vs Mendeley masks	0.222 / 0.316 / 0.472
Capsule debris masks	patch-token probe (2,160 expert-masked frames)	held-out pixel accuracy / IoU	0.922 / 0.625
Thyroid (TN3K), detected (A1)	visual audit, 40 images	detection lies on a real caliper / measurement mark	38/40 (95 %); 2 border false positives; 1 dotted zoom box (overlay)
Thyroid, marker-free (A0)	visual audit, 40 images	no visible caliper	38–39/40 (95–97 %); depth-scale ticks present in all images (not label-related)
Ovary (MMOTU), detected (A1)	visual audit, 40 images	an on-screen overlay is present	40/40; caliper specifically 31/40 (others: dotted zoom arcs, focus markers, thumbnail inset)
Ovary, marker-free (A0)	visual audit, 40 images	no visible caliper	37/40 (92 %)
Ovary, marker-free (A0)	visual audit, 40 images	no overlay of any kind (arrows, boxes)	~29/40 (72 %); text labels common

3.5.2 The crossover with real artifacts

Table 25 and Figure 11. With DINOv2, masking is worth less for in-ROI artifacts in every cohort:

- **Dermoscopy hair.** Masking *harms* when the hair lies on the lesion (-0.049 [$-0.083, -0.018$]) and helps when it lies outside ($+0.098$ [$+0.060, +0.136$]); crossover $+0.147$ [$+0.111, +0.185$] (as in Stage 1). After masking the Trap-A model still separates the correlated test (0.917) far better than the reversed one (0.460).
- **Thyroid calipers.** Masking helps in both traps but far more when the calipers lie outside the nodule ($+0.348$ [$+0.307, +0.388$] versus $+0.109$ [$+0.085, +0.133$]; crossover $+0.239$ [$+0.196, +0.280$]). In Trap A the masked model remains caliper-driven: correlated 0.908, reversed 0.398 (Table 28); in Trap B masking removes the gap (0.785 versus 0.752).
- **Ovarian calipers.** Crossover $+0.170$ [$+0.087, +0.252$] (Trap A $+0.132$ [$+0.072, +0.196$], Trap B $+0.302$ [$+0.225, +0.378$]).
- **Capsule debris.** The largest crossover ($+0.377$ [$+0.327, +0.426$]): masking recovers almost all of the reversed-test loss when the debris lies outside the lesion boxes ($+0.461$ [$+0.415, +0.506$]) and little when it lies on them ($+0.084$ [$+0.055, +0.112$]).

Other encoder families The crossover replicates with MedSigLIP (thyroid +0.195 [+0.153, +0.238], ovary +0.178 [+0.109, +0.248], capsule +0.335 [+0.286, +0.383]) and with the CNN ConvNeXt (thyroid +0.275 [+0.231, +0.320], capsule +0.320 [+0.268, +0.372]). With ConvNeXt, masking is useless inside the nodule (−0.004 [−0.029, +0.020]) and strongly helpful outside it (+0.271 [+0.228, +0.314]). The effect is not a property of vision transformers or of self-supervised training.

Harm or loss of benefit? Only for dermoscopy hair does masking actually harm in Trap A. For calipers and debris it still helps a little in Trap A — it removes clutter and overlays elsewhere on the screen — but a large residual shortcut remains. The universal real-data statement is therefore the *crossover*; harm appears where masking also removes disease context (dermoscopy, and in Stage 2 the controlled sweeps).

Table 25: Real-artifact traps: change in reversed-test AUROC from ROI masking, per trap, and the crossover (5 seeds $\times$ 5 grouped folds; crossed 95% CIs).

Cohort	Encoder	ERM reversed AUROC A / B	mask − ERM, Trap A (in ROI)	mask − ERM, Trap B (outside)	Crossover B − A
Dermoscopy hair	DINOv2	0.510 / 0.510	−0.049 [−0.083, −0.018]	+0.098 [+0.060, +0.136]	+0.147 [+0.111, +0.185]
Dermoscopy hair	DermLIP	0.665 / 0.735	−0.104 [−0.143, −0.067]	−0.090 [−0.135, −0.046]	+0.014 [−0.037, +0.063]
Thyroid calipers	DINOv2	0.289 / 0.405	+0.109 [+0.085, +0.133]	+0.348 [+0.307, +0.388]	+0.239 [+0.196, +0.280]
Thyroid calipers	MedSigLIP	0.289 / 0.481	+0.053 [+0.032, +0.073]	+0.248 [+0.200, +0.293]	+0.195 [+0.153, +0.238]
Ovarian calipers	DINOv2	0.413 / 0.423	+0.132 [+0.072, +0.196]	+0.302 [+0.225, +0.378]	+0.170 [+0.087, +0.252]
Capsule debris	DINOv2	0.581 / 0.429	+0.084 [+0.055, +0.112]	+0.461 [+0.415, +0.506]	+0.377 [+0.327, +0.426]
Capsule debris	MedSigLIP	0.570 / 0.464	+0.117 [+0.088, +0.147]	+0.452 [+0.407, +0.495]	+0.335 [+0.286, +0.383]
Thyroid calipers	ConvNeXt	0.357 / 0.429	−0.004 [−0.029, +0.020]	+0.271 [+0.228, +0.314]	+0.275 [+0.231, +0.320]
Ovarian calipers	MedSigLIP	0.426 / 0.551	+0.002 [−0.048, +0.051]	+0.180 [+0.120, +0.238]	+0.178 [+0.109, +0.248]
Capsule debris	ConvNeXt	0.528 / 0.431	+0.084 [+0.049, +0.118]	+0.403 [+0.358, +0.448]	+0.320 [+0.268, +0.372]

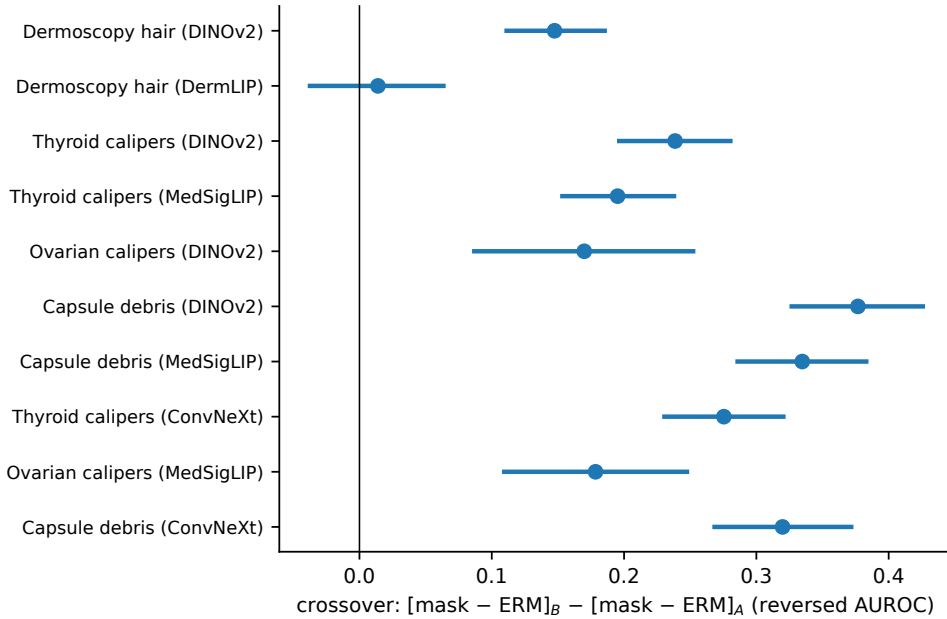


Figure 11: Crossover with real artifacts, per cohort and encoder (crossed 95% CIs).

3.5.3 The four location tests

Table 26. Each cohort’s crossover test was registered with the cohort; they are grouped here into one Holm family. All 4 of 4 are supported ($p \leq 0.0001$ before adjustment).

Table 26: The four pre-specified location tests (P1: crossover > 0), crossed bootstrap, two-sided bootstrap p (floored at 10^{-4}), Holm-adjusted over the four cohorts.

Cohort (DINOv2)	Crossover [95% CI]	p	p (Holm, 4)	Supported
ISIC hair (DINOv2)	+0.147 [+0.111, +0.185]	0.0001	0.0004	supported
Thyroid (DINOv2)	+0.239 [+0.196, +0.280]	0.0001	0.0004	supported
Capsule (DINOv2)	+0.377 [+0.327, +0.426]	0.0001	0.0004	supported
Ovary (DINOv2)	+0.170 [+0.087, +0.252]	0.0001	0.0004	supported

3.5.4 Every arm, and the environments

Tables 27 and 28. Three observations matter.

- 1. Label-based methods fix both traps.** Group balancing and DFR, which need image-level artifact labels, help in both traps of every cohort (e.g. thyroid DFR +0.463 [+0.436, +0.492] in Trap A), and their benefit does not depend on location. This replicates the pilot’s third claim on three new cohorts.
- 2. Oracle inpainting fails for capsule debris.** Inpainting the debris pixels *lowers* reversed AUROC (−0.172 [−0.202, −0.142] in Trap A). Erosions carry yellow-white fibrin that resembles debris; inpainting what the probe calls debris removes disease evidence. Removing artifact pixels is only safe when the artifact does not resemble the disease.
- 3. Two arms “help” by flipping the shortcut.** Unpaired LEACE and prevalence calibration show the largest reversed-test gains (+0.515 [+0.491, +0.538], +0.602 [+0.580, +0.625] for thyroid), but both remove or invert the artifact–label association using the artifact label of the *training* set (and prevalence calibration also at test time), so their correlated AUROC falls far below their reversed one — the same failure as in Stage 1. They are reported, not recommended. Paired LEACE, the sound variant, recovers little (thyroid +0.052 [+0.045, +0.059]).

Table 27: Every baseline arm in the real-artifact traps (DINOv2 @518, 5 seeds $\times$ 5 folds): change in reversed-test AUROC versus ERM, crossed 95% CIs. Inpainting needs the artifact mask; balancing, DFR, unpaired LEACE and prevalence calibration need image-level artifact labels (prevalence calibration also at test time).

Cohort	Arm – ERM	Trap A (in ROI)	Trap B (outside)
Dermoscopy hair	masking	−0.049 [−0.083, −0.018]	+0.098 [+0.060, +0.136]
	oracle inpainting	+0.098 [+0.084, +0.112]	+0.100 [+0.083, +0.117]
	group-balanced	+0.213 [+0.183, +0.243]	+0.204 [+0.166, +0.245]
	DFR	+0.196 [+0.157, +0.235]	+0.196 [+0.147, +0.251]
	paired LEACE	+0.074 [+0.062, +0.086]	+0.078 [+0.059, +0.097]
	unpaired LEACE	+0.320 [+0.280, +0.361]	+0.295 [+0.253, +0.336]
	prevalence calibration	+0.415 [+0.377, +0.455]	+0.429 [+0.391, +0.466]
Thyroid calipers	masking	+0.109 [+0.085, +0.133]	+0.348 [+0.307, +0.388]
	oracle inpainting	+0.102 [+0.092, +0.112]	+0.039 [+0.025, +0.053]
	group-balanced	+0.388 [+0.360, +0.418]	+0.254 [+0.218, +0.292]
	DFR	+0.463 [+0.436, +0.492]	+0.312 [+0.262, +0.362]
	paired LEACE	+0.052 [+0.045, +0.059]	+0.023 [+0.014, +0.032]
	unpaired LEACE	+0.515 [+0.491, +0.538]	+0.338 [+0.302, +0.375]
	prevalence calibration	+0.602 [+0.580, +0.625]	+0.501 [+0.463, +0.538]
Ovarian calipers	masking	+0.132 [+0.072, +0.196]	+0.302 [+0.225, +0.378]
	oracle inpainting	+0.160 [+0.135, +0.186]	+0.046 [+0.032, +0.061]
	group-balanced	+0.290 [+0.248, +0.332]	+0.256 [+0.190, +0.320]
	DFR	+0.310 [+0.263, +0.357]	+0.278 [+0.209, +0.349]
	paired LEACE	+0.142 [+0.119, +0.166]	+0.031 [+0.020, +0.043]
	unpaired LEACE	+0.369 [+0.320, +0.418]	+0.305 [+0.230, +0.375]
	prevalence calibration	+0.514 [+0.468, +0.561]	+0.507 [+0.440, +0.571]
Capsule debris	masking	+0.084 [+0.055, +0.112]	+0.461 [+0.415, +0.506]
	oracle inpainting	−0.172 [−0.202, −0.142]	−0.115 [−0.144, −0.087]
	group-balanced	+0.179 [+0.150, +0.205]	+0.244 [+0.216, +0.272]
	DFR	+0.234 [+0.199, +0.273]	+0.287 [+0.239, +0.335]

Cohort	Arm – ERM	Trap A (in ROI)	Trap B (outside)
	paired LEACE	+0.012 [+0.004, +0.020]	+0.010 [+0.001, +0.024]
	unpaired LEACE	+0.282 [+0.247, +0.316]	+0.416 [+0.374, +0.457]
	prevalence calibration	+0.347 [+0.316, +0.376]	+0.465 [+0.430, +0.498]

Table 28: AUROC of ERM, masking, balancing and DFR in the three test environments (DINOv2, mean over 5 seeds). A large correlated–reversed gap is reliance on the artifact.

Cohort	Trap	Arm	clean	correlated	reversed
Dermoscopy hair	Trap A	erm	0.734	0.920	0.510
	Trap A	mask	0.712	0.917	0.460
	Trap A	balanced	0.785	0.850	0.722
	Trap A	dfr	0.715	0.740	0.705
	Trap B	erm	0.716	0.894	0.510
	Trap B	mask	0.701	0.795	0.608
	Trap B	balanced	0.752	0.800	0.714
	Trap B	dfr	0.683	0.669	0.706
	Trap A	erm	0.608	0.879	0.289
	Trap A	mask	0.689	0.908	0.398
Thyroid calipers	Trap A	balanced	0.717	0.757	0.677
	Trap A	dfr	0.708	0.669	0.752
	Trap B	erm	0.600	0.789	0.405
	Trap B	mask	0.767	0.785	0.752
	Trap B	balanced	0.657	0.660	0.659
	Trap B	dfr	0.653	0.582	0.716
	Trap A	erm	0.635	0.826	0.413
	Trap A	mask	0.690	0.828	0.545
	Trap A	balanced	0.710	0.725	0.703
	Trap A	dfr	0.708	0.689	0.723
Ovarian calipers	Trap B	erm	0.636	0.815	0.423
	Trap B	mask	0.733	0.752	0.725
	Trap B	balanced	0.694	0.710	0.679
	Trap B	dfr	0.671	0.632	0.701
	Trap A	erm	0.796	0.941	0.581
	Trap A	mask	0.860	0.972	0.665
	Trap A	balanced	0.847	0.916	0.760
	Trap A	dfr	0.842	0.866	0.816
	Trap B	erm	0.714	0.922	0.429
	Trap B	mask	0.932	0.962	0.890
Capsule debris	Trap B	balanced	0.784	0.879	0.673
	Trap B	dfr	0.762	0.819	0.716

3.5.5 What carries the hair shortcut?

Hair presence and location go with patient and body site (Table 29): in HAM, hair-free lesions are more often from women and from the lower extremity; in BCN, in-lesion hair goes with older patients and the head and neck. Two diagnostics show that the hair shortcut is largely carried by these correlates rather than by hair pixels:

- **Pixel share.** Removing the hair pixels at test time only (oracle inpainting of the ERM model’s test images) closes 47% (Trap A) and 44% (Trap B) of ERM’s clean–reversed gap: more than half of what ERM exploits is not in the hair pixels.
- **Oracle inpainting** gains about +0.098 [+0.084, +0.112] in Trap A, half of what balancing gains (+0.213 [+0.183, +0.243]), which removes the association for hair and everything correlated with it.

Two archived sensitivity designs point the same way (**point estimates, not re-estimated**): an author-independent reconstruction of the traps (strictly hair-free comparison group, artifact-free clean test) gives a crossover of +0.166 but *no* Trap-A harm (+0.012); traps matched on site, sex and age band give +0.169 (Trap A +0.030). The crossover is robust to these choices; the in-lesion harm depends on including lightly haired images and on the strict hair-free group. Hair is therefore

the *correlate-carried* regime: masking cannot remove a shortcut that is carried by who the patient is and where the lesion is.

Table 29: Patient and site metadata of the hair groups (ISIC 2019): hair presence and location go with age, sex and body site.

Source	Group	Mean age	Share female	Share head/neck	Share lower extremity
HAM	hair-free	57.9	0.57	0.10	0.42
	Trap A (on lesion)	52.1	0.42	0.12	0.16
	Trap B (beside)	49.9	0.45	0.14	0.26
BCN	hair-free	61.3	0.59	0.18	0.32
	Trap A (on lesion)	58.9	0.45	0.33	0.13
	Trap B (beside)	51.8	0.45	0.22	0.18
MSK	hair-free	51.2	0.41	0.07	0.14
	Trap A (on lesion)	54.5	0.33	0.09	0.08
	Trap B (beside)	46.9	0.45	0.09	0.11

Table 30: Dermoscopy hair, Trap A, by image source (DINOv2).

Source	mask – ERM, Trap A
HAM	−0.011 [−0.064, +0.043]
BCN	−0.067 [−0.110, −0.025]

Hair area relative to the lesion In Trap A the in-lesion hair covers a median 0.033 of the lesion’s area — similar to the synthetic ruler of Stages 1–2. Masking harms in every tertile of that ratio (Table 31); unlike the synthetic sweep, no gradient is visible — consistent with a shortcut that is not mainly in the hair pixels.

Table 31: Dermoscopy hair, Trap A, by the ratio of in-lesion hair area to lesion area.

Hair/lesion area tertile	Median ratio	mask – ERM, Trap A
tertile 1	0.006	−0.057 [−0.095, −0.022]
tertile 2	0.033	−0.046 [−0.082, −0.013]
tertile 3	0.113	−0.037 [−0.071, −0.005]

3.5.6 The exception among encoders: DermLIP

With DermLIP masking harms at *both* hair locations (Trap A −0.104 [−0.143, −0.067], Trap B −0.090 [−0.135, −0.046]) and the crossover vanishes (+0.014 [−0.037, +0.063]). DermLIP was trained on whole dermatology images and loses disease signal when the background is removed, whatever the hair does (Stage 1: clean AUROC falls with masking in both traps).

3.5.7 Chest radiography: the boundary

Device traps Lung masking helped for both device types in all four registered findings (Table 32); the registered claim that masking harms — or helps less — for the in-lung catheter was supported for one finding only. A confound noted before the follow-up was fitted: the catheter-free group carries many more out-of-lung tubes than the catheter group (endotracheal tubes 77% versus 39%, nasogastric tubes 83% versus 42%), because CLiP’s catheter-free images were selected for other devices; masking removes those tubes. A follow-up matched within label on view and on the presence of the other devices (Infiltration) still shows no crossover (Trap A +0.089, Trap B +0.042, crossover −0.047; point estimates).

Table 32: Chest radiographs with real devices (NIH ChestX-ray14 linked to RANZCR-CLiP; RAD-DINO): Trap A = central venous catheter inside the lungs, Trap B = endotracheal tube outside them. Point estimates only, **not re-estimated** with the corrected bootstrap (per-image predictions were not saved; the old intervals are withdrawn).

Finding (NIH, RANZCR-CLiP devices)	mask – ERM, Trap A	Trap B	Crossover
Atelectasis	+0.045	+0.105	+0.060
Consolidation	+0.070	+0.068	−0.002
Effusion	+0.104	+0.100	−0.004
Infiltration	+0.094	+0.063	−0.032

Chest drains Table 33. The drain is an extreme shortcut: ERM’s reversed AUROC is 0.191 (RAD-DINO) and 0.280 (DINOv2), against clean 0.886 and 0.823. Lung masking barely helps (0.239, 0.292), because the drain lies in the lungs. Label-based DFR is best (0.683, 0.549): a drain comes with a treated, re-expanded lung, so the shortcut is carried by the treatment’s consequences, not only by the tube — a second correlate-carried case.

Table 33: Real chest drains (NIH ChestX-ray14 pneumothorax, 29,687 images, patient-grouped folds; drains lie inside the lungs, so this is an in-ROI trap only). AUROC, mean over 5 folds; **point estimates, not re-estimated** with the crossed bootstrap (per-image predictions were not saved).

Encoder	Arm	clean	correlated	reversed
RAD-DINO	ERM	0.886	0.948	0.191
	lung masking	0.870	0.949	0.239
	group-balanced	0.853	0.870	0.538
	DFR	0.800	0.738	0.683
	masking + DFR	0.801	0.770	0.646
DINOv2	ERM	0.823	0.922	0.280
	lung masking	0.834	0.932	0.292
	group-balanced	0.781	0.841	0.399
	DFR	0.770	0.808	0.549
	masking + DFR	0.788	0.827	0.566

Reading Together with the synthetic-tube sweep of Stage 2 (RAD-DINO ignores an out-of-lung tube), chest radiography is a boundary: in-lung devices are either confounded with other devices or carried by correlates, and the radiograph encoder does not use extra-pulmonary tubes at all. It is reported as a boundary case, not as a fifth confirmation.

3.5.8 Datasets considered and not used

Table 34: Cohorts evaluated for a real-artifact trap and rejected, with the reason (decided before any model was fitted on them).

Dataset	Artifact	Reason
Breast ultrasound (BUSI, BUS-BRA), attempt 1	calipers	the thyroid detector misses GE-style “x” calipers; a two-detector design left the artifact-free group only $\approx 70\%$ pure on audit
Breast ultrasound, attempt 2	calipers	a patch-token caliper probe (AUROC 0.98 on thyroid/ovary) found only 19 of 2,522 images free of on-screen markings: no artifact-free group exists
CANDID-PTX	chest drains	access not granted
CXR cardiomegaly lines	lines	too few labelled images
TCGA / GrandQC pathology	pen marks	pen masks unreliable
ISIC ink	ink marks	ink on the lesion in only 70 images
EAD endoscopy	specular reflections	no disease labels
BKAI-IGH NeoPolyp	specular highlights	no specular-free images (0), 41 in-polyp
colonoscopy		
DINOv3 encoder	—	weights gated, access refused

3.5.9 Sensitivity to the interval estimator

On the traps of this stage (Table 35) the crossed intervals are wider by a median factor of 1.62; 6 of 159 contrasts lose significance and 3 gain it (Table 36); none of them is a crossover or a masking effect on the reversed test.

Table 35: Real-artifact traps of the new cohorts under the per-seed and the crossed estimator (same predictions; baseline arms only).

Result group	Intervals	Excluded with the estimator	0 only per-seed	Excluded with the estimator	0 only crossed	Median width ratio (crossed / per-seed)
capsule/convnext384_universal	17	0		0		1.38
capsule/dino518_main	25	0		1		1.59
capsule/medsiglip448_main	25	0		0		1.69
ovary/dino518_main	25	2		0		1.67
thyroid/convnext384_universal	17	3		0		1.68
thyroid/dino518_main	25	1		1		1.62
thyroid/medsiglip448_main	25	0		1		1.57
all	159	6		3		1.62

Table 36: Contrasts of the real-artifact traps whose verdict depends on the estimator.

Result	Contrast	Crossed estimate [95% CI]	Verdict vs per-seed estimator
thyroid/dino518_main	trapB / dfr / clean / dfr / erm	+0.053 [+0.002, +0.105]	now excludes 0
thyroid/dino518_main	trapB / leace_paired / clean / leace_paired / erm	+0.008 [−0.001, +0.018]	no longer excludes 0
thyroid/medsiglip448_main	trapA / inpaint / clean / inpaint / erm	+0.015 [+0.003, +0.025]	now excludes 0
thyroid/convnext384_universal	trapA / mask / clean / mask / erm	+0.021 [−0.000, +0.041]	no longer excludes 0
thyroid/convnext384_universal	trapB / balanced / clean / balanced / erm	+0.039 [−0.002, +0.080]	no longer excludes 0
thyroid/convnext384_universal	trapB / dfr / clean / dfr / erm	+0.052 [−0.001, +0.103]	no longer excludes 0
ovary/dino518_main	trapB / dfr / clean / dfr / erm	+0.035 [−0.018, +0.089]	no longer excludes 0
ovary/dino518_main	trapB / leace_paired / clean / leace_paired / erm	+0.008 [−0.000, +0.015]	no longer excludes 0
capsule/dino518_main	trapB / leace_paired / test_rev / leace_paired / erm	+0.010 [+0.001, +0.024]	now excludes 0

3.6 What failed or changed

- **In-ROI masking harms only for dermoscopy hair**; for calipers and debris it helps less. The registered “harm” claims (T2, O2, C2) were not supported for thyroid, ovary and capsule.
- **DermLIP has no hair crossover**; “the crossover holds for every encoder” is not claimed.
- **Chest-radiograph device traps**: the registered in-ROI penalty was not supported (crossover in 1 of 4 findings); the device-matched follow-up does not change this. Their intervals and those of the drain runs could not be re-estimated with the crossed bootstrap (per-image predictions were not saved and RANZCR-CLiP is not on the current machine) and are shown as point estimates.
- **Oracle inpainting fails on capsule debris**, where the artifact resembles the disease.
- **The hair shortcut is mostly correlate-carried**, and the in-lesion harm is sensitive to how the hair-free group is defined.
- **Breast ultrasound was attempted twice and abandoned** (no artifact-free group).
- **The artifact masks were audited visually by a non-clinician**; no clinician has checked them yet.

3.7 Anticipated questions

Why not move the real artifact, as in the sweeps? A real artifact’s position is part of the image. The two-trap design compares images whose artifact happens to lie inside with images whose artifact lies outside; both are completed with the same artifact-free images, so the only designed difference is location. That the two groups of images may differ in other ways is a real limitation of this design.

Why $r \geq 0.5$ and $r < 0.1$, and what about the images in between? The thresholds come from the pilot and leave a gap so that the two traps are clearly separated; the in-between images are used by no trap.

Why does masking still help in Trap A for calipers and debris? Ultrasound and capsule frames carry other on-screen marks and clutter outside the ROI, and removing them helps. What masking cannot do is remove the caliper on the nodule; the masked Trap-A model’s correlated–reversed gap stays large.

Why is thyroid benign-heavy in Trap A? Calipers are drawn on nodules that are measured, and in this cohort measured nodules are more often benign; the environments fix the association at 0.9/0.1 whatever the natural rate.

Why are balancing and DFR not simply the answer? They need image-level artifact labels for every training image, which clinical datasets rarely have; masking needs only an ROI. They are the reference that shows what is achievable when labels exist.

Why keep DermLIP if it contradicts the law? It is a pre-registered replication, and its failure is informative: an encoder that needs the surroundings of a lesion loses disease signal with any mask, so the comparison between locations is dominated by that loss.

Why is chest radiography not counted as a confirmation? In the device traps the in-lung catheter is confounded with other devices, and in the drain trap the shortcut is carried by the treated lung. Neither isolates location; counting them either way would be unjustified.

Is Holm over four tests enough? The four p values are at the bootstrap floor (10^{-4}); any multiplicity correction over any plausible family leaves them significant.

How reliable are the capsule masks at IoU 0.625? Pixel IoU is modest, but the traps use thresholds on the *fraction* of the frame and of the lesion box covered; small boundary errors move few frames across 3% or 10%. Errors dilute effects rather than create them (Section on mask validity).

3.8 Conclusion

Supported. With real artifacts, masking is worth less when the artifact lies inside the ROI: the crossover is positive in all four cohorts with DINOv2 (Holm over the four location tests), with MedSigLIP in three cohorts and with a CNN in two. Label-based methods fix both traps.

Qualified. Only for dermoscopy hair does masking actually harm in the in-ROI trap, and that harm depends on how the hair-free group is defined; hair is a correlate-carried shortcut. **Not supported.** A crossover with DermLIP; an in-ROI penalty for real chest-radiograph devices; pixel removal for debris that resembles the disease.

3.9 Questions for the supervisor

1. Is a visual audit of the automatic artifact masks by a non-clinician acceptable for the thesis, or should a clinician check a sample before the real-artifact results are written up?
2. Should chest radiography stay in the main text as a boundary case with point estimates only, or move to an appendix?
3. How should the thesis present the correlate-carried nature of the hair shortcut: as a limitation of the hair cohort, or as a finding about real shortcuts in its own right?
4. Is it acceptable to count calipers and debris as confirmations when masking still helps (less) inside the ROI, i.e. to define the law by the crossover rather than by harm?

4 Stage 4 — Is it location, or which images carry the artifact?

In one sentence. The images whose artifact lies inside the ROI differ strongly from those whose artifact lies outside it (standardised differences up to 2.574), but the real-artifact crossover survives covariate matching and doubly robust regression adjustment in all three cohorts where matching is feasible (3 of 3); pasting the same real artifact inside or outside the ROI of the same images reproduces the location effect, although a neutral paste of the same shape reproduces most of it, so only a smaller part — significant for hair and thyroid calipers — is attributable to the artifact’s content; the crossover is also robust to the artifact-label definition, to stricter leakage groups and to a full rebuild of the data on a second machine.

4.1 Where this stage starts

Stage 3 showed a positive crossover with real artifacts in four cohorts and three encoder families. Its design compares two different sets of images: those whose artifact happens to lie inside the ROI (Trap A) and those whose artifact lies outside it (Trap B). Real artifacts are not placed at random. In dermoscopy, hair lies on large lesions more often than on small ones; in thyroid ultrasound, calipers inside the nodule are drawn on larger, more heavily measured nodules. Artifact-debiasing studies in dermoscopy make the same point (Bissoto et al., 2020; Nauta et al., 2022). The crossover could therefore reflect *which images* carry in-ROI artifacts rather than *where* the artifact lies. This stage separates the two, and then tests how fragile the result is to the choices that define it.

4.2 Questions

- Q4a** Do Trap-A and Trap-B images differ, and is the crossover explained by measured differences between them?
- Q4b** Does the location effect survive when the artifact is moved within the *same* image — and how much of that is due to the artifact rather than to pasting something into the image?
- Q4c** Is the crossover robust to how the artifact label is defined, to stricter leakage groups, and to rebuilding the whole pipeline?

Why three causal checks Matching removes measured imbalance but depends on the chosen covariates; regression on the matched sample (a doubly robust combination) protects against residual imbalance and against misspecification of either step alone; transplantation removes image-level confounding altogether but introduces its own artifact — the paste. The neutral paste (same mask shape, same compositing, filled with artifact-free tissue from the donor) measures that paste effect. *Rejected alternatives:* inverse-probability weighting (standardised differences above 2 in dermoscopy and capsule give extreme weights) and synthetic control images (they would repeat Stage 2, not test real artifacts).

4.3 Methods

Balance and matching (R1) Covariates fixed before any matching: dermoscopy — lesion area, hair amount, age, sex, body site, source, diagnosis within label; thyroid — nodule area, caliper pixels (log), native width and height (scanner proxy), mean grey level (gain proxy), nodule depth; ovary — tumour area, caliper pixels, image size, grey level, subtype within label; capsule — lesion-box area, contamination fraction, brightness, frame block (acquisition proxy). Standardised mean differences between the Trap-A and Trap-B artifact images, within label and pooled. Greedy 1:1 nearest-neighbour matching on the logit of a propensity model within label (exactly within source for dermoscopy), without replacement, calliper 0.2 SD; 0.05 SD as a stricter sensitivity analysis (R7). A cohort with fewer than 30 matched images per label is infeasible. The matched images replace the artifact cells of the traps; everything else is unchanged.

Regression adjustment (A2) For each reversed-test image of the matched sample, the *placement value* of the masked minus the ERM score (for a positive: the weighted share of negatives ranked below it; for a negative: the complement of the share of positives ranked above it), so that the class means reproduce the AUROC and the unadjusted difference between traps equals the crossover exactly (verified per seed). Within each seed, weighted least squares of this outcome on an intercept, the Trap-B indicator, the label and all matching covariates; the adjusted crossover is the Trap-B coefficient averaged over seeds. Crossed bootstrap with placement values and regression recomputed in every replicate (2,000 replicates); Holm over the three cohorts.

Transplant (R2) and neutral paste (R5) For each artifact-free recipient image, a real artifact instance is cut from a donor image of the same cohort (the donor’s mask pixels, largest connected component) and alpha-composited at a position fully inside ($\geq 95\%$ overlap) and at a position fully outside (0%) the recipient’s ROI, with a random dihedral transform; the positions are restricted to the field of view. The environments are then built as in the controlled sweeps with the pasted instance as the artifact; 5 presence seeds $\times$ 5 grouped folds. The neutral paste keeps the mask shape, transform, positions and compositing but fills the mask with tissue from the donor image at a window that does not touch its artifact.

Robustness (i) Artifact-label definitions: large detections only (≥ 50 caliper pixels) and strict location ($r \geq 0.7$ / $r < 0.05$) for thyroid and ovary; for capsule, a Trap B that also requires debris to cover less than 5% of the lesion box. (ii) Leakage: embedding-based near-duplicate groups (below). (iii) Rebuild: every dataset re-downloaded and every derived object — lesion U-Net, caliper detector, debris probe, leakage groups, feature caches — rebuilt on a second machine.

4.4 Experiments

Table 37: Experiments of this stage and their registration documents (docs/PREREGISTRATION_<name>.md or docs/<name>.md). Commit times are set by the committing machine and are not independent proof of order; the REVIEW2, REVIEW3 and FINAL registrations were also pushed to GitHub before their runs.

Experiment	Status	Registration	First commit (local time)	Support criterion	
Covariate balance (SMD) and matched traps, M1	supported registered	REVIEW2 (R1)	29cd0da, 18:56	2026-09-27	matched crossover CI > 0, Holm over 3 feasible cohorts
Stricter calliper 0.05 SD (R7)	supported registered	REVIEW3	85da735, 23:43	2026-09-27	sensitivity, descriptive
Real-artifact transplant, T1	supported registered	REVIEW2 (R2)	29cd0da, 18:56	2026-09-27	interaction CI > 0, Holm over 4
Neutral paste (paste-edge control), N3	supported registered	REVIEW3 (R5)	85da735, 23:43	2026-09-27	artifact – neutral interaction CI > 0
Regression-adjusted crossover, A2	supported registered	FINAL (A2)	c45d482, 00:35	2026-09-28	adjusted crossover CI > 0, Holm over 3
Artifact-label sensitivity	supported registered	SENSITIVITY_ARTIFACT_LABELS	c1bdfdf, 09:46	2026-09-27	crossover keeps sign and CI > 0 in every variant
Embedding-based leakage groups	supported registered	EMBEDDING_GROUPS	b0eda23, 13:55	2026-09-27	same sign and CI verdict as with pHash groups
Rebuild on a second machine, R0	supported registered	REVIEW2 (R0)	29cd0da, 18:56	2026-09-27	same sign, CI > 0, $ \Delta \leq 0.03$
Capsule matched analysis	not supported descriptive	REVIEW2 (R1)	29cd0da, 18:56	2026-09-27	infeasible under the registered rule

4.5 Results

4.5.1 The traps contain different images

The in-ROI and out-of-ROI artifact images differ strongly (Table 38): in dermoscopy the largest SMD is 2.127 (lesion area: hair lies inside large lesions), in capsule 2.574 (roi area), in thyroid 0.527 (caliper px log) and in ovary 0.389. Stage 3’s crossover is therefore not a comparison of like with like.

Table 38: Balance of pre-specified covariates between the in-ROI (Trap A) and out-of-ROI (Trap B) artifact images, before and after 1:1 nearest-neighbour matching within label (calliper 0.2 SD, pooled standardised mean differences). Registered rule: fewer than 30 matched images per label in either trap makes a cohort infeasible; target max $|\text{SMD}| \leq 0.1$ after matching.

Cohort	Least balanced covariate	max $ \text{SMD} $ before	after matching	min matched pairs per label	feasible (rule)
ISIC hair	lesion_area	2.127	0.108	398	yes
Thyroid	caliper_px_log	0.527	0.045	93	yes
Ovary	caliper_px_log	0.389	0.079	86	yes
Capsule	roi_area	2.574	0.231	19	no

Table 39: Per-covariate balance, thyroid and ovary (pooled over labels). The dermoscopy and capsule tables (17 and 20 covariates) are in `results/rerun_2026-09-28/review2/R1_balance.csv`.

Cohort	Covariate	SMD before	SMD after
Thyroid	roi_area	+0.395	<i>n/a</i>
Thyroid	caliper_px_log	+0.527	<i>n/a</i>
Thyroid	width	−0.143	<i>n/a</i>
Thyroid	height	−0.123	<i>n/a</i>
Thyroid	mean_grey	+0.183	<i>n/a</i>
Thyroid	roi_centroid_row	+0.060	<i>n/a</i>
Ovary	roi_area	+0.320	<i>n/a</i>
Ovary	caliper_px_log	+0.389	<i>n/a</i>
Ovary	width	−0.298	<i>n/a</i>
Ovary	height	+0.047	<i>n/a</i>
Ovary	mean_grey	+0.107	<i>n/a</i>
Ovary	subtype=0.0	+0.149	−0.079
Ovary	subtype=1.0	+0.059	+0.072
Ovary	subtype=2.0	−0.159	−0.012
Ovary	subtype=3.0	−0.054	+0.000
Ovary	subtype=4.0	−0.022	+0.024
Ovary	subtype=6.0	+0.024	+0.056
Ovary	subtype=7.0	−0.020	−0.062

4.5.2 Matching

Matching brings the maximum pooled SMD to 0.108 (dermoscopy), 0.045 (thyroid) and 0.079 (ovary); capsule matching is infeasible (19 pairs in the smaller label; SMD after matching 0.231). The crossover is essentially unchanged (Table 40): dermoscopy +0.154 [+0.123, +0.186], thyroid +0.238 [+0.186, +0.290], ovary +0.163 [+0.050, +0.273] (M1 supported in all three after Holm); with the stricter calliper it shrinks by 9–16% and survives: +0.124 [+0.086, +0.162], +0.218 [+0.165, +0.270], +0.152 [+0.041, +0.265]. Capsule, matched descriptively, gives +0.230 [+0.036, +0.434] — lower than unmatched (+0.377 [+0.327, +0.426]) and imprecise, so capsule’s real-trap crossover is reported as descriptive.

Table 40: Crossover before and after covariate matching (reversed-test AUROC, crossed 95% CIs). The 0.05 SD calliper is a stricter, registered sensitivity analysis (R7).

Cohort	Crossover, all images	matched (0.2 SD)	matched (0.05 SD)	M1 (Holm, 3)
ISIC hair	+0.147 [+0.111, +0.185]	+0.154 [+0.123, +0.186]	+0.124 [+0.086, +0.162]	supported
Thyroid	+0.239 [+0.196, +0.280]	+0.238 [+0.186, +0.290]	+0.218 [+0.165, +0.270]	supported
Ovary	+0.170 [+0.087, +0.252]	+0.163 [+0.050, +0.273]	+0.152 [+0.041, +0.265]	supported
Capsule	+0.377 [+0.327, +0.426]	+0.230 [+0.036, +0.434]	+0.328 [+0.087, +0.573]	descriptive (infeasible)

4.5.3 Regression adjustment on the matched sample (A2)

Within-label imbalance up to about 0.16 remained after matching. The doubly robust adjusted crossover equals the unadjusted one to the third decimal in every cohort (Table 41): dermoscopy +0.156 [+0.127, +0.187] (unadjusted +0.154 [+0.124, +0.186]), thyroid +0.238 [+0.184, +0.291] (+0.238 [+0.185, +0.291]), ovary +0.165 [+0.056, +0.273] (+0.163 [+0.058, +0.268]); Holm-adjusted $p = 0.002$, 0.002 and 0.002. All 3 registered tests are supported. Residual measured imbalance does not carry the effect. What this cannot exclude is an *unmeasured* difference.

Table 41: Regression-adjusted crossover on the matched traps (A2): per-image placement values of the masked minus the ERM score regressed on the Trap-B indicator, the label and all matching covariates, within seed; crossed bootstrap (2,000 replicates). Capsule is not analysed (matching infeasible; descriptive only).

Cohort	Test images per seed	Covariate columns	Post-matching max SMD pooled / per label	Unadjusted crossover	Adjusted (doubly robust)	p (Holm, 3)	A2
ISIC hair	2960	17	0.108 / 0.164	+0.154 [+0.124, +0.186]	+0.156 [+0.127, +0.187]	0.002	supported
Thyroid	2150	6	0.045 / 0.150	+0.238 [+0.185, +0.291]	+0.238 [+0.184, +0.291]	0.002	supported
Ovary	576	11	0.079 / 0.140	+0.163 [+0.058, +0.268]	+0.165 [+0.056, +0.273]	0.002	supported

Table 42: Post-matching balance table of the A2 sample (pooled over labels; test images of the reversed environment).

Cohort	Covariate	Trap A mean	Trap B mean	SMD
ISIC hair	lesion_area	0.297	0.292	+0.033
ISIC hair	hair_amount	0.018	0.015	+0.108
ISIC hair	age	57.119	57.332	−0.012
ISIC hair	age_missing	0.019	0.026	−0.048
ISIC hair	sex=female	0.504	0.521	−0.035
ISIC hair	sex=male	0.480	0.457	+0.046
ISIC hair	sex=missing	0.016	0.022	−0.040
ISIC hair	site=head/neck	0.174	0.179	−0.013
ISIC hair	site=lower extremity	0.219	0.213	+0.015
ISIC hair	site=missing	0.076	0.072	+0.015
ISIC hair	site=oral/genital	0.001	0.002	−0.015
ISIC hair	site=palms/soles	0.027	0.030	−0.020
ISIC hair	site=torso	0.383	0.387	−0.008
ISIC hair	site=upper extremity	0.120	0.118	+0.008
ISIC hair	benign_dx=BKL	0.092	0.100	−0.026
ISIC hair	benign_dx=MEL	0.218	0.218	+0.000
ISIC hair	benign_dx=NV	0.442	0.413	+0.060
ISIC hair	benign_dx=other	0.248	0.270	−0.050
ISIC hair	source=BCN	0.545	0.545	+0.000
ISIC hair	source=HAM	0.386	0.386	+0.000
ISIC hair	source=MSK	0.069	0.069	+0.000
Thyroid	roi_area	0.117	0.118	−0.008
Thyroid	caliper_px_log	5.495	5.448	+0.037
Thyroid	width	450.288	441.291	+0.045
Thyroid	height	373.576	371.744	+0.017
Thyroid	mean_grey	60.457	60.609	−0.006
Thyroid	roi_centroid_row	0.367	0.366	+0.007
Ovary	roi_area	0.141	0.142	−0.001
Ovary	caliper_px_log	6.518	6.481	+0.032
Ovary	width	864.648	856.939	+0.037
Ovary	height	550.073	549.944	+0.003
Ovary	mean_grey	57.307	56.935	+0.022
Ovary	subtype=0.0	0.218	0.251	−0.079
Ovary	subtype=1.0	0.201	0.173	+0.072
Ovary	subtype=2.0	0.296	0.302	−0.012
Ovary	subtype=3.0	0.089	0.089	+0.000
Ovary	subtype=4.0	0.061	0.056	+0.024
Ovary	subtype=6.0	0.106	0.089	+0.056
Ovary	subtype=7.0	0.028	0.039	−0.062

4.5.4 The same artifact inside and outside the ROI

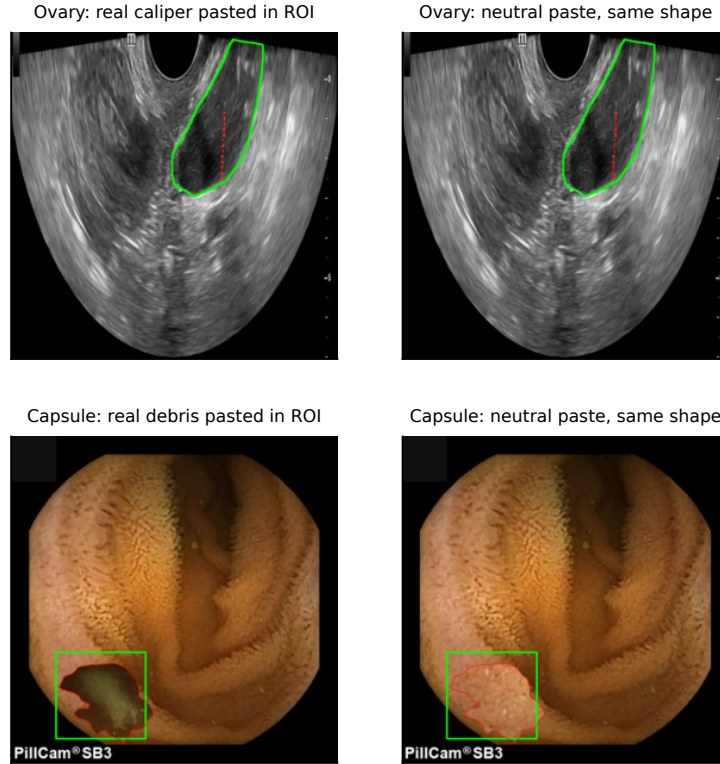


Figure 12: Transplant and neutral paste on the same recipient images (openly licensed cohorts; MMOTU, SEE-AI CC BY 4.0). Left: a real artifact instance pasted inside the ROI (green; the pasted region outlined in red). Right: the neutral paste — same mask shape and compositing, filled with artifact-free tissue from the donor.

Pasting the same real instance outside versus inside the ROI yields a positive location interaction in every cohort (Table 43; dermoscopy +0.253 [+0.214, +0.293], thyroid +0.277 [+0.256, +0.298], ovary +0.074 [+0.044, +0.104], capsule +0.497 [+0.427, +0.567]; Holm over four). Table 44 shows the mechanism of Stage 2 on real artifacts: outside the ROI masking removes the pasted artifact completely — the masked model’s correlated and reversed AUROC coincide (e.g. thyroid 0.846 and 0.846) and $|\Delta p| = 0$ — whereas inside it the artifact survives, and for thyroid, ovary and capsule the masked model reacts to it *more* than ERM (capsule 0.551 versus 0.238). Group balancing has a small location interaction of the opposite sign (e.g. -0.052 [$-0.071, -0.033$] for hair), much smaller than masking’s: it is nearly location-invariant.

Table 43: Real-artifact transplant (T1): the same real artifact instance, cut from a donor image, pasted outside and inside the ROI of the same artifact-free recipient images. These effects combine the artifact with the paste itself (seam, occluded tissue); Table 45 separates them.

Cohort	mask – ERM, pasted outside ROI	pasted inside ROI	Location interaction	Balanced: interaction
ISIC hair	+0.178 [+0.128, +0.227]	−0.075 [−0.114, −0.036]	+0.253 [+0.214, +0.293]	−0.052 [−0.071, −0.033]
Thyroid	+0.349 [+0.321, +0.379]	+0.072 [+0.043, +0.100]	+0.277 [+0.256, +0.298]	−0.016 [−0.025, −0.007]
Ovary	+0.064 [+0.001, +0.126]	−0.010 [−0.075, +0.056]	+0.074 [+0.044, +0.104]	−0.011 [−0.025, +0.002]
Capsule	+0.322 [+0.262, +0.384]	−0.175 [−0.241, −0.109]	+0.497 [+0.427, +0.567]	−0.032 [−0.068, +0.004]

Table 44: Transplant, per location: correlated and reversed AUROC of ERM and of the masked model, and the same-head counterfactual sensitivity to the pasted artifact (T3). Outside the ROI the masked model’s correlated and reversed AUROC coincide and $|\Delta p| = 0$: the pasted artifact is gone.

Cohort	Recipients	ERM in: corr / rev	mask in: corr / rev	mask out: corr / rev	$ \Delta p $ in: ERM / mask	$ \Delta p $ out: mask
ISIC hair	860	0.904 / 0.454	0.842 / 0.379	0.716 / 0.716	0.340 / 0.297	0.000
Thyroid	1800	0.851 / 0.487	0.906 / 0.559	0.846 / 0.846	0.206 / 0.271	0.000
Ovary	348	0.771 / 0.657	0.825 / 0.647	0.750 / 0.750	0.037 / 0.116	0.000
Capsule	340	0.906 / 0.508	0.956 / 0.333	0.914 / 0.914	0.238 / 0.551	0.000

How much of it is the artifact? The neutral paste reproduces 52% to 92% of the interaction (Table 45). Pasting *anything* into the ROI changes what masking does, because the environments make the pasted region label-correlated whatever it contains: the neutral paste becomes a shortcut itself (ERM learns it), and masking keeps it only when it lies inside the ROI. Only the difference between the two — N3, the interaction with the real artifact minus the interaction with the neutral paste — is attributable to the artifact’s content: dermoscopy +0.049 [+0.026, +0.072], thyroid +0.132 [+0.116, +0.148], ovary +0.022 [−0.002, +0.048] and capsule +0.037 [−0.017, +0.093] (2 of 4 excluding zero). By the registered decision rule (N3 significant *and* neutral share below one quarter) the transplant signal is not attributable to the artifact alone in any cohort.

What the transplant does and does not show It shows that on *fixed images*, where a label-correlated pattern sits decides what masking removes: the image-population explanation of Q4a is excluded. It does *not* give the harm of a real artifact in its natural context: the effect of masking with pasted hair inside the lesion combines the hair with the paste and is not attributed to real hair. For hair specifically, the transplant constructs a pixel-carried shortcut, whereas natural hair is largely correlate-carried (Stage 3).

Table 45: Paste-edge control (R5): the same mask shape and compositing, filled with the donor’s own artifact-free tissue. Only the N3 column — the interaction with the real artifact minus the interaction with the neutral paste — is attributable to the artifact’s content.

Cohort	mask − ERM, neutral outside	neutral inside	Neutral interaction	Artifact − neutral (N3)	Neutral / artifact interaction
ISIC hair	+0.129 [+0.079, +0.181]	−0.075 [−0.116, −0.035]	+0.204 [+0.164, +0.244]	+0.049 [+0.026, +0.072]	81%
Thyroid	+0.200 [+0.175, +0.226]	+0.055 [+0.028, +0.081]	+0.145 [+0.130, +0.161]	+0.132 [+0.116, +0.148]	52%
Ovary	+0.072 [+0.009, +0.135]	+0.019 [−0.041, +0.080]	+0.052 [+0.021, +0.084]	+0.022 [−0.002, +0.048]	71%
Capsule	+0.215 [+0.154, +0.280]	−0.244 [−0.314, −0.174]	+0.459 [+0.383, +0.535]	+0.037 [−0.017, +0.093]	92%

4.5.5 Robustness of the crossover

Artifact-label definition Table 46. With only large caliper detections, which are almost certainly real calipers, the crossover is +0.244 [+0.199, +0.289] (thyroid) and +0.188 [+0.108, +0.266] (ovary); with strict location thresholds +0.212 [+0.164, +0.259] and +0.196 [+0.117, +0.277]. It keeps its sign and significance in every variant and grows for the ovary, as expected if detector errors dilute rather than create the effect. For capsule, a Trap B that also excludes debris covering part of the lesion box gives +0.255 [+0.153, +0.349] — smaller than the main +0.377 [+0.327, +0.426] (the strict Trap B has far fewer benign images) but clearly positive; the expectation that the main definition was conservative was not confirmed.

Table 46: Sensitivity of the crossover to the definition of the artifact label (registered before running): only large detections, stricter location thresholds, and for capsule a Trap B that also requires debris to cover less than 5% of the lesion box.

Cohort	Artifact-label definition	Artifact images A / B	Crossover
Thyroid	main	1150 / 326	+0.239 [+0.196, +0.280]
Thyroid	large markers (≥ 50 px)	1083 / 274	+0.244 [+0.199, +0.289]
Thyroid	strict location ($r \geq 0.7$ / $r < 0.05$)	919 / 298	+0.212 [+0.164, +0.259]
Ovary	main	617 / 194	+0.170 [+0.087, +0.252]
Ovary	large markers (≥ 50 px)	609 / 183	+0.188 [+0.108, +0.266]
Ovary	strict location ($r \geq 0.7$ / $r < 0.05$)	571 / 186	+0.196 [+0.117, +0.277]
Capsule	main	816 / 1240	+0.377 [+0.327, +0.426]
Capsule	strict Trap B (lesion coverage $< 5\%$)	816 / 373	+0.255 [+0.153, +0.349]

Leakage without patient identifiers TN3K, MMOTU and SEE-AI publish no patient or video identifiers, so the main analysis groups near-duplicates by perceptual hash (and, for capsule, frame blocks). A stricter grouping joins images whose DINOv2 embeddings are near-identical (Table 47). Raw embedding cosine was dominated by the shared ultrasound appearance (median nearest-neighbour similarity 0.99 in the ovary), so embeddings were mean-centred per cohort — a change recorded before any model was refitted. The crossover is unchanged: thyroid +0.235 [+0.193, +0.275], ovary +0.157 [+0.053, +0.255], capsule +0.353 [+0.296, +0.410]. Embedding groups remain a proxy for patient identity: they remove visually near-identical images, not different-looking images of one patient.

Table 47: Stricter leakage groups for the cohorts without patient identifiers: images joined whenever their (mean-centred) DINOv2 embeddings have cosine $\geq \tau$, united with the existing groups; τ is the most aggressive grid value that keeps the largest group below 5% of the cohort (chosen without labels).

Cohort	Leakage groups	cosine τ	largest group	Crossover, pHash groups	Crossover, embedding groups
Thyroid	3,455 $\rightarrow$ 3,081	0.80	4.6%	+0.239 [+0.196, +0.280]	+0.235 [+0.193, +0.275]
Ovary	1,059 $\rightarrow$ 969	0.80	2.1%	+0.170 [+0.087, +0.252]	+0.157 [+0.053, +0.255]
Capsule	137 $\rightarrow$ 135	0.93	3.7%	+0.377 [+0.327, +0.426]	+0.353 [+0.296, +0.410]

Rebuild on a second machine Table 48. After every dataset was downloaded again and every derived object rebuilt on different hardware, each crossover reproduces within 0.01 (registered tolerance 0.03). The re-trained lesion U-Net changed hair-trap membership by about 3%, and the retrained debris probe moved a handful of frames across the 3%/10% thresholds; neither matters. All intervals in Stages 3–4 come from the rebuilt data.

Table 48: Reproduction of the four crossovers after every dataset was re-downloaded and every derived object (lesion U-Net, detectors, probe, groups, caches) rebuilt on a second machine (registered criterion: same sign, CI excluding 0, $|\Delta| \leq 0.03$).

Cohort	First build (point)	Rebuilt data, second machine	$ \Delta $
ISIC hair	+0.145	+0.147 [+0.111, +0.185]	0.003
Thyroid	+0.230	+0.239 [+0.196, +0.280]	0.009
Ovary	+0.170	+0.170 [+0.087, +0.252]	0.001
Capsule	+0.368	+0.377 [+0.327, +0.426]	0.008

4.5.6 Sensitivity to the interval estimator

On the analyses of this stage (Table 49, Figure 13) the crossed intervals are wider by a median factor of 1.63; 3 of 204 contrasts lose significance (Table 50), none of them a crossover, matched

crossover or transplant interaction of masking.

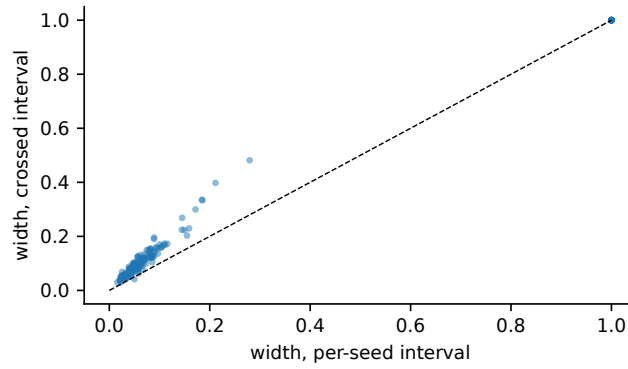


Figure 13: Width of each crossed interval against the per-seed interval of the same contrast (dashed: equal width).

Table 49: The analyses of this stage under the per-seed and the crossed estimator (same predictions; baseline arms only).

Result group	Intervals	Excluded with the estimator	0 only per-seed	Excluded with the estimator	0 only crossed	Median width ra- tio (crossed / per- seed)
capsule/dino518_emb_groups	9	0		0		2.02
capsule/dino518_matched	5	0		0		1.75
capsule/dino518_matched05	5	0		0		1.72
capsule/dino518_repro	5	0		0		1.50
capsule/dino518_sens_strictB	9	0		0		1.66
ovary/dino518_emb_groups	9	0		0		1.55
ovary/dino518_matched	5	0		0		1.51
ovary/dino518_matched05	5	0		0		1.52
ovary/dino518_repro	5	0		0		1.55
ovary/dino518_sens_px50	9	0		0		1.62
ovary/dino518_sens_strictloc	9	0		0		1.59
review2/transplant	60	1		0		1.63
spec_e13/dino518_matched	9	0		0		1.65
spec_e13/dino518_matched05	9	1		0		1.63
spec_e13/dino518_repro	9	0		0		1.71
thyroid/dino518_emb_groups	9	0		0		1.56
thyroid/dino518_matched	5	0		0		1.67
thyroid/dino518_matched05	5	0		0		1.69
thyroid/dino518_repro	5	0		0		1.76
thyroid/dino518_sens_px50	9	0		0		1.58
thyroid/dino518_sens_strictloc	9	1		0		1.63
all	204	3		0		1.63

Table 50: Contrasts of this stage whose verdict depends on the estimator.

Result	Contrast	Crossed estimate [95% CI]	Verdict vs per-seed estimator
thyroid/dino518_sens_strictloc	trapB / balanced / clean / balanced / erm	+0.043 [−0.008, +0.091]	no longer excludes 0
spec_e13/dino518_matched05	trapA / mask / clean / all / mask / erm	−0.026 [−0.053, +0.000]	no longer excludes 0
review2/transplant/ ovary_dino518	0.0 / balanced / erm / test_rev	+0.019 [−0.006, +0.046]	no longer excludes 0

4.6 What failed or changed

- **The transplant is mostly a paste effect.** The neutral paste reproduces 52–92% of the interaction; by the registered rule the transplant is not attributable to the artifact alone in any

cohort. No harm figure from the transplant is attributed to real artifacts.

- **Capsule cannot be matched** under the registered rule; its real-trap crossover is descriptive, and its location evidence rests on the transplant.
- **Stricter matching does not improve within-label balance** (up to 0.18–0.19) and shrinks the crossover by 9–16%.
- **The capsule strict-Trap-B variant** gives a smaller crossover, contrary to the expectation registered with it.

4.7 Anticipated questions

Why not simply adjust for lesion size in the original traps? Adjustment without matching extrapolates where the traps do not overlap (hair SMD above 2); matching first restricts the comparison to images that exist in both traps, and regression then removes what imbalance remains. Doing both is doubly robust: either step alone being correct is enough.

Why placement values? AUROC is not a sum over images, so it cannot be regressed directly. Placement values decompose it exactly into per-image contributions whose class means reproduce the AUROC, so a regression on them adjusts the crossover itself.

Why is capsule infeasible? In-ROI debris goes with large lesion boxes (SMD 2.6); among polyp-like lesions only 19 matched pairs exist, below the registered minimum of 30.

If a neutral paste gives most of the effect, is the transplant useless? No. It answers the question it was designed for — whether the location effect exists on fixed images — and the neutral paste answers a second question: the effect is about *any* label-correlated pattern inside the ROI, of which a real artifact is one example. What it cannot give is an estimate of harm for a particular real artifact.

Why does the neutral paste become a shortcut? Because the environments make its presence label-correlated, and a pasted patch leaves a visible seam and occludes the tissue beneath; the encoder can detect both.

Could unmeasured confounding still explain the real-trap crossover? In principle, yes; matching and regression only handle measured covariates. The transplant shows that location alone produces the effect on fixed images, which makes an unmeasured-confounder explanation unnecessary, but it cannot rule one out for the natural traps.

Why group by embeddings rather than ask for patient identifiers? The data providers do not publish them. Embedding groups are the strictest available proxy; they reduced thyroid groups from 3,455 to 3,081 without changing the result.

Why rebuild everything instead of re-running the models? Re-running on cached objects tests only the last step. Rebuilding tests the whole chain — downloads, masks, detectors, groups, features — and found two gaps in the reproduction recipe, which were fixed.

4.8 Conclusion

Supported. The real-artifact crossover is not explained by measured covariates: it survives matching (3 of 3 feasible cohorts) and doubly robust adjustment (3 of 3, Holm). Moving the same real

artifact inside the ROI of fixed images reproduces the location effect in all four cohorts, and for hair and thyroid calipers the artifact's content adds to the effect of the paste. The crossover is robust to the artifact-label definition, to stricter leakage groups and to a full rebuild. **Qualified.** Most of the transplant interaction is produced by the paste itself; unmeasured confounding of the natural location cannot be excluded; capsule is descriptive. **Not supported.** A harm figure for real artifacts from the transplant; the capsule matched analysis.

4.9 Questions for the supervisor

1. Is matching plus doubly robust adjustment plus the transplant sufficient causal evidence for the thesis, or should a negative-control analysis be attempted as well?
2. How should the paste effect be presented: as a limitation of the transplant, or as a finding in itself (any label-correlated pattern inside the ROI defeats masking)?
3. Should capsule endoscopy remain a primary cohort given that its real-trap crossover is descriptive?
4. Is the embedding-group analysis an acceptable answer to the absence of patient identifiers, or must the limitation be stated more strongly?

5 Stage 5 — Practice, breadth and theory

In one sentence. On unaltered test sets masking harms exactly the cases the location law says it should — shortcut-conflicting images with an in-ROI artifact — and on the patient-disjoint thyroid split it lowers sensitivity at every validation-fixed operating point; the crossover replicates in 17 of 18 frozen encoder-cohort pairs, at three model sizes, in 3 of 5 fine-tuned networks, in the zero-shot scores of vision-language models, and on new patients (ISIC 2020, +0.250 [+0.174, +0.327]); a two-parameter linear-Gaussian model predicts held-out reversed AUROC with mean error 0.052, less than half the error of the best heuristic, and explains why the law holds.

5.1 Where this stage starts

Stages 1–4 established the location law in constructed environments — a strong artifact-label association in training and its reversal at test — showed its mechanism (retention with amplified reliance), and showed that it is not explained by which images carry the artifact. A clinician will ask two questions that this cannot answer: does it matter on data as collected, without an imposed association, and at the operating point a clinic would use? A methodologist will ask two more: is the result a property of one encoder and one training regime, and can it be predicted rather than only observed? This stage answers all four.

5.2 Questions

- Q5a** On natural test sets, does masking harm the shortcut-conflicting cases, and does that change clinical sensitivity at fixed operating points?
- Q5b** How broadly does the crossover replicate: encoder families and sizes, end-to-end fine-tuning, untrained (zero-shot) models, new patients?
- Q5c** Can a simple model predict a model’s reversed-test AUROC from what is observable without the reversed test — and so explain when masking helps?

Design decisions *Hard pairs.* On a natural test set the artifact-label association is weak, so the overall AUROC barely moves. Positive-negative pairs are therefore split by whether their artifact status conflicts with the test set’s own association (hard) or follows it (easy); the law predicts harm on hard pairs, where the retained in-ROI artifact points the wrong way. *Operating points fixed on validation data* (never on test) at five clinically motivated targets, with thresholds re-estimated in every bootstrap replicate. *A theory judged by held-out error, not by fit:* the model is fitted to each cell’s clean and correlated AUROC and must predict the reversed AUROC it never saw, against two parameter-free heuristics. Rejected: judging the theory by agreement with simulations drawn from its own assumptions (circular), and by the correlation of predicted and observed crossovers (a heuristic reaches almost the same correlation).

5.3 Methods

Natural test sets Models are trained on the natural training distribution (no imposed association). Thyroid: the official patient-disjoint TNCD split (Gong et al., 2022) (614 test images; in-ROI calipers on 37.6% of benign and 24.1% of malignant training images). Dermoscopy: each source (BCN, HAM, MSK) held out in turn (in-lesion hair on 31% of melanomas and 20% of benign lesions). Capsule: held-out frame-block groups. 5 seeds of a group-safe train/validation split. A = in-ROI artifact. Arms ERM, masking and group balancing.

Clinical metrics and operating points AUROC, AUPRC, Brier score, expected calibration error; sensitivity and specificity at OP1–OP5, overall and among shortcut-conflicting positives (for thyroid: malignant nodules with a caliper on the nodule).

Replication Frozen DINOv2 ViT-S/B/L (21/86/304 M parameters), MedSigLIP-448, ConvNeXt-384 and DermLIP with linear heads; ResNet-50 and ViT-S/16 fine-tuned end to end (ImageNet initialisation, 224 px, 8 epochs, AdamW, one-cycle learning rate 10^{-4} or $3 \cdot 10^{-5}$, head $\times 10$, thresholds on clean validation; the five folds of seed 42 are the bootstrap clusters, 15 clusters for the ovary); zero-shot vision–language scores (cosine to a positive minus a negative class prompt) of MedSigLIP and DermLIP.

External validation ISIC 2020 (Rotemberg et al., 2021) was run only if a gate registered before any external label was read held: direct download without an agreement, a research licence, patient identifiers, masks producible by frozen models, and at least 50 images in every cell. Lesion masks: the existing U-Net (ISIC 2018 Task 1 only). Hair masks: a segmenter trained only on the ISIC 2019 hair masks and frozen before any ISIC 2020 image was scored. Protocol of the ISIC 2019 traps unchanged; groups = patient identifier.

Theory A frozen encoder maps an image to a disease block with signal-to-noise ratio S , an artifact block with signal-to-noise ratio A , and noise. Training uses $P(A=1 | Y=1) = p_1 = 0.9$ and $P(A=1 | Y=0) = p_0 = 0.1$. An LDA-optimal linear head (which logistic regression approximates) weights each block by its training signal; the artifact’s effective training signal is

$$\tilde{A} = \frac{(p_1 - p_0)^2 A}{1 + p_1(1 - p_1)A}, \quad \text{AUROC}_{\text{rev}} = \Phi \left(\frac{S - \tilde{A}}{\sqrt{2(S + \tilde{A})}} \right),$$

which rises with S and falls with $\tilde{A}$; for any test composition $q_y = P(A=1 | Y=y)$ the same model gives the AUROC in closed form (`docs/THEORY.md`). Its consequences: (1) out-of-ROI masking sets $\tilde{A} \rightarrow 0$ and helps unless it destroys most of the disease signal; (2) in-ROI masking keeps $\tilde{A}$ and changes S to S_m , so it *harms iff it removes disease-informative context* ($S_m < S$); (3) for the same S_m the crossover is positive for every $\tilde{A} > 0$; (4) group balancing sets the effective association to zero for the artifact and everything correlated with it, independently of location; (5) a shortcut carried by correlates that the clean and correlated environments do not reveal is invisible to the model. *Test*: for every cell (result $\times$ trap or overlap $\times$ arm) (S, A) is fitted to the observed clean and correlated AUROC and the reversed AUROC is predicted; heuristics “reversed = clean” and “symmetric reversal” (reversed = clean – (correlated – clean)).

5.4 Experiments

Table 51: Experiments of this stage and their registration documents (docs/PREREGISTRATION_<name>.md). Commit times are set by the committing machine and are not independent proof of order.

Experiment	Status	Registration	First commit (local time)	Support criterion
Zero-shot vision-language scores, Z1	partly supported registered (descriptive); crossover test post hoc	TEXT_PROMPT	7877e8c, 12:21	2026-09-27 correlated — reversed gap reported
Natural test sets, N3 (hard pairs)	supported registered	NATURAL	74bd243, 09:39	2026-09-27 mask — ERM < 0 on hard pairs
Operating points S1–S3	supported registered	REVIEW2 (R3)	29cd0da, 18:56	2026-09-27 “masking lowers sensitivity” only if S1 and ≥ 3 of OP2–OP5
Thyroid subgroup, n=78 (S3)	partly supported registered test in a post hoc subgroup	REVIEW2 (R3)	29cd0da, 18:56	2026-09-27 sensitivity loss in the subgroup, CI < 0
Crossed CIs for the subgroup (R6)	supported registered	REVIEW3 (R6)	85da735, 23:43	2026-09-27 report the crossed CI
Fine-tuned hair model and literature comparison (R4)	supported registered	REVIEW2 (R4)	29cd0da, 18:56	2026-09-27 crossover CI > 0; comparison descriptive
Encoder scale (S/B/L)	supported registered	SCALE	13303b0, 14:13	2026-09-27 crossover CI > 0 at every scale
Fine-tuned networks	supported registered	FINETUNE	7c8b466, 07:06	2026-09-27 crossover CI > 0 (secondary)
DermLIP hair traps	supported registered	ISIC2019_TRAPS	76e25d4, 13:14	2026-09-26 crossover CI > 0
Theory as held-out predictor, T1–T4	supported registered	THEORY_PREDICTION	4f78653, 13:46	2026-09-27 MAE below both heuristics; crossover signs
External validation, X1 (ISIC 2020)	supported registered	FINAL (A4)	c45d482, 00:35	2026-09-28 gate, then crossover CI > 0

5.5 Results

5.5.1 Natural test sets

Table 52. On all pairs, masking changes little. On shortcut-conflicting pairs it harms in thyroid (−0.095 [−0.165, −0.022]), BCN (−0.025 [−0.043, −0.008]) and MSK (−0.065 [−0.111, −0.019]), and helps in HAM (+0.050 [+0.030, +0.070]); in capsule the interval includes zero (+0.021 [−0.003, +0.048]). Where masking harms the hard pairs, it helps or is neutral on the easy ones (MSK +0.100 [+0.056, +0.147]) — the signature of a model that still uses an in-ROI artifact after masking: it gets the aligned cases right for the wrong reason and the conflicting cases wrong. The registered claim N3 is supported in three of five natural test sets. Group balancing improves on masking for the hard pairs of thyroid (+0.136 [+0.063, +0.207]), BCN (+0.043 [+0.026, +0.059]) and MSK (+0.100 [+0.055, +0.144]), but not on all pairs and not in capsule (−0.086 [−0.110, −0.065] on all pairs).

Table 52: Natural test sets: AUROC change from masking on all positive–negative pairs and on the pairs whose artifact status conflicts with (“hard”) or follows (“easy”) the test set’s own artifact–label association (DINOv2, crossed CIs).

Test set (no imposed correlation)	$P(A=1 Y=1) / P(A=1 Y=0)$	mask — ERM, all pairs	shortcut-conflicting pairs	shortcut-aligned pairs
Thyroid (official patient-disjoint split)	0.331 / 0.434	−0.004 [−0.048, +0.041]	−0.095 [−0.165, −0.022]	+0.042 [−0.011, +0.097]
Dermoscopy, BCN held out	0.356 / 0.239	−0.025 [−0.035, −0.016]	−0.025 [−0.043, −0.008]	−0.022 [−0.035, −0.008]
Dermoscopy, HAM held out	0.274 / 0.184	+0.031 [+0.017, +0.045]	+0.050 [+0.030, +0.070]	+0.004 [−0.019, +0.027]
Dermoscopy, MSK held out	0.109 / 0.074	+0.016 [−0.007, +0.038]	−0.065 [−0.111, −0.019]	+0.100 [+0.056, +0.147]
Capsule (held-out frames)	0.101 / 0.232	+0.077 [+0.059, +0.097]	+0.021 [−0.003, +0.048]	+0.071 [+0.040, +0.108]

5.5.2 Clinical consequences: thyroid operating points

On the official, patient-disjoint thyroid test split the AUROC of ERM and of the masked model is almost the same (0.735 and 0.731), but masking lowers sensitivity at every operating point (Table 53, Figure 14): at OP1 by -0.275 $[-0.435, -0.138]$ (from 0.699 to 0.424) and at OP2–OP5 with intervals below zero in 4 of 4. By the decision rule registered before the operating points were computed (S1 and a loss at ≥ 3 of OP2–OP5), the thesis may say that masking lowers sensitivity on this test set. Specificity rises: masking moves the operating point, and the loss falls on the malignant nodules whose caliper lies on the nodule.

The subgroup of 78 malignant nodules **Registration status: a pre-registered test inside a post hoc subgroup — exploratory confirmation.** The subgroup (malignant nodules with an in-ROI caliper) was defined after the natural-distribution AUROC results were known; the operating-point test in it was registered before the operating points were computed. The same 78 nodules enter every seed. At OP2, sensitivity in this subgroup falls from 0.549 to 0.177, a change of -0.372 $[-0.499, -0.236]$. The effect is large and consistent across operating points; its confirmatory weight is limited by the post hoc subgroup definition.

Table 53: Thyroid, official patient-disjoint test split: effect of masking at five validation-fixed operating points (crossed bootstrap with thresholds re-estimated in every replicate). The subgroup is the malignant nodules with an in-ROI caliper (shortcut-conflicting).

Operating point (fixed on validation)	Sensitivity ERM → mask	Δ sensitivity	Δ specificity	Sensitivity, conflicting malignant (ERM → mask)	Δ in that subgroup
OP1 max. balanced accuracy	0.699 → 0.424	-0.275 $[-0.435, -0.138]$	$+0.202$ $[+0.115, +0.330]$	0.703 → 0.223	-0.479 $[-0.640, -0.313]$
OP2 specificity ≥ 0.80	0.557 → 0.355	-0.202 $[-0.300, -0.100]$	$+0.117$ $[+0.059, +0.172]$	0.549 → 0.177	-0.372 $[-0.499, -0.236]$
OP3 specificity ≥ 0.90	0.371 → 0.219	-0.152 $[-0.253, -0.043]$	$+0.075$ $[+0.029, +0.122]$	0.336 → 0.100	-0.236 $[-0.383, -0.089]$
OP4 sensitivity ≥ 0.80	0.753 → 0.380	-0.373 $[-0.469, -0.270]$	$+0.278$ $[+0.203, +0.359]$	0.744 → 0.197	-0.546 $[-0.671, -0.421]$
OP5 sensitivity ≥ 0.90	0.869 → 0.588	-0.281 $[-0.391, -0.178]$	$+0.339$ $[+0.236, +0.425]$	0.862 → 0.400	-0.462 $[-0.614, -0.310]$

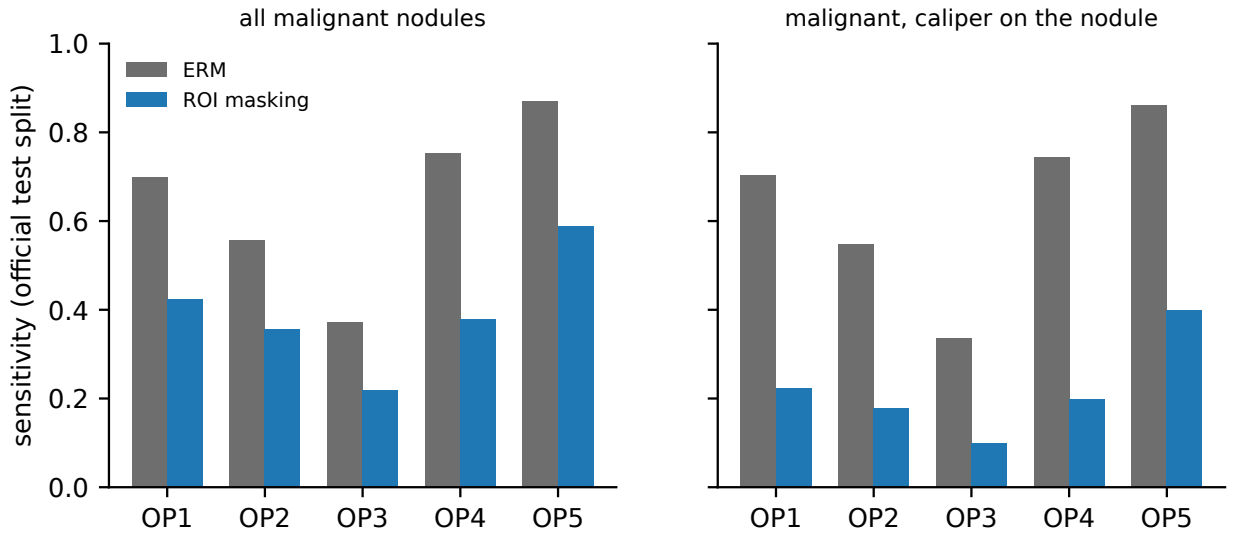


Figure 14: Thyroid, official patient-disjoint test split: sensitivity of ERM and of the masked model at five validation-fixed operating points, for all malignant nodules (left) and for the 78 malignant nodules with a caliper on the nodule (right; post hoc subgroup, exploratory).

Other test sets Table 54. At OP2 masking lowers sensitivity, including for the shortcut-conflicting positives, in the HAM and MSK hold-outs, where it raises specificity; in BCN it lowers specificity; in capsule it raises sensitivity without a significant change for the conflicting erosions. The thyroid pattern — a threshold that no longer catches the positives whose supporting artifact is gone — recurs in two of the four other test sets. Table 55 gives the threshold-free and calibration

metrics: in thyroid and BCN masking worsens Brier score and calibration; in HAM, MSK and capsule it improves every metric, matching the AUROC results.

Table 54: Other natural test sets at OP2 (validation specificity ≥ 0.80), mask – ERM, crossed CIs.

Test set	Conflicting positives	Δ sensitivity	Δ specificity	Δ sensitivity, conflicting
Dermoscopy BCN	1832	+0.017 [−0.003, +0.036]	−0.110 [−0.135, −0.079]	+0.019 [−0.004, +0.042]
Dermoscopy HAM	808	−0.062 [−0.110, −0.015]	+0.092 [+0.064, +0.121]	−0.055 [−0.104, −0.007]
Dermoscopy MSK	492	−0.074 [−0.127, −0.026]	+0.083 [+0.064, +0.110]	−0.094 [−0.149, −0.044]
Capsule	38	+0.140 [+0.082, +0.222]	+0.041 [−0.029, +0.107]	+0.022 [−0.041, +0.123]

Table 55: Clinical metrics on the natural test sets (seed means; threshold at maximal balanced accuracy on validation data).

Test set	Arm	AUROC	AUPRC	Brier	ECE	Sensitivity	Specificity
Capsule (held-out frames)	ERM	0.889	0.897	0.130	0.042	0.770	0.844
	mask	0.967	0.971	0.069	0.029	0.856	0.933
	balanced	0.881	0.889	0.136	0.051	0.757	0.838
ISIC, BCN held out	ERM	0.786	0.587	0.238	0.279	0.842	0.521
	mask	0.761	0.564	0.287	0.330	0.842	0.443
	balanced	0.782	0.587	0.229	0.260	0.835	0.509
ISIC, HAM held out	ERM	0.752	0.355	0.161	0.233	0.575	0.789
	mask	0.782	0.396	0.138	0.199	0.569	0.831
	balanced	0.751	0.350	0.154	0.225	0.495	0.845
ISIC, MSK held out	ERM	0.770	0.477	0.171	0.160	0.600	0.790
	mask	0.785	0.497	0.149	0.120	0.587	0.823
	balanced	0.767	0.464	0.175	0.165	0.628	0.759
Thyroid (official split)	ERM	0.735	0.604	0.209	0.088	0.699	0.653
	mask	0.731	0.629	0.215	0.119	0.424	0.856
	balanced	0.714	0.580	0.220	0.115	0.707	0.612

5.5.3 Replication breadth

Table 59 and Figure 15. The crossover’s interval lies above zero for 17 of 18 frozen encoder–cohort pairs; the exception is DermLIP (Stage 3).

Model scale Table 56. The law holds at every size in every cohort. Larger models do not escape the shortcut; they exploit an in-ROI caliper *more*: ERM’s correlated–reversed gap in the thyroid Trap A grows from 0.537 (ViT-S) to 0.590 (ViT-B) and 0.622 (ViT-L), and in the ovary from 0.320 to 0.564; capsule is flat.

Table 56: Encoder scale (DINOv2 ViT-S/14 21 M, ViT-B/14 86 M, ViT-L/14 304 M parameters): ERM’s shortcut gap in the in-ROI trap and the crossover.

Cohort	DINOv2	ERM clean AUROC (Trap A)	ERM gap corr – rev (Trap A)	Crossover
Thyroid	ViT-S/14	0.629	0.537	+0.247 [+0.205, +0.288]
Thyroid	ViT-B/14	0.608	0.590	+0.239 [+0.196, +0.280]
Thyroid	ViT-L/14	0.601	0.622	+0.263 [+0.218, +0.307]
Ovary	ViT-S/14	0.698	0.320	+0.296 [+0.208, +0.384]
Ovary	ViT-B/14	0.635	0.413	+0.170 [+0.087, +0.252]
Ovary	ViT-L/14	0.608	0.564	+0.207 [+0.127, +0.286]
Capsule	ViT-S/14	0.803	0.363	+0.370 [+0.325, +0.415]
Capsule	ViT-B/14	0.796	0.360	+0.377 [+0.327, +0.426]
Capsule	ViT-L/14	0.820	0.347	+0.417 [+0.369, +0.464]

End-to-end fine-tuning Table 57. The crossover replicates in 3 of 5 fine-tuned networks: dermoscopy hair with ResNet-50 (+0.143 [+0.054, +0.232]), thyroid with ViT-S and capsule with ResNet-50, where masking harms strongly inside the ROI (−0.279 [−0.377, −0.185]) and helps

outside (+0.327 [+0.235, +0.416]). The thyroid ResNet-50 points the same way without significance. The fine-tuned ovary network is *not evaluable*: with 15 clusters its ERM clean AUROC is 0.60 (Trap A) and 0.50 (Trap B) — it barely learns the task — so its null says nothing about the law. Fine-tuning does not remove the effect; it is not an artefact of frozen features.

Table 57: End-to-end fine-tuned networks (ImageNet initialisation, 224 px, 8 epochs; folds as bootstrap clusters): effect of masking per trap (crossed CIs).

Fine-tuned network	mask – ERM, Trap A	mask – ERM, Trap B	mask+balanced – balanced, Trap A
Dermoscopy hair, ResNet-50	−0.002 [−0.054, +0.049]	+0.141 [+0.071, +0.214]	−0.008 [−0.072, +0.059]
Thyroid, ResNet-50	+0.078 [+0.032, +0.123]	+0.142 [+0.073, +0.211]	+0.085 [+0.035, +0.137]
Thyroid, ViT-S/16	+0.122 [+0.068, +0.170]	+0.244 [+0.163, +0.329]	+0.059 [+0.003, +0.122]
Capsule, ResNet-50	−0.279 [−0.377, −0.185]	+0.327 [+0.235, +0.416]	−0.020 [−0.107, +0.074]
Ovary, ResNet-50 (15 clusters)	+0.152 [+0.070, +0.232]	+0.211 [+0.090, +0.328]	<i>n/a</i>

Untrained models Table 58. Zero-shot scores of vision–language models are already swayed by artifacts (capsule correlated–reversed gap −0.402), and masking changes them according to the law: crossover thyroid +0.103 [+0.064, +0.143], ovary +0.181 [+0.103, +0.261], capsule +0.045 [+0.001, +0.090] (borderline), dermoscopy (DermLIP) +0.040 [+0.021, +0.058] — 4 of 4 positive. The effect is therefore a property of the representation, not of the trained head. Caveat: MedSigLIP’s zero-shot thyroid score is at chance (clean AUROC 0.451), so that row shows artifact effects on an uninformative score.

Table 58: Zero-shot vision–language scores (no training: cosine to a positive minus a negative class prompt) on the trap test environments, unmasked and masked (crossed CIs). The gap shows the untrained score is already swayed by the artifact; the crossover shows that masking changes it according to the location law.

Cohort	Model	Zero-shot clean AUROC (Trap A)	Correlated – reversed gap	mask effect, Trap A	mask effect, Trap B	Crossover
Thyroid	MedSigLIP	0.451	+0.072	+0.017 [−0.011, +0.045]	+0.120 [+0.078, +0.162]	+0.103 [+0.064, +0.143]
Ovary	MedSigLIP	0.623	−0.126	−0.235 [−0.289, −0.180]	−0.054 [−0.135, +0.030]	+0.181 [+0.103, +0.261]
Capsule	MedSigLIP	0.631	−0.402	+0.004 [−0.021, +0.029]	+0.049 [+0.001, +0.098]	+0.045 [+0.001, +0.090]
Dermoscopy hair	DermLIP	0.697	+0.082	−0.079 [−0.114, −0.044]	−0.039 [−0.070, −0.008]	+0.040 [+0.021, +0.058]

New patients ISIC 2020 met every condition of the registered gate: 32,997 images of 2,056 patients, patient-disjoint folds; smallest cell 53 hair-free melanomas (gate 50). The frozen hair segmenter reaches a hair-free balanced accuracy of 0.93 and pixel IoU 0.60 on held-out ISIC 2019 images. Masking harms with hair on the lesion (−0.064 [−0.128, −0.002]) and helps with hair beside it (+0.186 [+0.118, +0.249]); the crossover is +0.250 [+0.174, +0.327] (registered claim X1, supported). Thyroid cohorts with patient identifiers either lack nodule masks (ThyUS2Path (Hou et al., 2024), whose scanner interface also triggers the caliper detector) or require a data-use agreement (TN-SCUI 2020, Stanford AIMI), and were not used; DDTI (134 images) is too small.

Table 59: Replication breadth of the real-artifact crossover (crossed CIs).

Cohort	Encoder	Training	Crossover [95% CI]
Dermoscopy hair	DINOv2 ViT-S/14	frozen	+0.163 [+0.130, +0.195]
Dermoscopy hair	DINOv2 ViT-B/14	frozen	+0.147 [+0.111, +0.185]
Dermoscopy hair	DINOv2 ViT-L/14	frozen	+0.153 [+0.110, +0.195]
Dermoscopy hair	DermLIP	frozen	+0.014 [−0.037, +0.063]
Thyroid	DINOv2 ViT-S/14	frozen	+0.247 [+0.205, +0.288]
Thyroid	DINOv2 ViT-B/14	frozen	+0.239 [+0.196, +0.280]
Thyroid	DINOv2 ViT-L/14	frozen	+0.263 [+0.218, +0.307]
Thyroid	MedSigLIP-448	frozen	+0.195 [+0.153, +0.238]
Thyroid	ConvNeXt-384	frozen	+0.275 [+0.231, +0.320]
Ovary	DINOv2 ViT-S/14	frozen	+0.296 [+0.208, +0.384]
Ovary	DINOv2 ViT-B/14	frozen	+0.170 [+0.087, +0.252]
Ovary	DINOv2 ViT-L/14	frozen	+0.207 [+0.127, +0.286]
Ovary	MedSigLIP-448	frozen	+0.178 [+0.109, +0.248]
Capsule	DINOv2 ViT-S/14	frozen	+0.370 [+0.325, +0.415]
Capsule	DINOv2 ViT-B/14	frozen	+0.377 [+0.327, +0.426]
Capsule	DINOv2 ViT-L/14	frozen	+0.417 [+0.369, +0.464]
Capsule	MedSigLIP-448	frozen	+0.335 [+0.286, +0.383]
Capsule	ConvNeXt-384	frozen	+0.320 [+0.268, +0.372]
Dermoscopy hair	ResNet-50	fine-tuned	+0.143 [+0.054, +0.232]
Thyroid	ResNet-50	fine-tuned	+0.063 [−0.022, +0.144]
Thyroid	ViT-S	fine-tuned	+0.122 [+0.039, +0.211]
Capsule	ResNet-50	fine-tuned	+0.606 [+0.495, +0.716]
Ovary	ResNet-50 (15 clusters)	fine-tuned	+0.058 [−0.062, +0.179]
Dermoscopy hair, ISIC 2020 (external)	DINOv2 ViT-B/14	frozen, new patients	+0.250 [+0.174, +0.327]

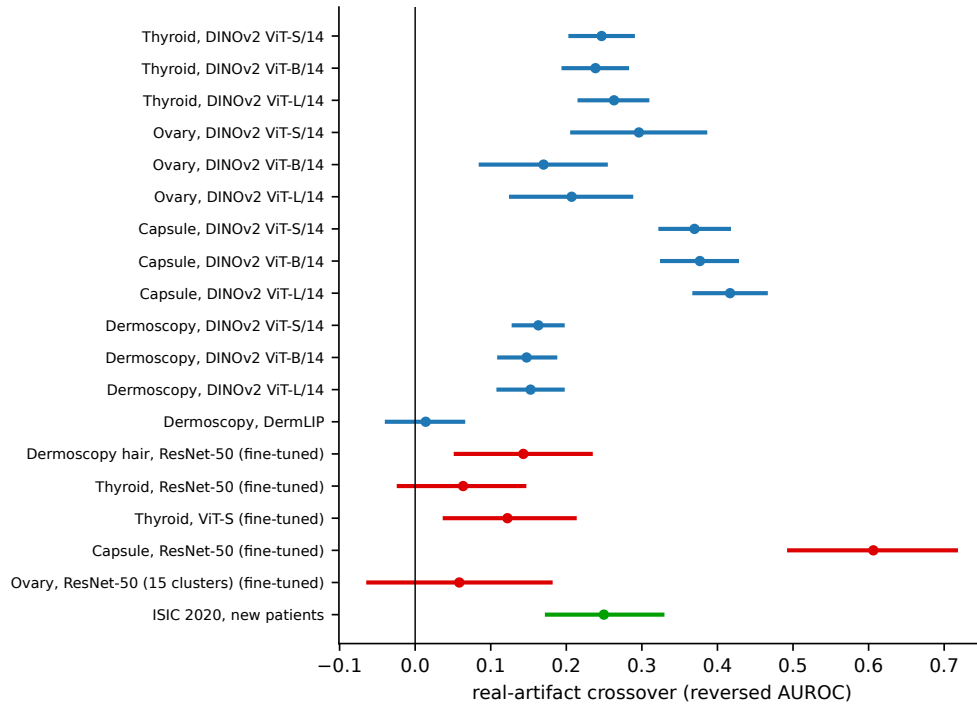


Figure 15: Crossover across encoders and scales (blue), fine-tuned networks (red) and new patients (green); crossed 95% CIs.

Absolute performance Table 60. On the one cohort with a published benchmark on the same patient-disjoint split, the frozen DINOv2 linear probe reaches an AUROC of 0.735, below published fine-tuned CNNs (0.77–0.80). The study is about the *relative* effect of masking on a fixed representation, not about state-of-the-art accuracy, but the gap is a limitation.

Table 60: Absolute performance: the frozen encoder with a linear head against published fine-tuned CNNs on the same patient-disjoint split.

Model (thyroid, official TNCD test split)	AUROC	Source
ResNet-18, cross-entropy (fine-tuned)	0.773	Gong et al. (2022)
DenseNet-121, cross-entropy (fine-tuned)	0.780	Gong et al. (2022)
ConvNeXt-T, cross-entropy (fine-tuned)	0.784	Gong et al. (2022)
ResNet-18, adaptive curriculum (fine-tuned)	0.799	Gong et al. (2022)
DINOv2 ViT-B/14 frozen + linear head, ERM (rebuilt run)	0.735	this work

5.5.4 The DermLIP exception

With DermLIP, masking harms at both hair locations (Trap A -0.104 $[-0.143, -0.067]$, Trap B -0.090 $[-0.135, -0.046]$). Two measurements explain it. First, masking costs DermLIP disease signal wherever the hair lies: clean-test AUROC changes by -0.066 $[-0.089, -0.043]$ and -0.063 $[-0.094, -0.033]$. Second, the pixel part of its hair shortcut is small at both locations (oracle hair inpainting $+0.033$ $[+0.025, +0.042]$ and $+0.040$ $[+0.015, +0.064]$), whereas group balancing, which also removes correlates of hair, gains $+0.146$ $[+0.122, +0.170]$ and $+0.110$ $[+0.089, +0.132]$: DermLIP carries the hair shortcut mostly through correlates that masking does not remove at either location. In the terms of the theory, masking lowers S at both locations and removes little of $\tilde{A}$ at either, so there is no gap to find. The theory anticipates this from the clean and correlated AUROC alone: predicted crossover $+0.013$, observed $+0.014$.

5.5.5 The theory, judged by held-out error

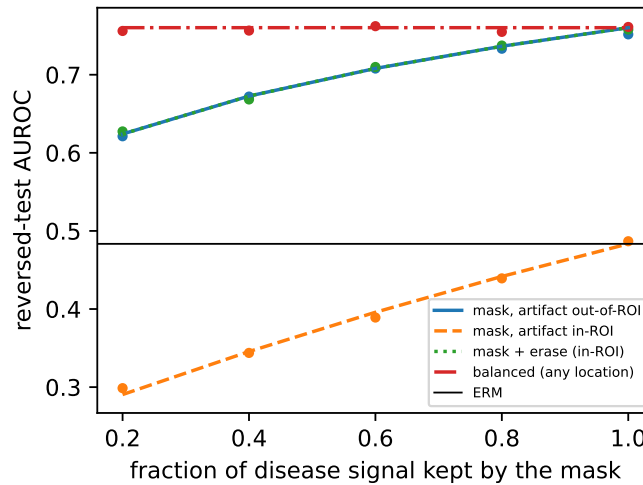


Figure 16: The closed form (lines) against Monte Carlo simulation of its own model (points): the formula is exact up to simulation noise (at most 0.009 AUROC over 29 settings). This checks the algebra, not the theory.

Table 61 and Figure 17. Over 136 primary cells the theory predicts the reversed AUROC — never used in fitting — with mean absolute error 0.052 (Pearson r 0.914; 66% within 0.05), against 0.129 for the symmetric-reversal heuristic and 0.237 for “reversed = clean”. It gets the sign of all 9 real-trap crossovers right (MAE 0.015) and of 3/3 fine-tuned crossovers — networks that violate its assumptions — and the sign of 44 of 50 controlled-sweep gains (heuristic: 38). From the fitted S alone, the sign of the in-ROI masking effect is right in 8 of 9 cells. Its clear failure is the chest-drain cohort of Stage 3: the clean and correlated environments do not reveal a shortcut carried by the treated lung, so the model cannot see it — the boundary its assumption (5) predicts.

The account is simple: a model’s reliance on an artifact is fixed by the ratio of artifact to disease signal it sees; masking lowers the disease signal (less context) and, when the artifact lies inside the ROI, leaves the artifact’s signal intact, so it raises that ratio exactly when the artifact is inside.

Table 61: Held-out prediction of the reversed-test AUROC (never used in fitting): the two-parameter theory fitted to each cell’s clean and correlated AUROC, against two parameter-free heuristics.

Cells	Predictor	n	MAE	Pearson r	within 0.05
Primary cells (ERM, mask)	linear-Gaussian theory	136	0.052	0.914	66%
	heuristic: symmetric reversal	136	0.129	0.743	40%
	heuristic: reversed = clean	136	0.237	0.550	18%
All ERM-head arms	linear-Gaussian theory	598	0.036	0.933	79%
	heuristic: symmetric reversal	598	0.076	0.823	65%
	heuristic: reversed = clean	598	0.151	0.584	40%
Fine-tuned (outside assumptions)	linear-Gaussian theory	12	0.027	0.980	92%

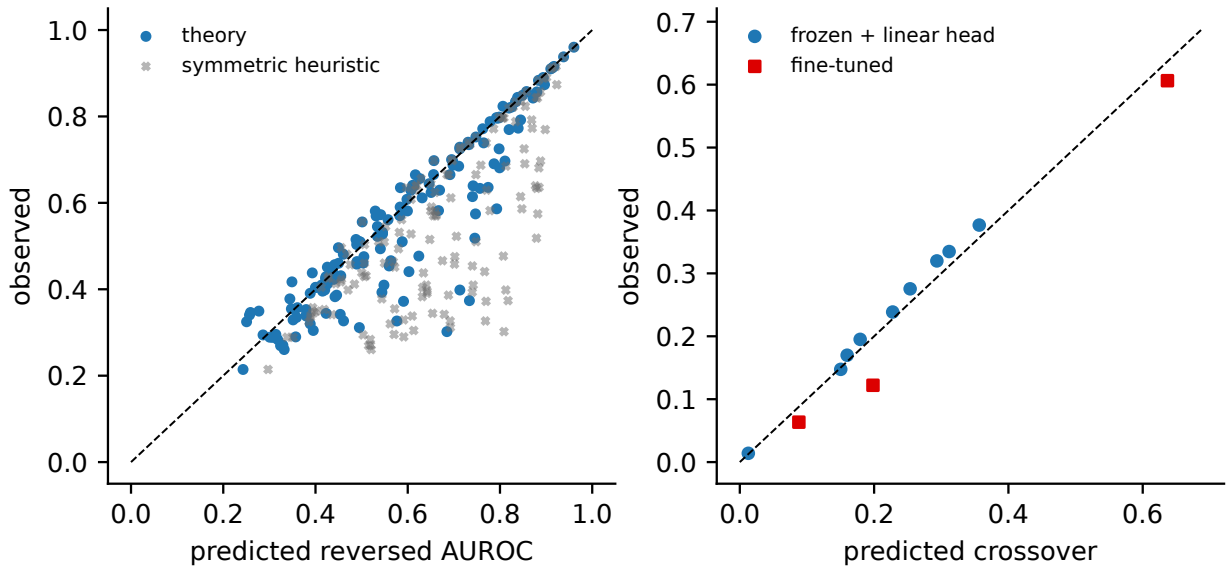


Figure 17: Left: predicted versus observed reversed AUROC (theory, blue; symmetric heuristic, grey). Right: predicted versus observed real-trap crossover (frozen encoders, blue; fine-tuned, red).

Verdict by the registered rule The rule called the theory *quantitatively predictive* only if every crossover sign agreed, both correlations were ≥ 0.7 , and its error beat “reversed = clean”. On the full registered cell set one near-zero chest-radiograph crossover (observed ≈ -0.002) had the wrong sign; the report therefore calls it a *qualitative account that closely matches held-out results* and does not say that it “predicts magnitudes”. The correlation of predicted and observed crossovers ($r = 0.999$) is not used as evidence, because the symmetric heuristic also correlates almost perfectly; the held-out error is the evidence.

5.5.6 Sensitivity to the interval estimator

On the analyses of this stage (Table 62) the crossed intervals are wider by a median factor of 1.53; 12 of 148 contrasts lose significance and 1 gains it (Table 63). The ones that matter for the text — masking’s gain on the easy thyroid pairs and on all MSK pairs — are therefore not claimed; natural-set intervals are about twice as wide as the per-seed ones, because every seed scores the same test images.

Table 62: The analyses of this stage under the per-seed and the crossed estimator (same predictions; baseline arms only).

Result group	Intervals	Excluded with the estimator	0 only per-seed	Excluded with the estimator	0 only crossed	Median width ra- tio (crossed / per- seed)
capsule/dinol518_scale	9	1		0		1.53
capsule/dinos518_scale	9	0		0		1.38
finetune/capsule	4	0		0		1.10
finetune/isc	4	0		0		1.20
finetune/ovary	2	0		0		1.23
finetune/thyroid	8	1		1		1.17
natural/capsule_dino518_repro	6	0		0		1.16
natural/isc_BCN_dino518_repro	6	0		0		1.72
natural/isc_HAM_dino518_repro	6	0		0		1.95
natural/isc_MSK_dino518_repro	6	2		0		2.01
natural/thyroid_dino518_repro	6	1		0		2.06
ovary/dinol518_scale	9	1		0		1.58
ovary/dinos518_scale	9	1		0		1.56
spec_e13/dinol518_spec_scale	17	2		0		1.44
spec_e13/dinos518_spec_scale	17	1		0		1.61
thyroid/dinol518_scale	9	1		0		1.59
thyroid/dinos518_scale	9	0		0		1.37
zero_shot/boot	12	1		0		2.00
all	148	12		1		1.53

Table 63: Contrasts of this stage whose verdict depends on the estimator.

Result	Contrast	Crossed estimate [95% CI]	Verdict vs per-seed estimator
zero_shot	medsiglip448 / ovary / trapB / zs_mask-zs	-0.054 [-0.135, +0.030]	no longer excludes 0
thyroid/dinol518_scale	trapB / balanced / clean / balanced / erm	+0.036 [-0.002, +0.073]	no longer excludes 0
ovary/dinos518_scale	trapB / balanced / clean / balanced / erm	+0.038 [-0.000, +0.077]	no longer excludes 0
ovary/dinol518_scale	trapB / balanced / clean / balanced / erm	+0.043 [-0.002, +0.087]	no longer excludes 0
capsule/dinol518_scale	trapA / mask / clean / mask / erm	+0.012 [-0.006, +0.031]	no longer excludes 0
spec_e13/dinos518_spec_scale	trapB / mask / clean / all / mask / erm	+0.028 [-0.000, +0.055]	no longer excludes 0
spec_e13/dinol518_spec_scale	trapA / mask / test_rev / HAM / mask / erm	-0.044 [-0.090, +0.002]	no longer excludes 0
spec_e13/dinol518_spec_scale	trapB / mask / clean / all / mask / erm	-0.025 [-0.055, +0.003]	no longer excludes 0
natural/thyroid_dino518_repro	easy / mask-erm	+0.042 [-0.011, +0.097]	no longer excludes 0
natural/isc_MSK_dino518_repro	all / mask-erm	+0.016 [-0.007, +0.038]	no longer excludes 0
natural/isc_MSK_dino518_repro	all / balanced-mask	-0.018 [-0.041, +0.005]	no longer excludes 0
finetune/thyroid/resnet50	trapB / mask_balanced / balanced	+0.069 [-0.004, +0.143]	no longer excludes 0
finetune/thyroid/	trapA / mask_balanced / balanced	+0.059 [+0.003, +0.122]	now excludes 0
vit_small_patch16_224.augreg_in21k_ft_in1k			

5.6 What failed or changed

- **Natural test sets:** N3 holds in three of five; HAM goes the other way and capsule is inconclusive.
- **The thyroid subgroup** is a post hoc subgroup; its test was registered only after its AUROC result was known.
- **Fine-tuning:** thyroid ResNet-50 is not significant; the ovary network is not evaluable.
- **DermLIP:** no crossover; explained, and anticipated by the theory, but still an exception.
- **The theory** is a qualitative account by its registered rule and fails for correlate-carried shortcuts (chest drains); the chest-radiograph cells could not be re-estimated and are excluded from its current error figures.
- **Absolute performance** is below task-specific fine-tuned models on thyroid.
- **External validation is dermoscopy only;** no ultrasound or capsule cohort passed the gate.

5.7 Anticipated questions

Why does AUROC barely change on natural data while sensitivity drops by a quarter? AUROC averages over all positive–negative pairs, and most pairs do not involve the shortcut. A fixed threshold, however, is set on validation data where the caliper helps; after masking the model’s scores for malignant nodules without a supporting caliper fall below it. The hard-pair AUROC and the subgroup sensitivity isolate exactly those cases.

Why does masking help on the hard HAM pairs? This is not explained, and it is reported as a failure of N3 for that test set. A possible reading — that for HAM-style images the model trained on BCN and MSK relied more on hair beside the lesion than on hair on it, so that masking removed more shortcut than it kept — has not been tested.

Why do larger models exploit the caliper more? A plausible reading, not tested directly, is that they represent small, high-contrast structures such as caliper crosses more distinctly (a larger A in the theory), so the linear head finds the shortcut more easily. What is established is only that scale is not a defence.

Why include zero-shot scores whose thyroid AUROC is at chance? Only to show that the location effect lives in the representation: even without any head trained on the trap, masking changes the artifact’s influence differently inside and outside the ROI.

Why is ISIC 2020 external if it is also ISIC? It is a different acquisition campaign with different patients, patient identifiers and no image overlap; the hair masks and lesion masks were produced by models frozen before any ISIC 2020 image was seen. It is external for patients, not for modality.

Is a two-parameter model with two inputs per cell not trivially accurate? It has two inputs and predicts a third quantity it never saw; the heuristics use the same two inputs and are two to four times worse. It also predicts crossovers, which combine four cells, and signs of in-ROI effects.

Why keep the subgroup result if it is post hoc? It is the clinically meaningful case — a malignant nodule the sonographer measured — and the test in it was registered before it was computed. It is labelled exploratory everywhere.

5.8 Conclusion

Supported. On unaltered data masking harms the shortcut-conflicting cases in three of five test sets and lowers thyroid sensitivity at every validation-fixed operating point (registered decision rule met). The crossover replicates across encoder families and sizes, in three of the four evaluable fine-tuned networks, in untrained vision–language scores and on new patients. A two-parameter theory matches held-out reversed AUROC far better than heuristics, explains every regime observed so far, and anticipates the DermLIP exception. **Qualified.** The 78-nodule subgroup is exploratory confirmation; the theory is a qualitative account by its own rule. **Not supported.** Harm on natural HAM and capsule test sets; a crossover for DermLIP and for the thyroid ResNet-50.

5.9 Questions for the supervisor

1. Is a sensitivity loss on one patient-disjoint split (thyroid) enough for a clinical claim, or should the thesis lead with the AUROC on shortcut-conflicting cases?

2. Should the theory be a main contribution (with its held-out error as evidence) or a supporting section?
3. Is ISIC 2020 enough external validation, given that it is the same modality and artifact as the main dermoscopy analysis?
4. Should the below-benchmark absolute performance of frozen linear probes be addressed with fine-tuned models in the main text, or is the fine-tuned replication sufficient?

6 Stage 6 — What to do instead of masking, and the state of the thesis

In one sentence. What to use instead of masking depends on what carries the shortcut: an annotation-free mask-then-erase method (U-MtE) repairs artifacts that carry the shortcut in their own pixels (calipers: thyroid Trap A +0.333 [+0.302, +0.364] versus ERM, masking +0.109 [+0.085, +0.133]), needs a disease-protection step when the artifact resembles the pathology (capsule debris), and cannot help when the shortcut is carried by correlates of the artifact (hair, chest drains), where only label-based reweighting works; with image-level artifact labels, U-MtE with balancing or masking with DFR is the most robust choice; for fully fine-tuned networks erasure must happen during training; and no remedy improves on masking uniformly on unaltered data. The thesis manuscript, its number audit, the CLAIM checklist and a blinded clinician audit package are ready; the audit itself has not been performed.

6.1 Where this stage starts

Stages 1–5 established that ROI masking removes an artifact outside the ROI and concentrates the model on one inside it; that this is caused by location, not by which images carry the artifact; that it matters on unaltered data and at clinical operating points; and that a two-parameter model explains it. They also showed three facts that shape any remedy: label-based methods (balancing, DFR) fix both locations but need image-level artifact labels; removing the artifact’s pixels fails when the artifact resembles the disease (capsule debris); and some shortcuts are carried by correlates of the artifact rather than its pixels (hair, drains). This stage asks what a practitioner should do instead of masking, and reports the state of the thesis.

6.2 Questions

- Q6a** Can an in-ROI shortcut be removed from frozen foundation-model features *without* any artifact annotation — no artifact example, mask or label, only the ROI mask that masking already uses?
- Q6b** When does such a remedy fail, and which alternative fits which regime?
- Q6c** Can the same be done for networks fine-tuned end to end?
- Q6d** Does any remedy transfer to unaltered test data?
- Q6e** How will the automatic artifact labels be validated by a clinician, and what remains before submission?

6.3 Method: from insert-then-erase to U-MtE

Idea Masking leaves exactly one thing to remove: what lies inside the ROI and is not disease. If we knew the directions in feature space along which an artifact moves an image, we could project them out. The method estimates those directions by *inserting* artifacts into images and measuring how the features change.

Evolution

1. **Insert-then-erase (I2E).** Paste artifact *templates* (e.g. real hair strands cut from other images) into training images, collect the paired feature differences (with minus without), take the subspace U_k holding 90% of their held-out energy (cap 64 directions), and erase it: $x \mapsto x - U_k U_k^\top (x - \mu)$; a logistic head is trained on the erased features. Works, but needs examples of the artifact.
2. **Mask-then-erase (MtE).** Estimate the subspace on *masked* images (insert, then mask), so that erasure targets exactly what masking leaves; the head is trained and tested on masked, erased features.

3. **U-MtE (universal).** Replace the templates by one *generic* procedural overlay library — thin strokes, thick bands, solid and translucent patches, blobs, dashed lines, small glyphs; random colour (half grey), opacity, width and position, half centred in the ROI (Figure 18) — used *unchanged* for every artifact, dataset and modality. It contains no ruler, no caliper and no hair. U-MtE needs no artifact example, mask or label; only the ROI mask.
4. **Disease protection.** If the erased subspace overlaps the disease direction, erasure removes disease evidence. Protection estimates label directions W on the *artifact-free* training images and erases $\text{orth}((I - WW^\top)U_k)$, which leaves every protected direction untouched ($w^\top P(x) = w^\top x$). It needs diagnosis labels of artifact-free images (which any training set has) and no artifact label.
5. **U-MtE with balancing.** When image-level artifact labels exist, the head is group-balanced.

Why it should work (theory of Stage 5) Erasure sets the effective artifact signal $\tilde{A}$ to zero but scales the disease signal by $1 - \rho^2$, where ρ is the overlap between the erased subspace and the disease direction. It therefore beats in-ROI masking unless ρ is large — which is the case when the artifact resembles the disease, and which protection forces to zero along the protected directions. Overlay *augmentation* with the same overlays only decorrelates the label from the generated overlays, not from the real artifact; erasure removes the whole subspace of “foreign overlay” changes, which contains the real artifact’s direction when both move the representation in shared ways. The theory therefore predicts erase $\gg$ augment for overlay-like artifacts.

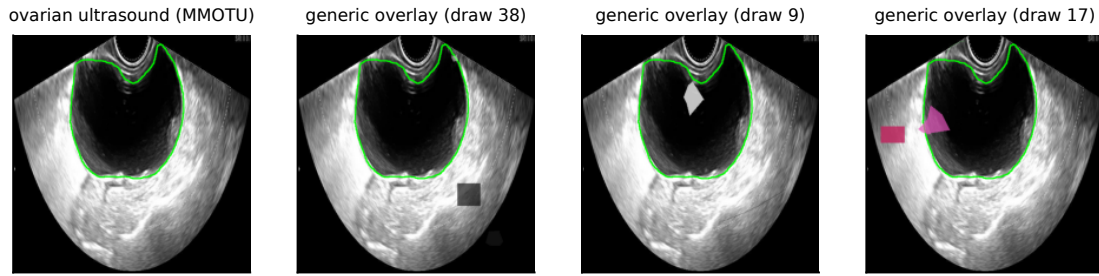


Figure 18: The generic overlay library (the same for every modality) drawn on an ovarian ultrasound image (MMOTU, CC BY 4.0; green: ROI). Overlays are inserted, then the image is masked, to estimate the subspace that U-MtE erases.

Baselines JTT (Liu et al., 2021) (label-free: up-weight the training errors of an ERM model); SPLINCE (Holstege et al., 2025) (covariance-preserving linear concept removal with artifact labels; faithful whitened implementation, verified against the paper’s theorem); LEACE (Stages 1 and 3); LaMa inpainting (Suvorov et al., 2022) of the artifact pixels (needs artifact masks); text-prompted erasure with vision-language models (the artifact direction from a text description, Chuang et al., 2023); SLAS, a few-shot patch-token artifact localiser (5–50 annotated images); pseudo-group balancing with groups inferred by an overlay detector; erasure followed by DFR; and two validation-driven selectors that pick, per fold, the candidate with the best worst-group validation score.

Fine-tuned networks `mte_ft`: masked images with and without a generic overlay, plus a penalty on the distance of their normalised penultimate features; `umte_ft` (erase-while-fine-tuning): a running overlay-shift subspace is estimated on the network’s own features and projected out before the head at every step, plus the invariance penalty; `cons_ft`: prediction consistency with and without an overlay; `umte_cons_ft`: all three; `mte_post`: fine-tune like masking, then apply U-MtE to the penultimate features.

Tool Both methods are packaged for any backbone: `python -m wtss.umte fit/embed` (images and ROI masks only) and `python -m wtss.slas fit/embed` (a few annotated images).

6.4 Experiments

Table 64: Experiments of this stage and their registration documents (docs/PREREGISTRATION_<name>.md or docs/<name>.md). Commit times are set by the committing machine and are not independent proof of order.

Experiment	Status	Registration	First commit (local time)	Support criterion
U-I2E / U-MtE, protection, balancing, U7–U14	supported registered (with amendments before each run)	UNIVERSAL_I2E	ee95f10, 20:14	2026-09-26 per claim, CI; no gaming (clean loss ≤ 0.02 , corr $\geq$ rev -0.02)
Remedy tests P2–P4 in the 16-test family	partly supported tests registered; family grouped post hoc	STATISTICAL_PLAN	4e9d44c, 09:37	2026-09-27 Holm $p < 0.05$, estimate > 0
Erase-rank ablation	supported registered	UMTE_ABLATION	30f19c8, 14:15	2026-09-27 descriptive
Encoder size (S2)	supported registered	SCALE	13303b0, 14:13	2026-09-27 protected U-MtE $>$ mask, CI
LaMa inpainting	supported registered	INPAINT_LAMA	715b790, 09:41	2026-09-27 reported with CI, no direction
Text-prompted erasure (V1–V3)	supported registered	TEXT_PROMPT	7877e8c, 12:21	2026-09-27 V1: text erasure $>$ mask
SLAS (S0–S6)	partly supported registration and localisation result share a commit	SLAS	2d1ba20, 23:33	2026-09-26 S1: mask-then-suppress $>$ token masking
Chest drains (D1–D5)	supported registered	CXR_DRAIN	3d0cffc, 03:29	2026-09-27 U-MtE $>$ mask, CI
Fine-tuning F1–F3	supported registered	FINETUNE	7c8b466, 07:06	2026-09-27 mte_ft $>$ mask, CI
Fine-tune, then erase	supported registered	FT_ERASE	8ac7a7d, 14:20	2026-09-27 mte_post $>$ mask_post
Erase while fine-tuning (G1–G4)	supported registered (amendments before running)	FT_UMTE	b7d7077, 14:52	2026-09-27 umte(_cons)_ft $>$ mask for both architectures
Leakage: embedding groups (P2–P4)	supported registered	EMBEDDING_GROUPS	b0eda23, 13:55	2026-09-27 same sign and CI verdict
Natural data (N1, N2, N4)	supported registered	NATURAL	74bd243, 09:39	2026-09-27 protected U-MtE $\geq$ mask
Clinician image audit (prepared)	supported registered	FINAL (Part B)	c45d482, 00:35	2026-09-28 precision, recall, location agreement, κ

Every remedy analysis was registered before it was run (docs/PREREGISTRATION_UNIVERSAL_I2E.md and its amendments U7–U14; fine-tuning in PREREGISTRATION_FT_ERASE.md and PREREGISTRATION_FT_UMTE.md; the rank ablation, LaMa, text prompts, SLAS and the chest drains in their own documents). Every outcome is reported whichever way it went.

6.5 Results

6.5.1 Remedies in the real traps

Figure 19 and Table 65 (DINOv2, Trap A); Table 66 gives every arm, both traps and three encoder families.

- **Calipers (pixel-carried).** Without any annotation, U-MtE repairs most of what masking leaves in thyroid (+0.333 [+0.302, +0.364] versus ERM; masking +0.109 [+0.085, +0.133]), with MedSigLIP (+0.299 [+0.270, +0.328]) and with ConvNeXt (+0.131 [+0.087, +0.171]), and in the ovary with MedSigLIP (+0.212 [+0.163, +0.260]). With DINOv2 in the ovary its gain is small (+0.152 [+0.098, +0.203]; protected +0.175 [+0.116, +0.235]).

- **Debris (pathology-like).** Unprotected U-MtE *lowers* capsule performance (-0.042 [$-0.076, -0.009$]; MedSigLIP -0.113 [$-0.146, -0.080$]): debris resembles erosion fibrin, so the overlay subspace overlaps the disease. Protection repairs this ($+0.078$ [$+0.047, +0.107$]) but does not beat masking; label-based methods are best.
- **Hair (correlate-carried).** No pixel-level method helps much (U-MtE -0.009 [$-0.043, +0.024$] versus ERM, i.e. $+0.040$ [$+0.024, +0.056$] over masking); balancing ($+0.213$ [$+0.183, +0.243$]) and DFR ($+0.196$ [$+0.157, +0.235$]) are required.
- **Outside the ROI (Trap B)** U-MtE neither helps nor harms relative to masking — masking has already removed the artifact.

Table 65: Remedies in the in-ROI trap: change in reversed-test AUROC versus ERM (DINOv2, 5 seeds $\times$ 5 folds, crossed CIs). U-MtE needs no artifact annotation; balancing and DFR need image-level artifact labels.

Cohort (Trap A, in-ROI)	ROI masking	U-MtE	U-MtE protected	group balancing	DFR	U-MtE + balancing
Dermoscopy hair	-0.049 [$-0.083, -0.018$]	-0.009 [$-0.043, +0.024$]	-0.008 [$-0.046, +0.028$]	$+0.213$ [$+0.183, +0.243$]	$+0.196$ [$+0.157, +0.235$]	$+0.185$ [$+0.145, +0.224$]
Thyroid calipers	$+0.109$ [$+0.085, +0.133$]	$+0.333$ [$+0.302, +0.364$]	$+0.393$ [$+0.347, +0.434$]	$+0.388$ [$+0.360, +0.418$]	$+0.463$ [$+0.436, +0.492$]	$+0.498$ [$+0.471, +0.524$]
Ovarian calipers	$+0.132$ [$+0.072, +0.196$]	$+0.152$ [$+0.098, +0.203$]	$+0.175$ [$+0.116, +0.235$]	$+0.290$ [$+0.248, +0.332$]	$+0.310$ [$+0.263, +0.357$]	$+0.321$ [$+0.271, +0.372$]
Capsule debris	$+0.084$ [$+0.055, +0.112$]	-0.042 [$-0.076, -0.009$]	$+0.078$ [$+0.047, +0.107$]	$+0.179$ [$+0.150, +0.205$]	$+0.234$ [$+0.199, +0.273$]	$+0.258$ [$+0.221, +0.293$]

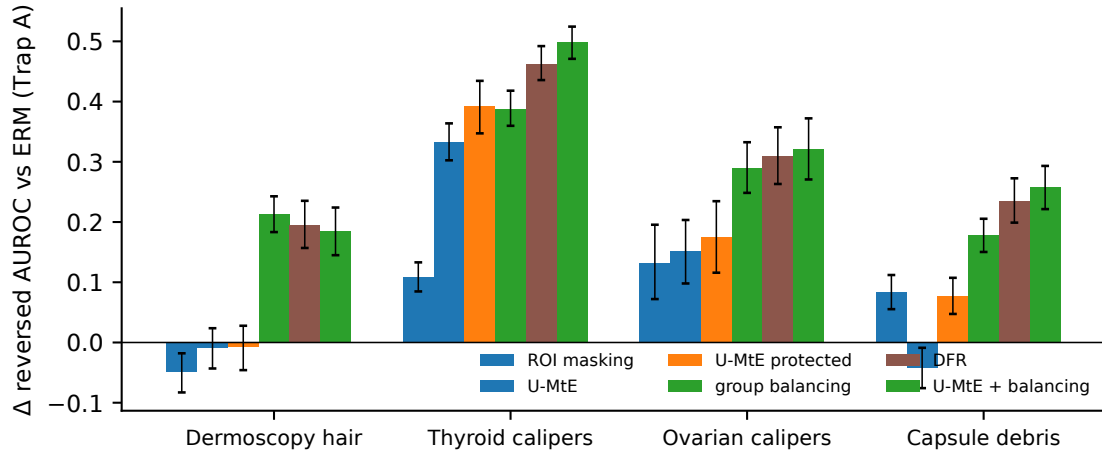


Figure 19: Remedies in the in-ROI trap (change in reversed AUROC versus ERM, DINOv2, crossed 95% CIs).

Table 66: Every remedy in every real-artifact trap and encoder: change in reversed-test AUROC versus ERM (5 seeds $\times$ 5 grouped folds, crossed CIs). U-MtE and the augmentation arm use only the ROI mask; balancing, DFR and their combinations need image-level artifact labels.

Cohort, encoder	Arm — ERM	Trap A (in ROI)	Trap B (outside)
Dermoscopy hair, DINOv2	masking	-0.049 [$-0.083, -0.018$]	$+0.098$ [$+0.060, +0.136$]
	U-MtE	-0.009 [$-0.043, +0.024$]	$+0.121$ [$+0.083, +0.160$]
	U-MtE protected	-0.008 [$-0.046, +0.028$]	$+0.124$ [$+0.077, +0.170$]
	same overlays as augmentation	-0.044 [$-0.078, -0.012$]	$+0.107$ [$+0.070, +0.145$]
	group balancing	$+0.213$ [$+0.183, +0.243$]	$+0.204$ [$+0.166, +0.245$]
	DFR	$+0.196$ [$+0.157, +0.235$]	$+0.196$ [$+0.147, +0.251$]
	masking + DFR	$+0.152$ [$+0.103, +0.202$]	$+0.206$ [$+0.142, +0.267$]
Thyroid calipers, DINOv2	U-MtE + balancing	$+0.185$ [$+0.145, +0.224$]	$+0.240$ [$+0.197, +0.283$]
	masking	$+0.109$ [$+0.085, +0.133$]	$+0.348$ [$+0.307, +0.388$]
	U-MtE	$+0.333$ [$+0.302, +0.364$]	$+0.347$ [$+0.308, +0.386$]
	U-MtE protected	$+0.393$ [$+0.347, +0.434$]	$+0.349$ [$+0.310, +0.389$]
	same overlays as augmentation	$+0.118$ [$+0.094, +0.143$]	$+0.345$ [$+0.305, +0.385$]
	group balancing	$+0.388$ [$+0.360, +0.418$]	$+0.254$ [$+0.218, +0.292$]

Cohort, encoder			Arm – ERM	Trap A (in ROI)	Trap B (outside)
Thyroid SigLIP	calipers,	Med-	DFR	+0.463 [+0.436, +0.492]	+0.312 [+0.262, +0.362]
			masking + DFR	+0.537 [+0.506, +0.567]	+0.357 [+0.313, +0.400]
			U-MtE + balancing	+0.498 [+0.471, +0.524]	+0.371 [+0.329, +0.412]
			masking	+0.053 [+0.032, +0.073]	+0.248 [+0.200, +0.293]
			U-MtE	+0.299 [+0.270, +0.328]	+0.276 [+0.232, +0.318]
Thyroid vNeXt	calipers,	Con-	U-MtE protected	+0.358 [+0.330, +0.388]	+0.271 [+0.224, +0.318]
			same overlays as augmentation	+0.074 [+0.053, +0.095]	+0.248 [+0.202, +0.293]
			group balancing	+0.409 [+0.382, +0.436]	+0.201 [+0.161, +0.241]
			DFR	+0.463 [+0.437, +0.489]	+0.243 [+0.207, +0.282]
			masking + DFR	+0.517 [+0.490, +0.545]	+0.279 [+0.234, +0.324]
Ovarian calipers, DINOv2			U-MtE + balancing	+0.505 [+0.478, +0.531]	+0.313 [+0.269, +0.358]
			masking	−0.004 [−0.029, +0.020]	+0.271 [+0.228, +0.314]
			U-MtE	+0.131 [+0.087, +0.171]	+0.276 [+0.231, +0.320]
			U-MtE protected	+0.179 [+0.112, +0.236]	+0.286 [+0.241, +0.332]
			same overlays as augmentation	+0.003 [−0.021, +0.027]	+0.270 [+0.227, +0.313]
Ovarian calipers, Med- SigLIP			group balancing	+0.326 [+0.305, +0.348]	+0.222 [+0.186, +0.258]
			DFR	+0.369 [+0.335, +0.405]	+0.260 [+0.212, +0.305]
			masking + DFR	+0.408 [+0.372, +0.442]	+0.307 [+0.259, +0.354]
			U-MtE + balancing	+0.397 [+0.362, +0.434]	+0.313 [+0.266, +0.358]
			masking	+0.132 [+0.072, +0.196]	+0.302 [+0.225, +0.378]
Capsule debris, DINOv2			U-MtE	+0.152 [+0.098, +0.203]	+0.317 [+0.242, +0.388]
			U-MtE protected	+0.175 [+0.116, +0.235]	+0.322 [+0.243, +0.398]
			same overlays as augmentation	+0.117 [+0.054, +0.178]	+0.305 [+0.224, +0.383]
			group balancing	+0.290 [+0.248, +0.332]	+0.256 [+0.190, +0.320]
			DFR	+0.310 [+0.263, +0.357]	+0.278 [+0.209, +0.349]
Capsule debris, Med- SigLIP			masking + DFR	+0.339 [+0.284, +0.393]	+0.347 [+0.264, +0.428]
			U-MtE + balancing	+0.321 [+0.271, +0.372]	+0.343 [+0.265, +0.419]
			masking	+0.002 [−0.048, +0.051]	+0.180 [+0.120, +0.238]
			U-MtE	+0.212 [+0.163, +0.260]	+0.191 [+0.136, +0.245]
			U-MtE protected	+0.155 [+0.098, +0.209]	+0.166 [+0.108, +0.224]
Capsule debris, Con- vNeXt			same overlays as augmentation	+0.018 [−0.027, +0.062]	+0.179 [+0.123, +0.235]
			group balancing	+0.302 [+0.254, +0.351]	+0.149 [+0.103, +0.197]
			DFR	+0.370 [+0.320, +0.420]	+0.210 [+0.154, +0.265]
			masking + DFR	+0.356 [+0.299, +0.411]	+0.229 [+0.161, +0.296]
			U-MtE + balancing	+0.364 [+0.312, +0.416]	+0.219 [+0.161, +0.277]
Capsule debris, Med- SigLIP			masking	+0.084 [+0.055, +0.112]	+0.461 [+0.415, +0.506]
			U-MtE	−0.042 [−0.076, −0.009]	+0.441 [+0.396, +0.483]
			U-MtE protected	+0.078 [+0.047, +0.107]	+0.442 [+0.397, +0.486]
			same overlays as augmentation	+0.096 [+0.067, +0.126]	+0.462 [+0.417, +0.505]
			group balancing	+0.179 [+0.150, +0.205]	+0.244 [+0.216, +0.272]
Capsule debris, Con- vNeXt			DFR	+0.234 [+0.199, +0.273]	+0.287 [+0.239, +0.335]
			masking + DFR	+0.312 [+0.275, +0.349]	+0.446 [+0.393, +0.497]
			U-MtE + balancing	+0.258 [+0.221, +0.293]	+0.471 [+0.428, +0.514]
			masking	+0.117 [+0.088, +0.147]	+0.452 [+0.407, +0.495]
			U-MtE	−0.113 [−0.146, −0.080]	+0.406 [+0.362, +0.450]
Capsule debris, Med- SigLIP			U-MtE protected	+0.088 [+0.055, +0.121]	+0.448 [+0.404, +0.491]
			same overlays as augmentation	+0.136 [+0.106, +0.166]	+0.451 [+0.407, +0.495]
			group balancing	+0.225 [+0.203, +0.248]	+0.240 [+0.211, +0.269]
			DFR	+0.254 [+0.222, +0.286]	+0.279 [+0.242, +0.320]
			masking + DFR	+0.345 [+0.312, +0.378]	+0.438 [+0.391, +0.483]
Capsule debris, Con- vNeXt			U-MtE + balancing	+0.269 [+0.232, +0.308]	+0.453 [+0.410, +0.496]
			masking	+0.084 [+0.049, +0.118]	+0.403 [+0.358, +0.448]
			U-MtE	+0.009 [−0.024, +0.040]	+0.378 [+0.334, +0.422]
			U-MtE protected	+0.108 [+0.074, +0.142]	+0.383 [+0.337, +0.429]
			same overlays as augmentation	+0.099 [+0.066, +0.131]	+0.406 [+0.361, +0.449]
Capsule debris, Med- SigLIP			group balancing	+0.197 [+0.172, +0.221]	+0.226 [+0.198, +0.256]
			DFR	+0.255 [+0.223, +0.291]	+0.293 [+0.234, +0.346]
			masking + DFR	+0.330 [+0.296, +0.364]	+0.453 [+0.403, +0.502]
			U-MtE + balancing	+0.273 [+0.231, +0.312]	+0.433 [+0.384, +0.480]
			masking	+0.002 [−0.048, +0.051]	+0.180 [+0.120, +0.238]

6.5.2 Which model is most robust?

A reversed-test gain can come from inverting the shortcut rather than removing it. Table 67 therefore reports min(reversed, correlated) AUROC. With image-level artifact labels, U-MtE with balancing is the most robust model for thyroid with all three encoders (e.g. 0.787 with DINOv2)

and for the ovary with MedSigLIP (0.790); masking with DFR is best for capsule (0.892) and the DINOv2 ovary (0.752); plain balancing for hair (0.722) and DFR for chest drains (0.683). DFR and masking + DFR over-correct on thyroid: their correlated AUROC falls below the reversed one (marked [†]). Without artifact labels, protected U-MtE is the most robust label-free arm for thyroid; for capsule the best label-free choice is U-MtE combined with JTT (Table 74).

Table 67: Robustness summary in the in-ROI trap: min(reversed, correlated) AUROC per arm (mean over seeds; best per row in bold; [†] = the arm flips the shortcut, correlated < reversed −0.02). Chest drains: point estimates, not re-estimated.

Trap A	ERM	mask	U-MtE	U-MtE prot.	overlay aug.	JTT	balanced	DFR	mask+DFR	U-MtE+bal.
Dermoscopy hair, DINOv2	0.510	0.460	0.500	0.502	0.466	0.575	0.722	0.705	0.662	0.695
Thyroid calipers, DINOv2	0.289	0.398	0.622	0.682	0.407	0.417	0.677	0.669 [†]	0.751 [†]	0.787
Thyroid calipers, MedSigLIP	0.289	0.341	0.587	0.647	0.362	–	0.698	0.722 [†]	0.772 [†]	0.793
Thyroid calipers, ConvNeXt	0.357	0.353	0.488	0.536	0.361	–	0.684	0.669 [†]	0.713 [†]	0.755
Ovarian calipers, DINOv2	0.413	0.545	0.565	0.588	0.530	0.440	0.703	0.689 [†]	0.752	0.734
Ovarian calipers, MedSigLIP	0.426	0.428	0.638	0.581	0.444	–	0.729	0.756 [†]	0.782	0.790
Capsule debris, DINOv2	0.581	0.665	0.539	0.659	0.678	0.596	0.760	0.816	0.892	0.839
Capsule debris, MedSigLIP	0.570	0.687	0.457	0.658	0.706	–	0.796	0.825	0.915	0.839
Capsule debris, ConvNeXt	0.528	0.612	0.537	0.636	0.627	–	0.725	0.783	0.858	0.801
Chest drains, RAD-DINO	0.191	0.239	0.268	0.269	0.237	0.175	0.538	0.683	0.646	0.460

6.5.3 The 16-test primary family

Table 68. The four location tests (P1) and twelve remedy tests — protected U-MtE > mask (P2), erase > augment with the same overlays (P3), U-MtE+balancing > balancing (P4), in each of the four cohorts — were grouped into one Holm family after the individual results were known (conservative bookkeeping, not a pre-registered family). 11 of 16 are supported: all four P1; P2 in hair, thyroid and ovary (not capsule); P3 in hair and thyroid (capsule significant in the opposite direction; ovary not after Holm); P4 in thyroid and capsule.

Table 68: The 16-test primary family (DINOv2, Trap A reversed test unless stated): the four location tests and the twelve remedy tests, Holm-corrected together.

Family	Cohort	Estimate [95% CI]	<i>p</i>	<i>p</i> (Holm, 16)	Supported
P1 location crossover	ISIC hair (DINOv2)	+0.147 [+0.111, +0.185]	0.0001	0.0016	supported
P1 location crossover	Thyroid (DINOv2)	+0.239 [+0.196, +0.280]	0.0001	0.0016	supported
P1 location crossover	Capsule (DINOv2)	+0.377 [+0.327, +0.426]	0.0001	0.0016	supported
P1 location crossover	Ovary (DINOv2)	+0.170 [+0.087, +0.252]	0.0001	0.0016	supported
P2 U-MtE (protected) $\hat{\iota}$ mask	ISIC hair	+0.042 [+0.024, +0.058]	0.0001	0.0016	supported
P2 U-MtE (protected) $\hat{\iota}$ mask	Thyroid	+0.283 [+0.237, +0.329]	0.0001	0.0016	supported
P2 U-MtE (protected) $\hat{\iota}$ mask	Capsule	−0.006 [−0.018, +0.006]	0.2946	0.3486	not supported
P2 U-MtE (protected) $\hat{\iota}$ mask	Ovary	+0.043 [+0.023, +0.065]	0.0001	0.0016	supported
P3 erase $\hat{\iota}$ augment	ISIC hair	+0.035 [+0.017, +0.055]	0.0002	0.0016	supported
P3 erase $\hat{\iota}$ augment	Thyroid	+0.215 [+0.189, +0.244]	0.0001	0.0016	supported
P3 erase $\hat{\iota}$ augment	Capsule	−0.139 [−0.158, −0.120]	0.0001	0.0016	not supported (opposite)
P3 erase $\hat{\iota}$ augment	Ovary	+0.035 [+0.003, +0.067]	0.0326	0.1304	not supported
P4 U-MtE+balanced $\hat{\iota}$ balanced	ISIC hair	−0.028 [−0.064, +0.007]	0.1162	0.3486	not supported
P4 U-MtE+balanced $\hat{\iota}$ balanced	Thyroid	+0.110 [+0.079, +0.142]	0.0001	0.0016	supported
P4 U-MtE+balanced $\hat{\iota}$ balanced	Capsule	+0.079 [+0.042, +0.115]	0.0002	0.0016	supported
P4 U-MtE+balanced $\hat{\iota}$ balanced	Ovary	+0.031 [−0.014, +0.077]	0.1730	0.3486	not supported

6.5.4 Remedies in the controlled sweeps

Tables 69 and 70. With DINOv2 and RAD-DINO, U-MtE beats masking at full overlap in every modality sweep (thyroid +0.278 [+0.214, +0.344], capsule +0.121 [+0.078, +0.170], chest tube +0.134 [+0.113, +0.156]). With MedSigLIP it fails on both the synthetic caliper and the synthetic debris; protection turns the debris failure into a gain (+0.114 [+0.022, +0.238]) but not the caliper failure (−0.041 [−0.095, +0.002]). U-MtE with balancing beats masking in every sweep and has no positive location interaction (dermoscopy: −0.083 [−0.114, −0.050], against masking’s large positive one). On the dermoscopy ruler, the insert-then-erase precursor on unmasked features helps (+0.197 [+0.146, +0.247]) while plain U-MtE does not (−0.007 [−0.085, +0.070]): with the synthetic ruler,

the subspace estimated from generic overlays evidently misses the direction of the retained ruler, and only the balanced variant repairs it.

Table 69: Remedies in the controlled sweeps at full overlap (reversed-test AUROC, crossed CIs).

Controlled sweep, $r = 1$	U-MtE – mask	U-MtE protected – mask	U-MtE+bal. – mask	U-MtE+bal. – balanced
Thyroid caliper, DINOv2	+0.278 [+0.214, +0.344]	+0.287 [+0.141, +0.464]	+0.470 [+0.391, +0.550]	+0.091 [+0.013, +0.172]
Thyroid caliper, MedSigLIP	−0.076 [−0.120, −0.034]	−0.041 [−0.095, +0.002]	+0.458 [+0.393, +0.525]	+0.069 [+0.002, +0.136]
Ovarian caliper, DINOv2	+0.140 [+0.061, +0.225]	+0.114 [+0.043, +0.192]	+0.251 [+0.157, +0.355]	−0.032 [−0.165, +0.104]
Capsule debris, DINOv2	+0.121 [+0.078, +0.170]	+0.130 [+0.083, +0.189]	+0.211 [+0.134, +0.297]	+0.057 [−0.008, +0.121]
Capsule debris, MedSigLIP	−0.190 [−0.265, −0.120]	+0.114 [+0.022, +0.238]	+0.299 [+0.195, +0.398]	−0.018 [−0.101, +0.065]
Chest tube, RAD-DINO	+0.134 [+0.113, +0.156]	+0.144 [+0.125, +0.164]	+0.168 [+0.145, +0.191]	−0.031 [−0.042, −0.021]

Table 70: The dermoscopy ruler sweep at full overlap: the insert-then-erase precursor (U-I2E, erasing on unmasked features) and the mask-then-erase variants, versus ERM (crossed CIs).

Dermoscopy ruler sweep, $r = 1$ (DINOv2)	U-I2E – ERM	U-MtE – ERM	U-MtE protected – ERM	U-MtE+bal. – ERM
change in reversed AUROC	+0.197 [+0.146, +0.247]	−0.007 [−0.085, +0.070]	+0.091 [−0.016, +0.185]	+0.243 [+0.167, +0.317]

6.5.5 Erasure versus augmentation, protection, and other contrasts

Table 71. The *same* overlays used as training augmentation instead of for erasure gain almost nothing over masking (Table 66, “same overlays as augmentation”), while erasure gains +0.225 [+0.202, +0.247] over augmentation with MedSigLIP and +0.128 [+0.093, +0.161] with ConvNeXt on thyroid: it is the subspace estimation, not exposure to overlays, that helps. Protection is a safety net: it costs nothing on hair (+0.001 [−0.016, +0.017]), helps thyroid (+0.059 [+0.019, +0.096]) and removes the capsule failure (protected – mask −0.006 [−0.018, +0.006] versus unprotected −0.126 [−0.146, −0.107]).

Table 71: Secondary remedy contrasts (reversed test, crossed CIs).

Cohort	Contrast (Trap A)	Estimate [95% CI]
hair	mte – mask	+0.040 [+0.024, +0.056]
hair	mte_protect – mte	+0.001 [−0.016, +0.017]
thyroid	mte_protect – mte	+0.059 [+0.019, +0.096]
thyroid	mte – jtt	+0.206 [+0.171, +0.241]
ovary	mte – jtt	+0.125 [+0.063, +0.189]
thyroid MedSigLIP	mte – mte_aug	+0.225 [+0.202, +0.247]
thyroid ConvNeXt	mte – mte_aug	+0.128 [+0.093, +0.161]
thyroid ConvNeXt	mte – mask	+0.135 [+0.102, +0.168]
capsule	mte – mask	−0.126 [−0.146, −0.107]
capsule	mte_protect – mask	−0.006 [−0.018, +0.006]

6.5.6 How many directions must be erased?

Table 72. On thyroid the gain is robust for every setting that erases ≥ 16 directions or $\geq 50\%$ of the overlay-shift energy; one to four directions recover only a third of it — the overlay shift is multi-dimensional. In these cohorts the 90% rule reaches the cap of 64 directions. Unprotected erasure harms capsule more the more directions it removes, as the theory’s ρ predicts; the protected variant never harms capsule and helps the ovary at every rank.

Table 72: Erasure-rank ablation (DINOv2, Trap A): reversed-AUROC difference to masking (mean over seeds) for the energy rule (share of held-out overlay-shift energy removed; cap 64) and for fixed ranks k .

Erased rank	Thyroid U-MtE	Ovary U-MtE	Ovary protected	Capsule U-MtE	Capsule protected
energy 0.50	+0.190	−0.003	+0.033	−0.067	−0.006
energy 0.90 (main)	+0.224	+0.020	+0.043	−0.126	−0.006
energy 0.99	+0.224	+0.020	+0.043	−0.126	−0.006
$k = 1$	+0.068	+0.010	+0.009	+0.001	+0.000
$k = 4$	+0.056	−0.038	+0.015	−0.010	−0.003
$k = 16$	+0.183	−0.011	+0.020	−0.058	−0.009
$k = 64$	+0.224	+0.020	+0.043	−0.126	−0.006

6.5.7 Encoder size

Table 73. Protected U-MtE beats masking in thyroid and ovary at every DINOv2 size; on capsule protection is less complete with ViT-L.

Table 73: U-MtE across encoder sizes (crossed CIs).

Cohort	DINOv2	U-MtE – mask (Trap A)	U-MtE protected – mask
Thyroid	ViT-S/14	+0.192 [+0.159, +0.227]	+0.225 [+0.168, +0.276]
Thyroid	ViT-B/14	+0.224 [+0.195, +0.254]	+0.283 [+0.237, +0.329]
Thyroid	ViT-L/14	+0.246 [+0.216, +0.279]	+0.274 [+0.243, +0.306]
Ovary	ViT-S/14	+0.121 [+0.084, +0.158]	+0.103 [+0.074, +0.134]
Ovary	ViT-B/14	+0.020 [−0.016, +0.054]	+0.043 [+0.023, +0.065]
Ovary	ViT-L/14	+0.094 [+0.061, +0.127]	+0.060 [+0.037, +0.084]
Capsule	ViT-S/14	−0.153 [−0.172, −0.134]	−0.002 [−0.018, +0.016]
Capsule	ViT-B/14	−0.126 [−0.146, −0.107]	−0.006 [−0.018, +0.006]
Capsule	ViT-L/14	−0.097 [−0.114, −0.079]	−0.044 [−0.056, −0.032]

6.5.8 Published baselines and alternatives

Table 74 (crossed intervals available) and Table 75 (point estimates of analyses whose predictions were not saved).

- **JTT** (label-free): U-MtE beats it on thyroid (+0.206 [+0.171, +0.241]) and ovary (+0.125 [+0.063, +0.189]); JTT wins on hair (min(rev, corr) 0.564 versus 0.506); on capsule U-MtE+JTT is the most robust label-free combination (0.787).
- **SPLINCE** (needs artifact labels): the faithful implementation removes much disease signal (clean AUROC 0.577 on thyroid) and still inverts the shortcut on capsule; U-MtE with balancing is more robust on thyroid, capsule and hair (0.787 versus 0.565; 0.839 versus 0.668; 0.695 versus 0.575).
- **LaMa inpainting** of the detected artifact pixels, followed by masking, is a strong baseline that needs artifact masks; protected U-MtE matches it on thyroid without masks and is below it on ovary and hair; U-MtE with balancing beats it everywhere.
- **Text-prompted erasure** with MedSigLIP and DermLIP is a near no-op ($\leq +0.008$ over masking): the text-space artifact direction does not coincide with the image-space direction along which the artifact moves the representation.
- **SLAS** localises hair from a few annotated images, but the gain it enables is capped by what hair pixels carry: removing *every* real hair patch (an oracle) gains only +0.015 over token-level masking.
- **Pseudo-group balancing** from an overlay detector helps on thyroid (reversed 0.663 versus 0.622 for U-MtE) and not on hair, where the detector cannot see hair.
- **Erasure + DFR** adds at most about 0.01 over masking + DFR: once a group-robust head is used, masking does the rest.

- **Validation-driven selectors** beat masking in every cohort and are within about 0.02 of the best candidate, but neither is uniformly best; admitting DFR as a candidate fixes chest drains and makes the choice noisier elsewhere.
- **Chest drains:** U-MtE gains only about 0.03 over masking with RAD-DINO and nothing with DINOv2; DFR is best.

Table 74: Published baselines: JTT (label-free error upweighting) and SPLINCE (faithful whitened implementation of the covariance-preserving concept projection, needs artifact labels), beside U-MtE (mean over seeds).

Cohort (Trap A)	Arm	clean	correlated	reversed	min(rev, corr)
Thyroid	JTT	0.629	0.829	0.417	0.417
Thyroid	mask	0.689	0.908	0.398	0.398
Thyroid	mask+JTT	0.699	0.888	0.455	0.455
Thyroid	U-MtE	0.729	0.824	0.622	0.622
Thyroid	U-MtE+JTT	0.629	0.848	0.363	0.363
Thyroid	balanced	0.717	0.757	0.677	0.677
Thyroid	mask+SPLINCE	0.603	0.599	0.610	0.599
Thyroid	U-MtE+bal.	0.798	0.804	0.787	0.787
Thyroid	U-MtE prot.	0.728	0.769	0.682	0.682
Thyroid	SPLINCE	0.577	0.565	0.587	0.565
Capsule	JTT	0.786	0.928	0.596	0.596
Capsule	mask	0.860	0.972	0.665	0.665
Capsule	mask+JTT	0.878	0.956	0.760	0.760
Capsule	U-MtE	0.815	0.972	0.539	0.539
Capsule	U-MtE+JTT	0.881	0.945	0.787	0.787
Capsule	balanced	0.847	0.916	0.760	0.760
Capsule	mask+SPLINCE	0.786	0.715	0.852	0.715
Capsule	U-MtE+bal.	0.899	0.939	0.839	0.839
Capsule	U-MtE prot.	0.855	0.970	0.659	0.659
Capsule	SPLINCE	0.741	0.668	0.788	0.668
Dermoscopy hair	JTT	0.745	0.898	0.564	0.564
Dermoscopy hair	mask	0.712	0.916	0.454	0.454
Dermoscopy hair	mask+JTT	0.711	0.880	0.509	0.509
Dermoscopy hair	U-MtE	0.735	0.913	0.506	0.506
Dermoscopy hair	U-MtE+JTT	0.729	0.885	0.541	0.541
Dermoscopy hair	balanced	0.785	0.850	0.722	0.722
Dermoscopy hair	mask+SPLINCE	0.598	0.572	0.621	0.572
Dermoscopy hair	U-MtE+bal.	0.778	0.860	0.695	0.695
Dermoscopy hair	U-MtE prot.	0.718	0.904	0.502	0.502
Dermoscopy hair	SPLINCE	0.600	0.575	0.637	0.575

Table 75: Remedy analyses whose per-image predictions were not saved, so their intervals could not be re-estimated with the crossed bootstrap: **point estimates only**.

Analysis	Cohort / run	Contrast (Trap A, reversed)	Point estimate
LaMa inpainting (oracle masks)	thyroid	lama – mask	+0.025
	thyroid	mask_lama – mask	+0.226
	thyroid	mte_protect – mask_lama	−0.009
	thyroid	mte_balanced – mask_lama	+0.162
	thyroid	lama – erm	+0.136
	ovary	lama – mask	+0.027
	ovary	mask_lama – mask	+0.123
	ovary	mte_protect – mask_lama	−0.080
	ovary	mte_balanced – mask_lama	+0.067
	ovary	lama – erm	+0.158
	isic	lama – mask	+0.145
	isic	mask_lama – mask	+0.127
	isic	mte_protect – mask_lama	−0.086
	isic	mte_balanced – mask_lama	+0.123
	isic	lama – erm	+0.107
text-prompted erasure	thyroid	mask_text_erase – mask	+0.005
	thyroid	mte – mask	+0.204
	thyroid	mask_text_erase_balanced – balanced	+0.056
	ovary	mask_text_erase – mask	+0.008

Analysis	Cohort / run	Contrast (Trap A, reversed)	Point estimate
SLAS	ovary	mte – mask	+0.215
	ovary	mask_text_erase.balanced – balanced	–0.010
	capsule	mask_text_erase – mask	+0.001
	capsule	mte – mask	–0.236
	capsule	mask_text_erase.balanced – balanced	+0.066
	hair (DermLIP)	mask_text_erase – mask	+0.004
	hair (DermLIP)	mte – mask	–0.076
	hair (DermLIP)	mask_text_erase.balanced – balanced	–0.112
	hair	tok_mask – tok_erm	+0.030
	hair	mts – tok_mask	+0.022
	hair	slas – tok_erm	+0.008
	hair	mts_oracle – tok_mask	+0.015
	hair	mts_balanced – tok_balanced	+0.051
	thyroid	tok_mask – tok_erm	+0.059
	thyroid	mts – tok_mask	+0.081
erasure + DFR (U10)	thyroid	slas – tok_erm	+0.007
	thyroid	mts_oracle – tok_mask	+0.047
	thyroid	mts_balanced – tok_balanced	+0.070
	thyroid	mte_dfr – dfr	+0.101
	thyroid	mte_dfr – mask_dfr	+0.007
	thyroid	mask_dfr – dfr	+0.095
	capsule	mte_dfr – dfr	+0.080
	capsule	mte_dfr – mask_dfr	+0.000
	capsule	mask_dfr – dfr	+0.080
	hair	mte_dfr – dfr	–0.039
	hair	mte_dfr – mask_dfr	+0.014
	hair	mask_dfr – dfr	–0.053
pseudo-group balancing (U8)	thyroid	umte_pbal reversed AUROC	0.663
	thyroid	mte reversed AUROC	0.622
	thyroid	pbal reversed AUROC	0.409
	thyroid	erm reversed AUROC	0.289
	hair	umte_pbal reversed AUROC	0.510
	hair	mte reversed AUROC	0.506
	hair	pbal reversed AUROC	0.498
	hair	erm reversed AUROC	0.491
chest drains, RAD-DINO	drains	mte – mask	+0.029
	drains	mte_protect – mask	+0.030
	drains	mte – mte_aug	+0.031
	drains	mte – jtt	+0.094
	drains	mte.balanced – balanced	–0.079
location-adaptive selection (U13)	thyroid	auto – mask	+0.395
	thyroid	auto – dfr	+0.057
	capsule	auto – mask	+0.168
	capsule	auto – dfr	+0.047
	hair	auto – mask	+0.251
split-validation selection (U14)	hair	auto – dfr	–0.038
	chest drains	auto – mask	+0.299
	chest drains	auto – dfr	–0.144
	thyroid	auto – mask	+0.398
	thyroid	auto – dfr	+0.060
	capsule	auto – mask	+0.187
	capsule	auto – dfr	+0.066
	ovary	auto – mask	+0.170
	ovary	auto – dfr	–0.007
	hair	auto – mask	+0.234
	hair	auto – dfr	–0.055
	chest drains	auto – mask	+0.417
	chest drains	auto – dfr	–0.026

6.5.9 Robustness of the remedy tests to leakage groups

Table 76. With the stricter embedding-based groups of Stage 4, the thyroid remedy tests are unchanged; in the ovary, erase > augment (P3) and U-MtE+balancing > balancing (P4) lose significance: “erase beats augment” is robust in thyroid and hair but fragile in the ovary.

Table 76: The remedy tests re-run with the stricter embedding-based leakage groups (cohorts without patient identifiers).

Cohort	Test	With embedding-based leakage groups	CI > 0
Thyroid	P2 U-MtE (protected) ζ mask	+0.251 [+0.215, +0.284]	yes
Thyroid	P3 erase ζ augment	+0.192 [+0.160, +0.223]	yes
Thyroid	P4 U-MtE+balanced ζ balanced	+0.152 [+0.127, +0.177]	yes
Ovary	P2 U-MtE (protected) ζ mask	+0.051 [+0.020, +0.079]	yes
Ovary	P3 erase ζ augment	+0.024 [-0.014, +0.062]	no
Ovary	P4 U-MtE+balanced ζ balanced	+0.038 [-0.005, +0.081]	no
Capsule	P2 U-MtE (protected) ζ mask	-0.006 [-0.022, +0.014]	no
Capsule	P3 erase ζ augment	-0.128 [-0.158, -0.098]	no
Capsule	P4 U-MtE+balanced ζ balanced	+0.101 [+0.071, +0.132]	yes

6.5.10 Fully fine-tuned networks

Table 77. Applying U-MtE *after* fine-tuning does nothing (-0.000 [-0.001 , $+0.001$]): unconstrained fine-tuning builds a task-specific caliper feature that generic overlays do not move. Remedies must act *during* training. Each component alone works for only one architecture on thyroid (invariance for ResNet-50, $+0.139$ [$+0.096$, $+0.184$]; prediction consistency for ViT-S, $+0.061$ [$+0.036$, $+0.084$]). Combining projection, invariance and consistency (`umte_cons_ft`, registered before it was run) improves on fine-tuned masking for both: ResNet-50 $+0.195$ [$+0.153$, $+0.237$], ViT-S $+0.119$ [$+0.036$, $+0.198$] — without any artifact annotation. It fails on capsule (-0.026 [-0.056 , $+0.006$]), the pathology-like regime, and is not significant on the ovary ($+0.045$ [-0.036 , $+0.130$] with 15 clusters), where the fine-tuned network barely learns the task.

Table 77: Remedies for end-to-end fine-tuned networks: change in reversed-test AUROC versus fine-tuned masking (last column: versus the same fine-tuned body without erasure). Folds as bootstrap clusters; crossed CIs.

Fine-tuned network (Trap A)	invariance (umte_ft)	erase-while-fine-tuning (umte_ft)	prediction consistency (cons_ft)	all three (umte_cons_ft)	fine-tune, then erase
Thyroid, ResNet-50	+0.139 [+0.096, +0.184]	+0.186 [+0.144, +0.230]	+0.042 [+0.011, +0.074]	+0.195 [+0.153, +0.237]	-0.000 [-0.001, +0.001]
Thyroid, ViT-S/16	-0.000 [-0.026, +0.026]	+0.032 [-0.093, +0.144]	+0.061 [+0.036, +0.084]	+0.119 [+0.036, +0.198]	-
Capsule, ResNet-50	-0.027 [-0.051, -0.006]	-	-	-0.026 [-0.056, +0.006]	-
Ovary, ResNet-50 (15 clusters)	-	-	-	+0.045 [-0.036, +0.130]	-

6.5.11 Unaltered data

Table 78. On the natural test sets, of 20 remedy–test-set comparisons with masking (all pairs), 5 favour the remedy and 8 favour masking; the rest are not significant. Protected U-MtE is slightly *worse* than masking on the natural thyroid split (the registered claims N1 and N2 were not supported). The remedies repair the constructed shortcut of the traps; on data where the shortcut is weak, they mostly cost a little.

Table 78: Remedies on the natural test sets (AUROC, all pairs, crossed CIs): no remedy improves on masking uniformly.

Natural test set	U-MtE – mask	U-MtE protected – mask	balancing – mask	U-MtE + balancing – mask
Thyroid (official split)	+0.004 [-0.012, +0.020]	-0.021 [-0.034, -0.008]	-0.017 [-0.064, +0.030]	-0.005 [-0.023, +0.013]
ISIC, BCN held out	+0.007 [+0.004, +0.010]	-0.019 [-0.024, -0.014]	+0.022 [+0.012, +0.031]	+0.010 [+0.006, +0.015]
ISIC, HAM held out	+0.013 [+0.009, +0.017]	-0.006 [-0.016, +0.004]	-0.032 [-0.049, -0.013]	+0.018 [+0.009, +0.028]
ISIC, MSK held out	-0.008 [-0.014, -0.002]	-0.003 [-0.012, +0.006]	-0.018 [-0.041, +0.004]	-0.016 [-0.024, -0.008]
Capsule (held-out frames)	-0.005 [-0.011, +0.001]	-0.023 [-0.029, -0.017]	-0.086 [-0.109, -0.065]	-0.008 [-0.013, -0.001]

6.5.12 The decision guide

Table 79: What carries the shortcut decides the fix.

Regime	Example	Recommended	How to diagnose it
Artifact outside the ROI	calipers or rulers beside the lesion	ROI masking	measure the overlap r of detected artifacts
In-ROI, shortcut in the artifact's pixels	calipers, rulers, synthetic tubes	mask, then U-MtE (no labels) or U-MtE with balancing (labels); LaMa if artifact masks exist	oracle removal of the artifact recovers most of the gap
In-ROI, artifact resembles the pathology	capsule debris versus fibrin	protected U-MtE (no labels); masking + DFR (labels)	large fitted artifact–disease overlap; unprotected erasure lowers clean AUROC
Shortcut carried by correlates	dermoscopic hair, chest drains	DFR or group balancing (labels); JTT without labels	oracle removal recovers little
Encoder ignores the out-of-ROI artifact	chest tube with RAD-DINO	masking unnecessary outside	reversed $\approx$ clean AUROC at no overlap
Fully fine-tuned network	ResNet-50, ViT-S	erase while fine-tuning (projection + invariance + consistency)	post hoc erasure is null

No row claims an improvement on unaltered data: the guide tells a practitioner which method fits which regime and how to diagnose the regime, not which method to use everywhere.

6.6 Validating the artifact labels: the clinician audit

Every trap depends on automatic artifact labels (a rule-based caliper detector, a debris probe, published semi-automatic hair masks) that have so far been checked only visually by a non-clinician. A blinded audit package is ready in `audit/`:

- **Sample.** 240 images: 4 cohorts $\times$ 3 cells (artifact inside the ROI, outside it, artifact-free, by the automatic labels) $\times$ 20 (10 positive, 10 negative diagnosis), fixed seed, the 518-px images the models saw.
- **Blinding.** Every image and its overlay (ROI outline, automatic artifact mask) carry a neutral identifier; the key is kept outside the repository and fixed in advance by its SHA-256 (4b5db1521f1fc214...).
- **Task** (about 30–40s per image, 2.5 h in total): artifact present?, location relative to the ROI?, is the red mask correct?, is the green ROI correct? The sheet is `audit/review_sheet.csv`.
- **Analysis.** `audit/analyse_audit.py` computes precision and recall of the automatic labels, location agreement, ROI correctness and, with a second reviewer, Cohen's κ , and writes Table 80.
- **Licences** (Table 81). Images whose redistribution is not confirmed (TN3K thyroid images; overlays that draw the ISIC 2019 hair masks) are regenerated locally from the original data by the sampling script instead of being committed.

Rated so far: 0 of 240.

Table 80: Image audit of the automatic labels (to be filled).

Audit not yet performed. The table of precision, recall, location agreement, Cohen's κ and ROI-mask correctness per cohort is written here by `audit/analyse_audit.py` once the review sheet has been filled.

Table 81: Licence decisions for publishing images (`audit/LICENCES.md`); unclear means not published.

Dataset	Licence (source)	Committed?	Attribution
ISIC 2018 Task 1/2, ISIC 2019 images (incl. HAM10000, BCN_20000, MSK)	CC BY-NC 4.0 (challenge.isic-archive.com/data)	Yes (raw images)	ISIC Challenge datasets 2018/2019; Tschandl et al. 2018 (HAM10000); Combalia et al. 2019 (BCN_20000); Codella et al. (MSK)
HAM10000 lesion segmentations	CC BY-NC 4.0 (Harvard Dataverse doi:10.7910/DVN/DBW86T)	Yes (as ROI outlines)	Tschandl et al. 2020
ISIC 2019 hair/ruler masks (Wegley et al., Scholars' Mine doi:10.71674/man1-qa33)	no licence stated on the repository page	No — every overlay that draws these masks stays local	—
TN3K images, nodule masks; TNCD labels	not stated for the data; the GitHub repositories (TRFE-Net, ACL) license their *code* under MIT only	No — all thyroid images local	—
MMOTU OTU_2d	CC BY 4.0 (figshare doi:10.6084/m9.figshare.25058690)	Yes	Zhao et al. 2022, MMOTU
SEE-AI project dataset	CC BY 4.0 (Kaggle 'capsuleyolo/kyucapsule', the project's account)	Yes	Yokote et al. 2023/2024, SEE-AI project
Capsule expert contamination masks	CC BY 4.0 (figshare 27645021)	Yes	figshare 27645021
NIH ChestX-ray14	not used in the audit or example figures	—	—

6.7 The thesis manuscript

Title *Masking cannot remove what is inside the region of interest: location-dependent shortcut mitigation across dermoscopy, ultrasound and capsule endoscopy.* Target: *Medical Image Analysis*. Main text 22 pages (review format), supplement 18 pages; abstract 249 words (limit 250). An earlier version centred on U-MtE in NeurIPS format exists (`paper_neurips/`) but is not the current manuscript.

Structure Introduction; data and setup; the location law (controlled, real, transplant, matched, adjusted, replication including new patients, chest-radiograph boundary); mechanism; theory (held-out prediction); consequences on unaltered data (operating points; the thyroid subgroup labelled as exploratory confirmation); the decision guide; robustness; discussion with limitations. U-MtE, its ablations and the remedy results are in the supplement, with the registration table, the operating points of every cohort, the literature comparison, the covariate balance and the CLAIM checklist.

Why U-MtE is not the headline It repairs the constructed shortcut in the traps and fails to transfer uniformly to unaltered data (above). The finding that holds everywhere is the location law; the remedies are best presented as a decision guide.

Number audit Every interval in the main text and supplement must match an interval of a result file produced with the crossed bootstrap, together with its estimate; every other three-decimal number must occur in a result file. Result: 367 intervals and 970 numbers checked, 0 failed to trace. The six progress reports are checked the same way.

CLAIM 2024 (Tejani et al., 2024) 43 items: 35 yes, 3 partial, 2 no, 3 not applicable (partial or no: items 13, 18, 21, 33, 36). The remaining gaps are acquisition protocols the providers did not release, the absence of external ultrasound and capsule cohorts, and the clinician audit.

Related work and novelty The closest prior work is the dermoscopy debiasing literature (Bisoto et al., 2020, 2023; Nauta et al., 2022), linear concept erasure (Belrose et al., 2023; Ravfogel et al., 2020), covariance-preserving erasure (Holstege et al., 2025) and biased-prompt projection (Chuang et al., 2023). What is new: the overlap as the controlled variable and the location law across modalities; the causal and theoretical account; generic overlays used to *estimate* a nuisance subspace rather than as augmentation; and the ordering mask-then-erase. Disease protection is an adaptation of covariance-preserving projection and is presented as such.

6.7.1 Draft submission statements (for the author to confirm)

Data availability All datasets are public: ISIC 2018, 2019 and 2020 (CC BY-NC 4.0), HAM10000 lesion masks (CC BY-NC 4.0), the published ISIC 2019 hair/ruler masks, TN3K with TNCD labels, MMOTU (CC BY 4.0), SEE-AI and its expert contamination masks (CC BY 4.0), NIH ChestX-ray14 and RANZCR-CLiP (competition terms). Derived masks, cohort files and all result files are in the repository; images are not redistributed where the licence does not allow it.

Code availability Code, registrations with their commit history, result files and the scripts that regenerate every table, figure and number are in the public repository.

Ethics Retrospective analysis of public, de-identified datasets released for research; no new data were collected and no patient was contacted. Whether a formal waiver is needed must be confirmed with the institution.

Use of AI assistance The experiments, analysis code, reports and parts of the manuscript text were produced with the assistance of an AI system (Claude, Anthropic) working under the author’s direction; the author designed the study, reviewed every result and is responsible for the content. The wording should follow the journal’s policy on generative-AI disclosure.

6.8 Limitations of the work as a whole

1. Artifact labels come from detectors and a probe, audited visually by a non-clinician; the clinician audit is pending.
2. Three cohorts publish no patient identifiers; near-duplicate and embedding grouping are proxies.
3. Reversed environments are constructed by subsampling; natural test sets carry weak shortcuts.
4. Frozen linear probes are below published fine-tuned models in absolute performance (thyroid).
5. External validation exists for dermoscopy only.
6. The theory is fitted per model and cannot see correlate-carried shortcuts.
7. Disease protection needs diagnosis labels of artifact-free images; balanced variants need image-level artifact labels.
8. Some analyses (chest-radiograph devices and drains, LaMa, text prompts, SLAS, selectors) could not be re-estimated with the crossed bootstrap and are reported as point estimates.
9. Commit times of registrations are machine-set; only the registrations pushed to GitHub before running carry an external time stamp.

6.9 What failed or changed

- **No remedy transfers uniformly to unaltered data;** protected U-MtE did not meet its natural-data claims.

- **U-MtE fails for pathology-like artifacts** without protection, and for correlate-carried shortcuts at all.
- **Plain U-MtE failed on the dermoscopy ruler and on both MedSigLIP sweeps**; protection fixed only the debris one.
- **Post hoc erasure on fine-tuned networks is null**; single-component fine-tuning remedies transfer to one architecture only.
- **Text-prompted erasure is a near no-op.**
- **Erase > augment is fragile in the ovary** (not significant after Holm, and not with embedding groups).
- **The clinician audit has not been done.**
- **Two committed figures** (`report/figures/data.thyroid.png`, `data_isic.png`) predate the licence check and conflict with it; they need a decision.

6.10 Anticipated questions

Why erase instead of simply augmenting with the overlays? Augmentation teaches the head that the *generated* overlays are uninformative; the real artifact is a different pattern and keeps its association. Erasure removes the whole subspace in which foreign overlays move the representation, which also contains the real artifact’s direction when the two share it. The ablation (erase $\gg$ augment on thyroid with three encoders) confirms this.

Why generic overlays and not artifact templates? Templates need examples of the artifact, and an artifact detector to find them; the aim is a method that needs only what masking needs. Templates do help when available (the insert-then-erase precursor), but the generic library works for calipers in every encoder family without any artifact example.

Why does U-MtE fail on capsule, and why does protection fix it only partly? Debris and erosion fibrin share colour and texture, so the overlay subspace overlaps the disease (ρ large). Protection keeps the label directions estimated on artifact-free images, which removes the damage; it cannot create evidence that masking and erasure together removed, so it only reaches masking’s level.

Why does nothing pixel-based help for hair? Because most of the hair shortcut is not in the hair pixels (Stage 3: oracle removal of every hair patch gains little). A method that removes pixels or pixel-induced directions cannot remove an association carried by age, sex and body site.

Why is fine-tune-then-erase null but erase-while-fine-tuning not? After unconstrained fine-tuning on correlated data the network encodes the caliper along task-specific directions that generic overlays do not move; estimating the overlay subspace on the network’s own features *during* training, and requiring overlay-invariant features and predictions, stops that feature from forming.

If no remedy helps on unaltered data, what is the practical contribution? The practical contribution is the diagnosis: know where the artifacts sit, do not rely on masking when they sit inside the ROI, and pick the remedy by what carries the shortcut. On data with a strong in-ROI shortcut — the situation the traps construct and the thyroid operating points show — the right remedy makes a large difference.

Why keep SLAS and text prompts if they do little? Both were registered; negative results tell a reader which modern alternatives do not solve the problem, and SLAS’s oracle ceiling is itself evidence that hair is correlate-carried.

6.11 Conclusion

Supported. For frozen foundation-model features, U-MtE removes in-ROI shortcuts carried by distinct overlays without artifact annotation (thyroid with three encoder families and three sizes; ovary with MedSigLIP); erasure beats augmentation with the same overlays; protection makes it safe for pathology-like artifacts; with image-level labels, U-MtE with balancing or masking with DFR is the most robust choice. For fine-tuned networks, erasing while fine-tuning works on thyroid for both architectures. **Qualified.** No remedy improves on masking uniformly on unaltered data; the remedies are therefore a decision guide, not a headline method. **Not supported.** U-MtE for correlate-carried shortcuts; post hoc erasure after fine-tuning; text-prompted erasure. **Not yet available.** A clinician’s audit of the artifact labels; external validation in ultrasound and capsule endoscopy.

6.12 Questions for the supervisor

1. Should the remedies stay in the thesis as a decision guide within the main work, or form a separate chapter (or a second paper) with U-MtE as its contribution?
2. Who should perform the image audit (a dermatologist, a sonographer and a gastroenterologist?), and must it be completed before submission?
3. Is *Medical Image Analysis* the right target, or would a clinical-AI venue suit the operating-point results better?
4. Are the draft data-availability, ethics and AI-assistance statements acceptable to the institution?

6.13 Remaining work before submission

(i) Run the clinician audit and replace Table 80 with the output of `audit/analyse_audit.py`; (ii) decide on the two figures that conflict with the licence rule; (iii) confirm the submission statements; (iv) settle the supervisor’s decisions above; (v) deposit a snapshot of the repository with its registrations and tag the submitted version.

7 Conclusion

7.1 What the work shows

1. **Masking cannot remove what is inside the region of interest.** When only the artifact’s position changes, masking’s benefit falls with its overlap with the ROI: in the controlled dermoscopy sweep from +0.184 [+0.117, +0.250] at no overlap to −0.129 [−0.195, −0.064] at full overlap (interaction +0.313 [+0.258, +0.370]), and the interaction is positive in 10 of 11 controlled sweeps across five modalities and four encoders (Stage 2). Whether the lost benefit turns into harm depends on how much disease context masking also removes: it harms at full overlap in 7 sweeps.
2. **The mechanism is retention with amplified reliance.** Occlusion alone does not harm; harm follows the visibility of the artifact under mask dilation; and on the same images the masked model reacts to an in-ROI ruler about 2.1 times as strongly as ERM (+0.255 [+0.221, +0.287]). Masking removes the competing evidence and leaves the in-ROI artifact as the strongest remaining predictor (Stage 2).
3. **The law holds for real artifacts.** With real hair, sonographer calipers and capsule debris, masking is worth far less when the artifact lies inside the ROI: crossover positive in 9 of 10 cohort–encoder pairs and in all four pre-specified location tests (Holm), with DINOv2, MedSigLIP and a CNN (Stage 3). Only for dermoscopy hair does in-ROI masking actually harm; for calipers and debris it helps less.
4. **It is caused by location, not by which images carry the artifact.** The in-ROI and out-of-ROI images differ strongly (SMD up to 2.574), yet the crossover survives covariate matching and doubly robust adjustment in all three feasible cohorts (thyroid +0.238 [+0.184, +0.291]), and pasting the same real artifact inside or outside the ROI of the same images reproduces the location effect in all four cohorts. A neutral paste reproduces most of that transplant effect, which shows that *any* label-correlated pattern inside the ROI defeats masking; the artifact’s own content adds a significant part for hair and calipers (Stage 4).
5. **It matters on unaltered data and at the bedside.** On unaltered test sets masking harms the shortcut-conflicting cases in three of five (thyroid −0.095 [−0.165, −0.022]); on the patient-disjoint thyroid split it lowers sensitivity at every validation-fixed operating point (OP2 −0.202 [−0.300, −0.100]), most for malignant nodules with a caliper on the nodule (−0.372 [−0.499, −0.236]; a post hoc subgroup, exploratory) (Stage 5).
6. **It is general.** The crossover replicates at three model sizes (larger models exploit the in-ROI caliper more), in fine-tuned networks, in the zero-shot scores of vision–language models, and on new patients (ISIC 2020, +0.250 [+0.174, +0.327]) (Stage 5).
7. **It is predictable.** A two-parameter linear-Gaussian model fitted to each model’s clean and correlated AUROC predicts the held-out reversed AUROC with mean error 0.052 against 0.129 for the best heuristic, gets every real-trap crossover sign right, and explains when masking harms (it removes disease context) and why the crossover is always positive. By its own pre-registered rule it is a qualitative account; it cannot see shortcuts carried by correlates of the artifact (Stage 5).
8. **What carries the shortcut decides the fix.** For frozen foundation-model features, the annotation-free U-MtE repairs overlay-like in-ROI artifacts (thyroid Trap A +0.333 [+0.302, +0.364] versus ERM, masking +0.109 [+0.085, +0.133]), needs disease protection when the artifact resembles the pathology (capsule), and cannot help when the shortcut is carried by correlates (hair, chest drains), where label-based reweighting is required. With image-level labels, U-MtE with balancing or masking with DFR is the most robust choice. For fine-tuned networks, erasure must happen during training. On unaltered data no remedy improves on masking uniformly (5 of 20 comparisons favour a remedy, 8 favour masking) (Stage 6).

7.2 Contributions

- The *location law* of ROI masking, stated as a crossover, shown under controlled overlap in five modalities and with real artifacts in four diseases and three modalities, with chest radiography as a principled boundary.
- Its mechanism (retention with amplified reliance), its causal identification on fixed images, and its clinical consequence at validation-fixed operating points.
- A closed-form account that matches held-out results and explains every regime observed.
- A decision guide for practitioners and an annotation-free remedy (U-MtE) with a public tool, tested against the relevant published baselines.
- A methodological correction: the per-seed bootstrap inherited from the pilot understates uncertainty whenever seeds share test images; the crossed estimator fixes it and is used throughout.

7.3 Claims the work does not make

- That masking always harms in-ROI (it helps a little for calipers and debris).
- That the theory predicts magnitudes (it is a qualitative account that closely matches held-out results).
- That U-MtE improves deployment performance (it does not on unaltered data), or works post hoc on fine-tuned networks.
- That the transplant measures the harm of a particular real artifact (most of it is the paste).
- That the cohorts without patient identifiers are patient-independent.

7.4 Limitations and open work

Artifact labels come from detectors and a probe that a non-clinician audited visually; the blinded clinician audit is prepared but not done. Three cohorts publish no patient identifiers. Reversed environments are constructed by subsampling. Frozen linear probes are below published fine-tuned models on thyroid. External validation exists for dermoscopy only. Some analyses could only be kept as point estimates. The open work before submission is the clinician audit, the institutional confirmation of the submission statements, and the decisions on thesis structure listed at the end of Stage 6.

7.5 Take-home message

Before masking an image to its region of interest to remove a shortcut, find out where the artifacts sit. If they sit outside the ROI, masking works. If they sit inside it, masking keeps the artifact and removes everything that competed with it, so the model relies on the artifact more; the remedy must then be chosen by what carries the shortcut — the artifact’s pixels, a pathology-like appearance, or correlates of the artifact.

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